

TMS320DM270

Digital Media Processor

Register Manual Ver. β 0.7C
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preliminary

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HISTORY

Revision	Date	Notes
B 0.0	25 Jun 02	Initial revision.
B 0.1	19 Aug 02	Added 3A, NTSC/PAL encoder
B 0.2	23 Aug 02	Revised NTSC/PAL encoder
B 0.3	05 Sep 02	Changed highlight color for internal registers/bits
B 0.4	05 Oct 17	Revised 3A Engine
B 0.5	25 Oct 02	Added GIO FSEL description, revised clock controller, MMCIF and 3A Engine.
B 0.6	30 Oct 02	Revised CCD controller, Preview Engine, NTSC/PAL encoder and Serial I/F.
B 0.7	28 Feb 03	Unified into English version.

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1.Timer

TMS320DM270 has four timers. Respective timer output is input to the interrupt, timer0 is connected to interrupt0, timer1 is connected to interrupt1 and similarly timer3 is connected to interrupt3. When a timer count reaches 0, an interrupt is generated. There are one shot mode and the free run mode in the timer and when a timer count reaches 0, only one interrupt is generated in one shot mode, and in the free run mode, after generating the interrupt, a timer count is loaded again and count operation is repeated. The clock source to the timer can select an ARM clock or an external clock by CLKC (0x30884), and an actual count operation is divided by the value set by the TMPSx register. The count value is set by the TMVALx register. Moreover, when the timer is set in one shot mode, a timer is started by writing "1" to a TMTRGx register. In this section, x is 0 to 3, and it shows the respective timer from 0 to 3.

TMMD0 (0003:0000) Offset : 0x0

TMMD1 (0003:0080) Offset : 0x0

TMMD2 (0003:0100) Offset : 0x0

TMMD3 (0003:0180) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TSTx5	TSTx4	TSTx3	TST2	TST1	TST0	TMS1	TMS0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:2	TST5-TST0	Bit for test. Set to "0"
1:0	TMS1-TMS0	Timer mode select 00: Stop 01: One shot 10: Free run 11: Reserved

TMPS0 (0003:0004) Offset : 0x2

TMPS1 (0003:0084) Offset : 0x2

TMPS2 (0003:0104) Offset : 0x2

TMPS3 (0003:0184) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	TPS9	TPS8	TPS7	TPS6	TPS5	TPS4	TPS3	TPS2	TPS1	TPS0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	1	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	TPS9-TPS0	Timer pre scalar value. The clock is divided by this value. The setting value + 1 shows a dividing ratio.

TMVAL0 (0003:0006) Offset : 0x3

TMVAL1 (0003:0086) Offset : 0x3

TMVAL2 (0003:0106) Offset : 0x3

TMVAL3 (0003:0186) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TV15	TV14	TV13	TV12	TV11	TV10	TV9	TV8	TV7	TV6	TV5	TV4	TV3	TV2	TV1	TV0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	TV15-TV0	Timer count value. The setting value + 1 shows count value. Interrupt is generated at clock cycle x (TMPS + 1) x (TMVAL + 1) cycle.

TMTRG0 (0003:0008) Offset : 0x4

TMTRG1 (0003:0088) Offset : 0x4

TMTRG2 (0003:0108) Offset : 0x4

TMTRG3 (0003:0188) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TTRG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use
0	TTRG	One shot trigger. The count operation of the timer begins when "1" is written while the timer is set to the one shot mode.

TMCNT0 (0003:000A) Offset : 0x5

TMCNT1 (0003:008A) Offset : 0x5

TMCNT2 (0003:010A) Offset : 0x5

TMCNT3 (0003:018A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CV15	CV14	CV13	CV12	CV11	CV10	CV9	CV8	CV7	CV6	CV5	CV4	CV3	CV2	CV1	CV0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	CV15-CV0	Current value of timer is shown.

2. Serial Interface

TMS320DM270 has a 2-channel clock synchronous 3 line serial interface. Data transfer unit is eight bits width, and its 3 line interface consists of the clock line, the output data line, and the input data line. Data is transmitted at falling edge of clock and received at rising edge. The data transfer clock speed can be set with ARM CLK/2 acting as reference clock. The selection of LSB first/MSB first is programmable. Moreover, the polarity of the clock can be selected when it is not transferring data. The serial I/F 0 can use DMA to transfer large quantities of data between SDRAM and SIF 0. After serial communication completed, interrupt occurs. Master mode; conducting serial communication outputting clock and enable from DM270. Slave mode; conducting serial communication where DM270 receives clock and enable as input. The x of this section is 0 to 1, and it shows the respective serial interface from 0 to 1.

TXDATA0 (0003:0200) Offset : 0x0

TXDATA1 (0003:0280) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TXD7	TXD6	TXD5	TXD4	TXD3	TXD2	TXD1	TXD0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:0	TXD7-TXD0	Transmitter data register. When the data is set to this register, transmitting/receiving starts. While reading, the written value can be read. When SIOEN is 0 or XMIT is 1, writing is not possible.

RXDATA0 (0003:0202) Offset : 0x1

RXDATA1 (0003:0282) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	XMIT	RXD7	RXD6	RXD5	RXD4	RXD3	RXD2	RXD1	RXD0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit. Not in use.
8	XMIT	Transmitting/Receiving flag bit. 0 : Transmitting/Receiving completed. 1 : Transmitting/Receiving in progress
7:0	RXD7-RXD0	Receiving data register. Set the data in TXDATA register. After Transmitting/Receiving completed (occurrence of interrupt), the received data can be read.

SIOEN0 (0003:0204) Offset : 0x2

SIOEN1 (0003:0284) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SIOEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
0	SIOEN	Transmitting permission bit. When set to "1", transmitting is permitted. When it is prohibited, the data can't be written to TXDATA register. 0 : Transmitting prohibited 1 : Transmitting permitted

SIOMODE0 (0003:0206) Offset : 0x3**SIOMODE1 (0003:0286) Offset : 0x3**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	SLVEN	SIOCLR	SCLKM	MSB	MSSEL	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
R/W	-	-	-	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	X	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:13	RSV	Reserved bit. Not in use.
12	SLVEN	Monitor the external enable. External enable: SDEN0 (SIF0) GIO22(when SDEN1 is set) (SIF1)
11	SIOCLR	Register clearing bit. Reset serial port inside. All register value is kept.
10	SCLKM	Serial clock mode. The polarity of clock when not transferring data is set. 0: Clock level is 1 in non-transfer mode 1: Clock level is 0 in non-transfer mode
9	MSB	LSB first・MSB first bit (Bit order of transmitted or received data) 0: LSB first 1: MSB first
8	MSSEL	Switch between master and slave operation. 0: Master 1: Slave
0	BR7-BR0	Transfer rate setting bit. Bit Rate = ARM Clock/((set value+1)x 2)

DMATR0 (0003:0208) Offset : 0x4

Register for SIF0 only

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TRG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	TRG	DMA transfer starting bit. DMA transfer starts when 1 is set. It is cleared to 0 automatically. Writing 0 has no meaning. It is no use to set 1 while transferring.

DMAMODE0 (0003:020A) Offset : 0x5

Register for SIF0 only

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRM1	PRM0	DIR	TXSZ10	TXSZ9	TXSZ8	TXSZ7	TXSZ6	TXSZ5	TXSZ4	TXSZ3	TXSZ2	TXSZ1	TXSZ0
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:14	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
13:12	PRM1-PRM0	DMA transfer byte order 00 = D3D2D1D0 -> D3D2D1D0 01 = D3D2D1D0 -> D0D1D2D3 10 = D3D2D1D0 -> D2D3D0D1 11 = D3D2D1D0 -> D1D0D3D2
11	DIR	DMA transfer direction 0 : SIF to SDRAM 1 : SDRAM to SIF
10:0	TXSZ10-TXSZ0	Number of transfer byte. Only even numbers can be specified.

DMASTAD0 (0003:020C) Offset : 0x6

Register for SIF0 only

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	STAD15	STAD14	STAD13	STAD12	STAD11	STAD10	STAD9	STAD8	STAD7	STAD6	STAD5	STAD4	STAD3	STAD2	STAD1	STAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	STAD15-STAD0	DMA starting address lower 16 bits of SDRAM. Only even numbers can be specified. This corresponds to relative address.

DMASTAD1 (0003:020E) Offset : 0x7

Register for SIF0 only

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	STAD26	STAD25	STAD24	STAD23	STAD22	STAD21	STAD20	STAD19	STAD18	STAD17	STAD16
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit. Not in use.
10:0	STAD26-STAD16	DMA starting address high 11 bits of SDRAM. This corresponds to relative address.

DMASTAT0 (0003:0210) Offset : 0x8

Register for SIF0 only

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RUN	RSV	RSV	RSV	RSV	REST10	REST9	REST8	REST7	REST6	REST5	REST4	REST3	REST2	REST1	REST0
R/W	R	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	RUN	DMA transfer flag 0: Transfer completed. 1: Transfer in progress
14:11	RSV	Reserved bit. Not in use.
10:0	REST10-REST0	It shows DMA transfer byte remaining amount

3. UART

TMS320DM270 supports a two-channel two line (RX, TX) start-stop synchronization serial port (UART). It has the FIFO of 32 bytes inside for each transmitting + receiving side, and interrupt is generated by various FIFO statuses and the factor of errors, etc. Data is 8 or 7 bits and can choose the even number and odd number with parity bit or not, and stop bit of 1 or 2-bits. The timing of transmitting/receiving uses the clock of ARM, and can set up the bit rate arbitrarily by using the integer multiples of 1/16 of an ARM clock. x in this section is 0 or 1, and shows UART0 or UART1, respectively. PLLIN (CLKC) can also be used as reference clock instead of ARM.

DTRR0 (0003:0300) Offset : 0x0

DTRR1 (0003:0380) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RVFx	BFx	FEx	ORFx	PEFx	DTRx7	DTRx6	DTRx5	DTRx4	DTRx3	DTRx2	DTRx1	DTRx0
R/W	-	-	-	R	R	R	R	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-13	RSV	Reserved bit. Not in use.
12	RVFx	Receive word valid flag. It shows whether the read receiving word is valid or invalid. 0: Invalid receiving word 1: Valid receiving word
11	BFx	Break flag 0: No break 1: Break is detected.
10	FEx	Framing error 0: No framing error 1: With framing error
9	ORFx	Overrun flag 0: No overrun flag 1: With overrun flag
8	PEFx	Parity error flag 0: No parity error 1: With parity error
7-0	DTRx7-DTRx0	Transmitting/Receiving data register. It can read the oldest data stored in receiving FIFO at the reading time. At the same time, the status of this data can be seen in each bit. When written to this register, the data is stored to transmitting FIFO and the data is transferred. While writing, it is necessary to confirm that the FIFO is not full, and then write to it.

BRSR0 (0003:0302) Offset : 0x1

BRSR1 (0003:0382) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BRx15	BRx14	BRx13	BRx12	BRx11	BRx10	BRx9	BRx8	BRx7	BRx6	BRx5	BRx4	BRx3	BRx2	BRx1	BRx0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	BRx15-BRx0	Bit rate setting register. The bit rate of data transfer is set up. The basic clock is 1/16 of input clock (Input clock can be ARM clock or PLLIN. This selection is done in CLKC register.), and it is divided by setting value of this register. Set up value + 1 is the dividing ratio.

MSR0 (0003:0304) Offset : 0x2**MSR1 (0003:0384) Offset : 0x2**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RFTIEx	TFTIEx	REIEx	RSV	TOICx1	TOICx0	RSV	RSV	RSV	RSV	RSV	PEBx	PSBx	SBLSx	RSV	CLSx
R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	-	R/W
Default	0	0	0	0	0	0	-	-	-	0	0	0	0	0	-	0

Bit	Register Name	Description
15	RFTIEx	Receiving FIFO trigger/interrupt enable. When the receiving FIFO is more than trigger level, the interrupt is generated. 0: Disable of the above mentioned interrupt 1: Enable of the above mentioned interrupt
14	TFTIEx	Transmitting FIFO trigger/interrupt enable. When the transmitting FIFO is less than trigger level, the interrupt is generated. 0: Disable of the above-mentioned interrupt 1: Enable of the above mentioned interrupt
13	REIEx	Receiving data error interrupt enable. Even if one of these, parity error, overrun, frame error or receiving word of break exists in receive FIFO, the interrupt is generated. 0: Disable of the above-mentioned interrupt 1: Enable of the above-mentioned interrupt
12	RSV	Reserved bit. Not in use.
11-10	TOICx1-TOICx0	Time out interrupt enable. When the number of receiving data does not reach to receiving FIFO trigger level and following time is passed, interrupt is generated. This is for getting last data not aligned FIFO trigger level. TOICx1 TOICx0 0 0 Time out interrupt disable 0 1 3 word time 1 0 7 word time 1 1 15 word time
9-5	RSV	Reserved bit. Not in use.
4	PEBx	Parity enable bit. Parity bit is applied at the end of data. 0: Disable (No parity) 1: Enable
3	PSBx	Parity select bit. The mode of parity is selected. 0: Even number 1: Odd number
2	SBLSx	Stop bit length select. The length of stop bit is selected. 0: 1 bit 1: 2 bit
1	RSV	Reserved bit. Not in use.
0	CLSx	Character length select. The size of data is specified. 0: 8 bit 1: 7 bit

RFCR0 (0003:0306) Offset : 0x3**RFCR1 (0003:0386) Offset : 0x3**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RFCBx	RDEFx	RSV	RSV	RSV	RTLx2	RTLx1	RTLx0	RSV	RSV	RWx5	RWx4	RWx3	RWx2	RWx1	RWx0
R/W	R/W	R/W	-	-	-	R/W	R/W	R/W	-	-	R	R	R	R	R	R
Default	0	0	-	-	-	0	0	0	-	-	0	0	0	0	0	0

Bit	Register Name	Description
15	RFCBx	Receive FIFO clear bit. When "1" is written, receive FIFO is cleared. After

		cleared, it returns to "0"
14	RDEFx	Receiving data error detecting flag. When the data causes the receiving error in receiving FIFO exists, it is set to "1". The data causes the receiving error doesn't exists from receive FIFO, flag is cleared to "0". It is the same as the bit 9 of the SRx register. When "1" is written to this bit, the receiving block is reset and after resetting, it is reset to "0".
13-11	RSV	Reserved bit. Not in use.
10-8	RTLx2-RTLx0	Setting the receiving FIFO trigger level. Set the trigger level of receiving FIFO trigger interrupt. RTLx2 RTLx1 RTLx0 1 1 1 Reserved 1 1 0 Reserved 1 0 1 32 1 0 0 24 0 1 1 16 0 1 0 8 0 0 1 4 0 0 0 1
7-6	RSV	Reserved bit. Not in use.
5-0	RW5-RW0	Receiving word counter. The number of receiving word stored in receiving FIFO is shown.

TFCR0 (0003:0308) Offset : 0x4**TFCR1 (0003:0388) Offset : 0x4**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TFCBx	RSV	RSV	RSV	RSV	TTLx2	TTLx1	TTLx0	RSV	RSV	TWx5	TWx4	TWx3	TWx2	TWx1	TWx0
R/W	R/W	-	-	-	-	R/W	R/W	R/W	-	-	R	R	R	R	R	R
Default	0	-	-	-	-	0	0	0	-	-	0	0	0	0	0	0

Bit	Register Name	Description
15	TFCBx	Transmitting FIFO clear bit. When "1" is written, transmitting FIFO is cleared. After cleared, it returns to "0".
14-11	RSV	Reserved bit. Not in use.
10-8	TTLx2-TTLx0	Transmitting FIFO trigger level setting. The trigger level of transmitting FIFO trigger interrupt is set. TTLx2 TTLx1 TTLx0 1 1 1 Reserved 1 1 0 Reserved 1 0 1 32 1 0 0 24 0 1 1 16 0 1 0 8 0 0 1 4 0 0 0 1
7-6	RSV	Reserved bit. Not in use.
5-0	TW5-TW0	Transmitting word counter. The number of receiving word stored in transmitting FIFO is shown.

LCR0 (0003:030A) Offset : 0x5**LCR1 (0003:038A) Offset : 0x5**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	UTSTx	RSV	RSV	RSV	RSV	RSV	BOCx	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-	R/W	-	-	-	-	-	R/W	-	-	-	-	-	-	-	-
Default	-	0	-	-	-	-	-	0	-	-	-	-	-	-	-	-

Bit	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	UTSTx	Test mode of our company. Please set to "0".
13-9	RSV	Reserved bit. Not in use.
8	BOCx	Break output control. Controls the TX output. 0: Normal data 1: TX is set to "0" compulsorily.
7-0	RSV	Reserved bit. Not in use.

SR0 (0003:030C) Offset : 0x6

SR1 (0003:038C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RFTIx	TFTIx	RFERx	TOIFx	RSV	RSV	RSV	RSV	RSV	RFNFx	TFEFx	TREFx
R/W	-	-	-	-	R	R	R	R	-	-	-	-	-	R	R	R
Default	-	-	-	-	0	1	0	0	-	-	-	-	-	0	1	1

Bit	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11	RTFIx	Receiving FIFO trigger indicator 0: The data within receiving FIFO is below trigger level. 1: The data within receiving FIFO is above trigger level.
10	TFTIx	Transmitting FIFO trigger indicator 0: The data within transmitting FIFO is above trigger level 1: The data within transmitting FIFO is below trigger level
9	RFERx	Receiving data error detecting flag. When the data causes the receiving error exists in the receiving FIFO, the flag is set to "1". The data causes receiving error from receiving FIFO doesn't exists, the flag is cleared to "0". It is the same as the bit 14 of the RFCRx register.
8	TOIFx	Time out interrupt flag. This shows that time out interrupt occurs in "1". This register is cleared to "0" when read.
7-3	RSV	Reserved bit. Not in use.
2	RFNFx	Receiving FIFO not empty flag. It shows that at least data more than 1 exists in receiving FIFO. 0: No data in receiving FIFO 1: Data in receiving FIFO
1	TFEFx	Transmitting FIFO empty flag. It shows that the transmitting FIFO is in empty status. 0: Transmitting FIFO is not empty 1: Transmit FIFO is in empty status
0	TREFx	Transmitting register empty flag. When both transmitting FIFO and transmitting shift out register are empty, it is set to "1". 0: Data in transmitting FIFO or shift out register 1: Both transmitting FIFO and shift out register are empty.

4. Watch Dog Timer

TMS320DM270 supports the watch dog timer for the detection of hang-up, etc. If the watch dog timer is not reset within the setting time, interrupt or reset is generated for ARM.

WDTMD (0003:0400) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TST5	TST4	TST3	TST2	TST1	OS	TST0	WDTEN
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:3	TST5-TST1	Bits for test. Set it to "0".
2	OS	Output select 0: Interruption 1: Reset
1	TST0	Bits for test. Set it to "0".
0	WDTEN	Watch dog timer enable bit 0: Disable 1: Enable

WDTRST (0003:0402) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	WDRT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	WDRT	Watch dog timer reset. When "1" is written, watch dog timer is reset.

WDTPRSCL (0003:0404) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	WPS9	WPS8	WPS7	WPS6	WPS5	WPS4	WPS3	WPS2	WPS1	WPS0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	1	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	WPS9-WPS0	Watch dog timer pre scalar value. Clock is divided by this value. Dividing ratio is setting value + 1.

WDTVALL (0003:0406) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WTV15	WTV14	WTV13	WTV12	WTV11	WTV10	WTV9	WTV8	WTV7	WTV6	WTV5	WTV4	WTV3	WTV2	WTV1	WTV0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	WTV15-WTV0	Watch dog timer count value. The setting value is counted, and interrupt or reset is generated for ARM. The detection time of watch dog timer is set by the combination with WDTPRSCL. Detection time = (WDTPRSCL+1) x (WTV15-WTV0) / (External clock input)

WDTEXRST (0003:0408) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	WERST
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	WERST	Watch dog timer reset signal External reset output pin (when GIO25 MRST is set) 0: External reset output pin is set to "Low" compulsorily. 1: Reset signal to ARM is output to External reset output pin.

5. MMC/SD Card Interface

The operation clock of MMC/SD Card Interface runs based on dividing of ARM clock. Refer to the section for clock controller for the details of clock setting.

Features of MMC/SD Card Interface

- Support MMC and SD cards by conforming to both MMC and SPI protocols
- One or four-bit width data communication
- Software oriented implementation for future extensions
- MMC peripheral runs at a divided rate than the CPU clock.
- Supports SDIO card

MMCCTL (0003:0480) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	PERMDX	PERMDR	RSV	DATEG1	DATEG0	SPIEN	CSEN	SPICRC	WIDTH	CMDRST	DATRST
R/W	-	-	-	-	-	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10	PERMDX	Endian select enable when writing 0: Little Endian is selected. 1: Big Endian is selected.
9	PERMDR	Endian select Enable when reading 0: Little Endian is selected. 1: Big Endian is selected.
8	RSV	Reserved bit. Not in use.
7-6	DATEG1-DATEG0	DAT3 Edge Detection Select 00: DAT3 edge detection is disabled. 01: DAT3 rising edge detection is enabled. 10: DAT3 falling edge detection is enabled. 11: DAT3 both edge detection is enabled.
5	SPIEN	SPI Mode Enable 0: MMC mode is selected. 1: SPI mode is selected.
4	CSEN	Card Select Enable when writing (in only SPI mode) 0: CS(DAT3) output is H (inactive). 1: CS(DAT3) output is L (active).
3	SPICRC	CRC Checking (in only SPI mode) 0: CRC checking is disabled. 1: CRC checking is enabled.
2	WIDTH	Data Bus Width (in only MMC mode) 0: Data bus has 1-bit, (only DAT0 is used). 1: Data Bus has 4-bit, (all DAT0-3 are used).
1	CMDRST	CMD Logic Reset 0: The CMD line portion is enabled. 1: The CMD line portion is disabled and in reset state.
0	DATRST	DAT Logic Reset 0: The DAT line portion is enabled. 1: The DAT line portion is disabled and in reset state.

MMCCLK (0003:0482) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CLKEN	CLKRT7	CLKRT6	CLKRT5	CLKRT4	CLKRT3	CLKRT2	CLKRT1	CLKRT0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8	CLKEN	CLK Enable 0: CLK pin is disabled and fixed L. 1: CLK pin is enabled.
7-0	CLKRT7-CLKRT0	Clock Pin Rate This indicates divider rate of the function clock MMC clock = function clock / 2*(ClockRate + 1)

MMCST0 (0003:0484) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	DATED	DRRDY	DXRDY	SPIERR	CRCRS	CRCRD	CRCWR	TOUTRS	TOUTRD	RSPDNE	BSYDNE	DATDNE
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11	DATED	DAT3 Edge Detect, bit is cleared when read by CPU 0: No DAT3 edge is detected. 1: DAT3 edge has been detected.
10	DRRDY	Data Receive Ready 0: MMCDRR is not ready. 1: MMCDRR is ready to be reading a new data by CPU or DMA. At DAT logic reset (DATRST=1), sending command with Data Receive/Transmit Clear(DCLR=1) or reading MMCDRR, the DRRDY bit is clear to 0.
9	DXRDY	Data Transmit Ready 0: MMCDXR is not ready. 1: MMCDXR is ready to be writing a new data by CPU or DMA. At DAT logic reset (DATRST=1), sending command with Data Receive/Transmit Clear(DCLR=1) or writing MMCDXR, the DXRDY bit is set to 1.
8	SPIERR	Data Error (in SPI mode) 0: No data error is received. 1: Data error token has been received.
7	CRCRS	CRC error (response) 0: No CRC error is detected. 1: CRC error has been detected.
6	CRCRD	CRC error (read data) 0: No CRC error is detected. 1: CRC error has been detected.
5	CRCWR	CRC error (write data) 0: No CRC error is detected. 1: CRC error has been detected.
4	TOUTRS	Time-Out (response) 0: No time-out occurs. 1: Timed-out has occurred during waiting response.
3	TOUTRD	Time-Out (read data) 0: No time-out occurs. 1: Timed-out has occurred during waiting data.
2	RSPDNE	Command / Response Done 0: No Receiving response done. 1: Response successfully has received or command has sent without response.
1	BSYDNE	Busy Done 0: No busy releasing done. 1: Released from busy state or expected busy not detected.

0	DATDNE	Data Done 0: No data transfer done. 1: Data transfer of specified length successfully done.
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Note1: The transition from 0 to 1 of each bit in MMCST0 may cause interrupt signal to the ARM. It depends on MMCIM setting, (see also MMCIM). Each status bit is cleared when reading by the ARM except DRRDY and DXRDY.

Note2: Because of independence of the command portion and the data portion, any command such as CMD0 (GO_IDLE_STATE) or CMD12 (STOP_TRANSMISSION) can be sent to the card, even if during block transfer. In this situation, the data portion will detect and wait releasing busy state only when the command sent was R1b (to be specifically, command with BSYEXP bit), otherwise it will keep on transferring data.

MMCST1 (0003:0486) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DAT3ST	DRFUL	DXEMP	CLKSTP	BUSY
R/W	-	-	-	-	-	-	-	-	-	-	-	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	0	0	0	1	0

Bit	Register Name	Description
15-5	RSV	Reserved bit. Not in use.
4	DAT3ST	DAT3 Status 0: The status of DAT3 is L level. 1: The status of DAT3 is H level.
3	DRFUL	DRRDY Data receive full 0: MMCDRR(data receive register) is not full. 1: MMCDRR(data receive register) is full.
2	DXEMP	DXRDY Data transmit empty 0: MMCDXR(data transmit register) is not empty. 1: MMCDXR(data transmit register) is empty.
1	CLKSTP	Clock Stop Status 0:CLK is active. 1:CLK is held L because of no sending data during CLKEN is 0, receive shift register is full, transmit shift register is empty.
0	BUSY	Busy 0: No busy signal is detected 1: Busy signal is detected after response with busy or during block write.

MMCIM (0003:0488) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	EDATED	EDRRDY	EDXRDY	ESPIERR	ECRCRS	ECRCRD	ECRCWR	ETOUTRS	ETOUTRD	ECMDNE	BSYDNE	DATDNE
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11	EDATED	DATED permission bit. 0: Prohibited 1: Permitted
10	EDRRDY	DRRDY permission bit. 0: Prohibited 1: Permitted
9	EDXRDY	DXRDY permission bit. 0: Prohibited 1: Permitted
8	ESPIERR	SPIERR permission bit. (in SPI mode) 0: Prohibited 1: Permitted

7	ECRCRS	CRCRS permission bit. (response) 0: Prohibited 1: Permitted
6	ECRCRD	CRCRD permission bit. (read data) 0: Prohibited 1: Permitted
5	ECRCWR	CRCWR permission bit. (write data) 0: Prohibited 1: Permitted
4	ETOUTRS	TOUTRS permission bit. (response) 0: Prohibited 1: Permitted
3	ETOUTRD	TOUTRD permission bit. (read data) 0: Prohibited 1: Permitted
2	ERSPDNE	RSPDNE permission bit. 0: Prohibited 1: Permitted
1	EBSYDNE	BSYDNE permission bit. 0: Prohibited 1: Permitted
0	EDATDNE	DATDNE permission bit. 0: Prohibited 1: Permitted

MMCTOR (0003:048A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TOR7	TOR6	TOR5	TOR4	TOR3	TOR2	TOR1	TOR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	TOR7-TOR0	Time-Out Count For Response (in MMC mode) 0000 0000: no time-out 0000 0001: 1 CLK clock 0000 0010: 2 CLK clocks 1111 1111: 255 CLK clocks Time-Out Count For Response (in SPI mode) 0000 0000: no time-out 0000 0001: 1 x 8 CLK clocks 0000 0010: 2 x 8 CLK clocks 1111 1111: 255 x 8 CLK clocks

MMCTOD (0003:048C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TOD15	TOD14	TOD13	TOD12	TOD11	TOD10	TOD09	TOD08	TOD07	TOD06	TOD05	TOD04	TOD03	TOD02	TOD01	TOD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	TOD15- TOD00	Time-Out Count For Data Read (in MMC mode) 0x0000: no time-out 0x0001: 1 CLK clock 0x0002: 2 CLK clocks

		 0xffff: 65535 CLK clocks Time-Out Count For Data Read (in SPI mode) 0x0000: no time-out 0x0001: 1 x 8 CLK clocks 0x0002: 2 x 8 CLK clocks 0xffff: 65535 x 8 CLK clocks
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MMCBLEN (0003:048E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	BLEN11	BLEN10	BLEN9	BLEN8	BLEN7	BLEN6	BLEN5	BLEN4	BLEN3	BLEN2	BLEN1	BLEN0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	BLEN11-BLEN0	Block Length Specifies the byte count of a data block. The value 0x000 is prohibited.

Note: MMCBLEN specifies the data block length. This value must be same as CSD register settings in the card, to be precise, READ_BL_LEN when block read, or WRITE_BL_LEN when block write.
When DMAM = '1', MMCBLEN must take multiples of 4.

MMCNBLK (0003:0490) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	NBLK15	NBLK14	NBLK13	NBLK12	NBLK11	NBLK10	NBLK9	NBLK8	NBLK7	NBLK6	NBLK5	NBLK4	NBLK3	NBLK2	NBLK1	NBLK0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	NBLK15-NBLK0	Number of Blocks. Specifies the total number of blocks to be transferred. The value 0x0000 means infinite blocks.

MMCNBLC (0003:0492) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	NBLC15	NBLC14	NBLC13	NBLC12	NBLC11	NBLC10	NBLC9	NBLC8	NBLC7	NBLC6	NBLC5	NBLC4	NBLC3	NBLC2	NBLC1	NBLC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-0	NBLC15-NBLC0	MMCNBLC is used in order to test the counter that operates with the value of MMCNBLK in a factory test or a chip evaluation. Reading is possible in order to check the number of blocks to be transferred. But writing to this register is prohibited by User.

MMCDRR (0003:0494) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DRR15	DRR14	DRR13	DRR12	DRR11	DRR10	DRR9	DRR8	DRR7	DRR6	DRR5	DRR4	DRR3	DRR2	DRR1	DRR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	DRR15-DRR0	This register is used for reading of receive data from card when block read.

		If an odd byte number of a data block is set up by MMCBLLEN, last byte is received to DRR[7:0].
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MMCDXR (0003:0496) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DXR15	DXR14	DXR13	DXR12	DXR11	DXR10	DXR9	DXR8	DXR7	DXR6	DXR5	DXR4	DXR3	DXR2	DXR1	DXR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	DXR15- DXR0	This register is used for writing of transmit data to card when block write. If an odd byte number of a data block is set up by MMCBLLEN, last byte is transmitted from DXR[15:8].

MMCCMD (0003:0498) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DCLR	INITCK	WDATX	STRMTP	DTRW	RSPFMT1	REP FMT0	BSYEXP	PPLEN	RSV	CMD5	CMD4	CMD3	CMD2	CMD1	CMD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	-	0	0	0	0	0	0

Bit	Register Name	Description
15	DCLR	Data Receive/Transmit Clear 0: No clear 1: Clear
14	INITCK	Initialize Clock. Before sending the command (specified in CMD), 80 cycles of dummy clock will be inserted. This is required for resetting a card after power-on. 0: No initialization clock 1: Initialization clock insertion
13	WDATX	With Data 0: No data transfer 1: With data transfer
12	STRMTP	Stream Transfer 0: Block data transfer 1: Stream data transfer
11	DTRW	Write 0: Data read 1: Data write
10-9	RSPFMT1- RSPFMT0	Response Format 00: No response 01: R1/R4/R5/R6 (48-bit length) 10: R2 (136-bit length) 11: R3 (48-bit length, without CRC)
8	BSYEXP	Busy Expected 0: No busy signal expected. 1: Busy signal expected or in case of transfer abort.
7	PPLEN	Push Pull Enable 0: Push Pull Driver of CMD line is disable. 1: Push Pull Driver of CMD line is enable.
6	RSV	Reserved bit. Not in use.
5-0	CMD5-CMD0	Command Index. The command to be sent to the CMD line.

Note: The commands to be sent to the CMD line are stored in MMCCMD. Write to MMCCMD triggers to start sending a command. Thus, the argument registers (MMCARGx) have to be written properly before MMCCMD writing. The content of this register will be kept after transfer to the shift register. Lower byte of MMCCMD is used for the command index, and upper byte is used for specifying type of command and response, etc. This upper byte is dealt as one pair with the lower byte.

MMCARGL (0003:049A) Offset : 0xD

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ARGL15	ARGL14	ARGL13	ARGL12	ARGL11	ARGL10	ARGL09	ARGL08	ARGL07	ARGL06	ARGL05	ARGL04	ARGL03	ARGL02	ARGL01	ARGL00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	ARGL15- ARGL0	Argument Register. MMCARGL and H is used for specifying the arguments to be sent with the command specified by MMCCMD. Writing to MMCCMD cause start event of sending operation, that is, MMCARG has to be set before writing to MMCCMD. The content of this register will be kept after transfer to the shift register. However, modification to MMCARG is not allowed during sending operation.

MMCARGH (0003:049C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ARGH15	ARGH14	ARGH13	ARGH12	ARGH11	ARGH10	ARGH09	ARGH08	ARGH07	ARGH06	ARGH05	ARGH04	ARGH03	ARGH02	ARGH01	ARGH00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	ARGH15- ARGH0	Argument Register. MMCARGL and H are used for specifying the arguments to be sent with the command specified by MMCCMD. Writing to MMCCMD cause start event of sending operation, that is, MMCARG has to be set before writing to MMCCMD. The content of this register will be kept after transfer to the shift register. However, modification to MMCARG is not allowed during sending operation.

Note: Bit position of MMCCMD[5-0], MMCARGL and MMCARGH.

MMRs	Bit position of command	
	[47]	start bit
	[46]	transmission bit
MMCCMD[5-0]	[45-40]	command index
MMCARGH	[39-24]	argument
MMCARGL	[23-8]	argument
	[7-1]	CRC7
	[0]	end bit

MMCRSP0 (0003:049E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP015	MMCR SP014	MMCR SP013	MMCR SP012	MMCR SP011	MMCR SP010	MMCR SP009	MMCR SP008	MMCR SP007	MMCR SP006	MMCR SP005	MMCR SP004	MMCR SP003	MMCR SP002	MMCR SP001	MMCR SP000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP015-000	Response Register0

MMCRSP1 (0003:04A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP115	MMCR SP114	MMCR SP113	MMCR SP112	MMCR SP111	MMCR SP110	MMCR SP109	MMCR SP108	MMCR SP107	MMCR SP106	MMCR SP105	MMCR SP104	MMCR SP103	MMCR SP102	MMCR SP101	MMCR SP100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP115-100	Response Register1

MMCRSP2 (0003:04A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP215	MMCR SP214	MMCR SP213	MMCR SP212	MMCR SP211	MMCR SP210	MMCR SP209	MMCR SP208	MMCR SP207	MMCR SP206	MMCR SP205	MMCR SP204	MMCR SP203	MMCR SP202	MMCR SP201	MMCR SP200
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP215-200	Response Register2

MMCRSP3 (0003:04A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP315	MMCR SP314	MMCR SP313	MMCR SP312	MMCR SP311	MMCR SP310	MMCR SP309	MMCR SP308	MMCR SP307	MMCR SP306	MMCR SP305	MMCR SP304	MMCR SP303	MMCR SP302	MMCR SP301	MMCR SP300
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP315-300	Response Register3

MMCRSP4 (0003:04A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP415	MMCR SP414	MMCR SP413	MMCR SP412	MMCR SP411	MMCR SP410	MMCR SP409	MMCR SP408	MMCR SP407	MMCR SP406	MMCR SP405	MMCR SP404	MMCR SP403	MMCR SP402	MMCR SP401	MMCR SP400
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP415-400	Response Register4

MMCRSP5 (0003:04A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP515	MMCR SP514	MMCR SP513	MMCR SP512	MMCR SP511	MMCR SP510	MMCR SP509	MMCR SP508	MMCR SP507	MMCR SP506	MMCR SP505	MMCR SP504	MMCR SP503	MMCR SP502	MMCR SP501	MMCR SP500
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP515-500	Response Register5

MMCRSP6 (0003:04AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP615	MMCR SP614	MMCR SP613	MMCR SP612	MMCR SP611	MMCR SP610	MMCR SP609	MMCR SP608	MMCR SP607	MMCR SP606	MMCR SP605	MMCR SP604	MMCR SP603	MMCR SP602	MMCR SP601	MMCR SP600
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP615-600	Response Register6

MMCRSP7 (0003:04AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCR SP715	MMCR SP714	MMCR SP713	MMCR SP712	MMCR SP711	MMCR SP710	MMCR SP709	MMCR SP708	MMCR SP707	MMCR SP706	MMCR SP705	MMCR SP704	MMCR SP703	MMCR SP702	MMCR SP701	MMCR SP700
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP715-700	Response Register7

Note: Received response is stored in MMCRSP0-MMCRSP7 as the following. MMCRSP0-MMCRSP7 will be updated when receiving, even if the CPU don't have read previous content yet.

In case of R1 or R3 (R4, R5, or R6) response in MMC mode

MMRs	Bit position of command
MMCCIDX	[47-40]
MMCRSP7	[39-24]
MMCRSP6	[23-8]
MMCRSP5-0	-

In case of R2 response in MMC mode

MMRs	Bit position of command
MMCCIDX	[135-128]
MMCRSP7	[127-112]
MMCRSP6	[111-96]
MMCRSP5	[95-80]
MMCRSP4	[79-64]
MMCRSP3	[63-48]
MMCRSP2	[47-32]
MMCRSP1	[31-16]
MMCRSP0	[15-0]

In case of R1 response in SPI mode

MMRs	Bit position of command
MMCRSP7	[7-0] It is stored into LSB byte of MMCRSP7
MMCRSP6-0	-

In case of R2 response in SPI mode

MMRs	Bit position of command
MMCRSP7	[15-0]
MMCRSP6-0	-

In case of R3 response in SPI mode

MMRs	Bit position of command
MMCRSP7	[39-32] It is stored into LSB byte of MMCRSP7
MMCRSP6	[31-16]
MMCRSP5	[15-0]
MMCRSP4-0	-

MMCDRSP (0003:04AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DRSP7	DRSP6	DRSP5	DRSP4	DRSP3	DRSP2	DRSP1	DRSP0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DRSP7-DRSP0	Data Response

		The data response byte is stored when it is received in SPI mode. (The CRC status token, consists of three 0 bit, start-bit, CRC status and end-bit, is stored when it is received in MMC mode.)
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MMCETOK (0003:04B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ETOK7	ETOK6	ETOK5	ETOK4	ETOK3	ETOK2	ETOK1	ETOK0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	ETOK7- ETOK0	Error Token. The error token byte is stored when it is received in SPI mode.

MMCCIDX (0003:04B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CIDX7	CIDX6	CIDX5	CIDX4	CIDX3	CIDX2	CIDX1	CIDX0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	CIDX7- CIDX0	Command Index Head 1 byte of a response, consists of start bit, transmission bit and command index, is stored when it is received in MMC mode.

MMCCCKC (0003:04B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CPHA7	CPHA6	CPHA5	CPHA4	CPHA3	CPHA2	CPHA1	CPHA0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	TEST bits. Not in use.
7-0	CPHA7 - CPHA0	CLK start phase control if CPHA = ClockRate(in MMCCLK), when MMCCLK register is set up, MMC clock start immediately. if CPHA < ClockRate when ClockRate - CPHA = N, MMC clock start N*Tcycle(internal function_clock) delay. if CPHA > ClockRate when CPHA - ClockRate = M, MMC clock start (256-M)*Tcycle(internal function_clock) delay.

MMCTORC (0003:04B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TORC7	TORC6	TORC5	TORC4	TORC3	TORC2	TORC1	TORC0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	TORC7- TORC0	MMCTORC is used in order to test the counter that operates with the value of MMCTOR in a factory test or a chip evaluation. Reading is possible in

		order to check the number of cycles remaining before Time-Out. But writing to this register is prohibited by User.
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MMCTODC (0003:04B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TODC15	TODC14	TODC13	TODC12	TODC11	TODC10	TODC09	TODC08	TODC07	TODC06	TODC05	TODC04	TODC03	TODC02	TODC01	TODC00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-0	TODC15- TODC0	MMCTODC is used in order to test the counter that operates with the value of MMCTOD in a factory test or a chip evaluation. Reading is possible in order to check the number of cycles remaining before Time-Out. But writing to this register is prohibited by User.

MMCBLNC (0003:04BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BLNC7	BLNC6	BLNC5	BLNC4	BLNC3	BLNC2	BLNC1	BLNC0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	1

Bit	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	BLNC7-BLNC0	MMCBLNC is used in order to test the counter that operates with the value of MMCBLN in a factory test or a chip evaluation. Reading is possible in order to check the byte count of a block to be transferred. But writing to this register is prohibited by User.

DMATRG (0003:04BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BRK	TRG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	BRK	DMA transmission is broken forcibly clear to 0 automatically when hit the BLK bit.
0	TRG	Start DMA transmission, clear to 0 automatically when hit the TRG bit.

DMAMODE (0003:04BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	DTOIM	DMAM	DIR	DPRMDX	DPRMDR	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-	-	-
Default	-	0	0	0	0	0	-	-	-	-	-	-	-	-	-	-

Bit	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	DTOIM	Interrupt mask for DMA transfer time_out 0: Interrupt is masked 1: Interrupt is unmasked
13	DMAM	DMA Mode Enable 0: Disable DMA transmission mode. 1: Enable DMA transmission mode.
12	DIR	Direction for DMA transmission

		0: MMCIF to SDRAM 1: SDRAM to MMCIF
11	DPRMDX	Word Swap Enable for DMA Write 0: 32bit width data from SDRAM is written to card straight. SDRAM_out(31:0) -> CARD_in(31:0) 1: Data from SDRAM is swapped words by 16bits. SDRAM_out(31:0) -> CARD_in(15:0,31:16)
10	DPRMDR	Word Swap Enable for DMA Read 0: 32bit width data to SDRAM is read from card straight. CARD_out(31:0) -> SDRAM_in(31:0) 1: Data to SDRAM is swapped words by 16bits. CARD_out(31:0) -> SDRAM_in(15:0,31:16)
9-0	RSV	Reserved bit. Not in use.

Note: All bits in DMAMODE are cleared when DMA transfer finished. (when RUN in DMASAT1 transit from 1 to 0).

DMASTAD0 (0003:04C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TAD15	TAD14	TAD13	TAD12	TAD11	TAD10	TAD09	TAD08	TAD07	TAD06	TAD05	TAD04	TAD03	TAD02	TAD01	TAD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	TAD15 - TAD0	Lsb 16bits of DMA start address. (Relative address)

DMASTAD1 (0003:04C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	TAD26	TAD25	TAD24	TAD23	TAD22	TAD21	TAD20	TAD19	TAD18	TAD17	TAD16
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	TAD26 - TAD16	Msb 10bits of DMA start address. (Relative address) Address is specified as offset from start of SDRAM, must be 4 byte aligned.

DMASTAT0 (0003:04C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	REST15	REST14	REST13	REST12	REST11	REST10	REST09	REST08	REST07	REST06	REST05	REST04	REST03	REST02	REST01	REST00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	REST15-REST0	Lsb 16bits of transfer byte rest

DMASTAT1 (0003:04C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	TOUTDT	RUN	REST27	REST26	REST25	REST24	REST23	REST22	REST21	REST20	REST19	REST18	REST17	REST16
R/W	-	-	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13	TOUTDT	Time-Out (DMA transfer acknowledge) 0: No time-out occurs.

		1: Time-out has occurred during waiting acknowledge from SDRAMC.
12	RUN	DMA transmission flag 0: Finish transmit. 1: DMA transmission is running.
11-0	REST27 - 16	Msb 12bits of transfer byte rest

DMATOR (0003:04C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DTOR15	DTOR14	DTOR13	DTOR12	DTOR11	DTOR10	DTOR09	DTOR08	DTOR7	DTOR6	DTOR5	DTOR4	DTOR3	DTOR2	DTOR1	DTOR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	DTOR15 - 0	Time-Out Count for DMA transfer acknowledge from SDRAM DTOR = 0000 0000 0000 0000 no time-out DTOR = 0000 0000 0000 0001 1 CLK clock DTOR = 0000 0000 0000 0010 2 CLK clocks : DTOR = 1111 1111 1111 1111 65535 CLK clocks

5.1 Registers specific to SDIO card**SDIOCTL (0003:04CA) Offset : 0x25**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RDWTCR	RDWTRQ
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	RDWTCR	Read Wait enable for CRC error. If this bit is set, Read Wait operation automatically starts in case a CRC error is detected during multiple block read access and while it's not the last block to be transferred. In this case, RDWTRQ (Read Wait request) is automatically set. To end the Read Wait operation, write '0' to RDWTRQ. (No need to clear RDWTCR.)
0	RDWTRQ	Read Wait request. 0: End Read Wait operation and release DAT[2]. 1: Set a Read Wait request. If this bit is set, Read Wait operation starts 2 clocks after the end of the read data block. MMCIF asserts low level on DAT[2] until this bit is cleared.

SDIOST0 (0003:04CC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RDWTST	INTPRD	DAT1
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	1

Bit	Register Name	Description
15-3	RSV	Reserved bit. Not in use.
2	RDWTST	Read Wait status. RDWTST becomes '1' during Read Wait operation.
1	INTPRD	Interrupt period. INTPRD becomes '1' during Interrupt period.

0	DAT1	DAT[1] status. DAT[1] can be monitored through this bit.
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SDIOIEN (0003:04CE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RWSEN	IOINTEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	RWSEN	Read Wait interrupt enable. 0: Disable the interrupt. 1: Enable the interrupt. Interrupt is generated to notify that Read Wait operation has been started.
0	IOINTEN	SDIO card interrupt enable. 0: Disable the interrupt. 1: Enable the interrupt.

SDIOIST (0003:04D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RWS	IOINT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	RWS	Read Wait interrupt status. 0: Not interrupted. 1: Interrupted. This bit is set to '1' when Read Wait operation starts and Read Wait interrupt is enabled. To clear this bit, write '1'.
0	IOINT	SDIO card interrupt status 0: Not interrupted. 1: Interrupted. This bit is set to '1' if the SDIO card interrupt is detected and card interrupt is enabled. To clear this bit, write '1' after the interrupt on the DAT[1] line is removed.

6. Interrupt Controller

The interrupt controller has a SW programmable configuration. FIQ0, FIQ1, FIQ2, IRQ0, IRQ1 and IRQ2 shows just condition. The priority level of interrupt is programmable at ARM side. It is possible to give permission or prohibition for each interrupt. Even when it is prohibited, IRQ0, IRQ1, IRQ2, FIQ0, FIQ1 and FIQ2 can observe the condition of interrupt as a flag.

FIQ0 (0003:0500) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FIQ15	FIQ14	FIQ13	FIQ12	FIQ11	FIQ10	FIQ09	FIQ08	FIQ07	FIQ06	FIQ05	FIQ04	FIQ03	FIQ02	FIQ01	FIQ00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	FIQ[15:0]	The condition of the interrupt request mapped by FIQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

FIQ1 (0003:0502) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FIQ31	FIQ30	FIQ29	FIQ28	FIQ27	FIQ26	FIQ25	FIQ24	FIQ23	FIQ22	FIQ21	FIQ20	FIQ19	FIQ18	FIQ17	FIQ16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	FIQ[31:16]	The condition of the interrupt request mapped by FIQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

FIQ2 (0003:0504) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	FIQ39	FIQ38	FIQ37	FIQ36	FIQ35	FIQ34	FIQ33	FIQ32
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Register Name	Description
7:0	FIQ[39:32]	The condition of the interrupt request mapped by FIQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

IRQ0 (0003:0508) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IRQ15	IRQ14	IRQ13	IRQ12	IRQ11	IRQ10	IRQ09	IRQ08	IRQ07	IRQ06	IRQ05	IRQ04	IRQ03	IRQ02	IRQ01	IRQ00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	IRQ[15:0]	The condition of the interrupt request mapped by IRQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

IRQ1 (0003:050A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IRQ31	IRQ30	IRQ29	IRQ28	IRQ27	IRQ26	IRQ25	IRQ24	IRQ23	IRQ22	IRQ21	IRQ20	IRQ19	IRQ18	IRQ17	IRQ16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	IRQ[31:16]	The condition of the interrupt request mapped by IRQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

IRQ2 (0003:050C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	IRQ39	IRQ38	IRQ37	IRQ36	IRQ35	IRQ34	IRQ33	IRQ32
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Register Name	Description
7:0	IRQ [39:32]	The condition of the interrupt request mapped by IRQ. 0 in corresponding bit indicates occurrence of interrupt. In a state of prohibition by EINT0 register, interrupt to ARM is not generated but each bit of this register is reflecting the condition of interrupt.

FIQENTRY0 (0003:0510) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FIQENTRY [15:0]															
R/W	R															
Default	0															

Bits	Register Name	Description
15:0	FIQENTRY[15:0]	Low 16 bit of entry address to the highest priority interrupt service routine of the effective FIQ interrupt

FIQENTRY1 (0003:0512) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	FIQENTRY [28:16]												
R/W	-	-	-	R												
Default	-	-	-	0												

Bits	Register Name	Description
12:0	FIQENTRY[28:16]	High 13 bit of entry address to the highest priority interrupt service routine of the effective FIQ interrupt

IRQENTRY0 (0003:0518) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IRQENTRY [15:0]															
R/W	R															
Default	0															

Bits	Register Name	Description
15:0	IRQENTRY[15:0]	Low 16bit of entry address to the highest priority interrupt service routine of the effective IRQ interrupt

IRQENTRY1 (0003:051A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	IRQENTRY [28:16]												
R/W	-	-	-	R												
Default	-	-	-	0												

Bits	Register Name	Description
12:0	IRQENTRY[28:16]	High 13 bit of entry address to the highest priority interrupt service routine of the effective IRQ interrupt

FISEL0 (0003:0520) Offset : 0x010

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FISEL [15:0]															
R/W	R/W															
Default	0															

Bits	Register Name	Description
15:0	FISEL[15:0]	The bit for interrupt factor which has to be set to FIQ is made as "1". "0" means the interrupt is IRQ type. b15 INT15 MTC0 b14 INT14 USB b13 INT13 UART1 b12 INT12 UART0 b11 INT11 DSP HINT b10 INT10 External HOST b9 INT9 SERIAL1 b8 INT8 SERIAL0 b7 INT7 OSD b6 INT6 CCDC VD2 b5 INT5 CCDC VD1 b4 INT4 CCDC VD0 b3 INT3 TIMER3 b2 INT2 TIMER2 b1 INT1 TIMER1 b0 INT0 TIMER0

FISEL1 (0003:0522) Offset : 0x011

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FISEL [31:16]															
R/W	R/W															
Default	0															

Bits	Register Name	Description
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Bits	Register Name	Description
15:0	FISEL[31:16]	The bit for the interrupt factor which has to be set to FIQ is made as "1". "0" means the interrupt is IRQ type. b15 INT31 GIO12 b14 INT30 GIO11 b13 INT29 GIO10 b12 INT28 GIO9 b11 INT27 GIO8 b10 INT26 GIO7 b9 INT25 GIO6 b8 INT24 GIO5 b7 INT23 GIO4 b6 INT22 GIO3 b5 INT21 GIO2 b4 INT20 GIO1 b3 INT19 GIO0 b2 INT18 "SD/MMC I/F 1" b1 INT17 "SD/MMC I/F 0" b0 INT16 MTC1

FISEL2 (0003:0524) Offset : 0x012

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	FISEL [39:32]							
R/W	-	-	-	-	-	-	-	-	R/W							
Default	-	-	-	-	-	-	-	-	0							

Bits	Register Name	Description
7:0	FISEL[39:32]	The bit for the interrupt factor which has to be set to FIQ is made as "1". "0" means the interrupt is IRQ type. b7 INT39 (RSV) b6 INT38 CLKC b5 INT37 I2C b4 INT36 WDT b3 INT35 PREVIEW b2 INT34 GIO15 b1 INT33 GIO14 b0 INT32 GIO13

EINT0 (0003:0528) Offset : 0x014

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EINT15	EINT14	EINT13	EINT12	EINT11	EINT10	EINT09	EINT08	EINT07	EINT06	EINT05	EINT04	EINT03	EINT02	EINT01	EINT00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	EINT[15:0]	Interruption permit bit. Corresponding to each bit of interrupt from INT0 to INT15 and permit or prohibit each interrupt. 0: Prohibit 1: Permit

EINT1 (0003:052A) Offset : 0x015

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EINT31	EINT30	EINT29	EINT28	EINT27	EINT26	EINT25	EINT24	EINT23	EINT22	EINT21	EINT20	EINT19	EINT18	EINT17	EINT16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:0	EINT[31:16]	Interruption permit bit. Corresponding to each bit from INT16 to INT31 and permit or prohibit each interrupt. 0: Prohibit 1: Permit

EINT2 (0003:052C) Offset : 0x016

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	EINT39	EINT38	EINT37	EINT36	EINT35	EINT34	EINT33	EINT32
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
7:0	EINT[39:32]	Interruption permit bit. Corresponding to each bit from INT32 to INT39 and permit or prohibit each interrupt. 0: Prohibit 1: Permit

INTRAW (0003:0530) Offset : 0x018

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	IRAW1	IRAW0
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1	IRAW1	To set whether the masked interrupt reflects in the address of IRQENTRY register or not. 0: Does not reflect. 1: Reflects
0	IRAW0	To set whether the masked interrupt reflects in the address of FIQENTRY register or not. 0: Does not reflect. 1: Reflects

EABASE0 (0003:0538) Offset : 0x01C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EABASE[15:3]													RSV	SIZE	
R/W	R/W													-	R/W	
Default	0													-	0	

Bits	Register Name	Description
15:3	EABASE[15:3]	Base address low 15:3 of interrupt entry table is set. The interrupt entry table is arranged from 8-byte boundary.
1:0	SIZE	The size of each entry in the table is set. 0: 4 byte 1: 8 byte 2: 16 byte 3: 32 byte

EABASE1 (0003:053A) Offset : 0x01D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	EABASE[28:16]												
R/W	-	-	-	R/W												
Default	-	-	-	0												

Bits	Register Name	Description
12:0	EABASE1[28:16]	Base address high 28:16 of interrupt entry table is set.

INTPRI0 (0003:0540) Offset : 0x020

It is a register to set the priority of each interrupt. The order of priority is 40-leveled and one interrupt factor is allocated to one level. The priority of interrupt mapped to FIQ is decided among other interrupts mapped to FIQ and that of IRQ is decided among other interrupts mapped to IRQ. Therefore, setting can be made regardless of mapping to FIQ and IRQ.

Table 1: INTERRUPT SOURCE LIST

Identify Number	Interrupt Source
0	TIMER0
1	TIMER1
2	TIMER2
3	TIMER3
4	CCDC VD0
5	CCDC VD1
6	CCDC VD2
7	OSD
8	SERIAL0
9	SERIAL1
10	External HOST
11	DSP HINT
12	UART0
13	UART1
14	USB
15	MTC0
16	MTC1
17	"SD/MMC I/F 0"
18	"SD/MMC I/F 1"
19	GIO0
20	GIO1
21	GIO2
22	GIO3
23	GIO4
24	GIO5
25	GIO6
26	GIO7
27	GIO8
28	GIO9
29	GIO10
30	GIO11
31	GIO12
32	GIO13
33	GIO14
34	GIO15
35	PREVIEW
36	WDT

Identify Number	Interrupt Source
37	I2C
38	CLKC
39	(RSV)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI1						RSV	RSV	PRI0					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	1						-	-	0					

Bits	Register Name	Description
13:8	PRI1[5:0]	Locate the interrupt factor number whose priority is high in the 2 nd .
5:0	PRI0[5:0]	Locate the interrupt factor number whose priority is the highest.

INTPRI1 (0003:0542) Offset : 0x021

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI3						RSV	RSV	PRI2					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	3						-	-	2					

Bits	Register Name	Description
13:8	PRI3[5:0]	Locate the interrupt factor number whose priority is high in the 4 th .
5:0	PRI2[5:0]	Locate the interrupt factor number whose priority is high in the 3 rd .

INTPRI2 (0003:0544) Offset : 0x022

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI5						RSV	RSV	PRI4					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	5						-	-	4					

Bits	Register Name	Description
13:8	PRI5[5:0]	Locate the interrupt factor number whose priority is high in the 6 th .
5:0	PRI4[5:0]	Locate the interrupt factor number whose priority is high in the 5 th .

INTPRI3 (0003:0546) Offset : 0x023

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI7						RSV	RSV	PRI6					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	7						-	-	6					

Bits	Register Name	Description
13:8	PRI7[5:0]	Locate the interrupt factor number whose priority is high in the 8 th .
5:0	PRI6[5:0]	Locate the interrupt factor number whose priority is high in the 7 th .

INTPRI4 (0003:0548) Offset : 0x024

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI9						RSV	RSV	PRI8					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	9						-	-	8					

Bits	Register Name	Description
13:8	PRI9[5:0]	Locate the interrupt factor number whose priority is high in the 10 th .
5:0	PRI8[5:0]	Locate the interrupt factor number whose priority is high in the 9 th .

INTPRI5 (0003:054A) Offset : 0x025

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI11						RSV	RSV	PRI10					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	11						-	-	10					

Bits	Register Name	Description
13:8	PRI11[5:0]	Locate the interrupt factor number whose priority is high in the 12 th .
5:0	PRI10[5:0]	Locate the interrupt factor number whose priority is high in the 11 th .

INTPRI6 (0003:054C) Offset : 0x026

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI13						RSV	RSV	PRI12					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	13						-	-	12					

Bits	Register Name	Description
13:8	PRI13[5:0]	Locate the interrupt factor number whose priority is high in the 14 th .
5:0	PRI12[5:0]	Locate the interrupt factor number whose priority is high in the 13 th .

INTPRI7 (0003:054E) Offset : 0x027

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI15						RSV	RSV	PRI14					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	15						-	-	14					

Bits	Register Name	Description
13:8	PRI15[5:0]	Locate the interrupt factor number whose priority is high in the 16 th .
5:0	PRI14[5:0]	Locate the interrupt factor number whose priority is high in the 15 th .

INTPRI8 (0003:0550) Offset : 0x028

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI17						RSV	RSV	PRI16					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	17						-	-	16					

Bits	Register Name	Description
13:8	PRI17[5:0]	Locate the interrupt factor number whose priority is high in the 18 th .
5:0	PRI16[5:0]	Locate the interrupt factor number whose priority is high in the 17 th .

INTPRI9 (0003:0552) Offset : 0x029

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI19						RSV	RSV	PRI18					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	19						-	-	18					

Bits	Register Name	Description
13:8	PRI19[5:0]	Locate the interrupt factor number whose priority is high in the 20 th .
5:0	PRI18[5:0]	Locate the interrupt factor number whose priority is high in the 19 th .

INTPRI10 (0003:0554) Offset : 0x02A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI21						RSV	RSV	PRI20					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	21						-	-	20					

Bits	Register Name	Description
13:8	PRI21[5:0]	Locate the interrupt factor number whose priority is high in the 22 nd .
5:0	PRI20[5:0]	Locate the interrupt factor number whose priority is high in the 21 st .

INTPRI11 (0003:0556) Offset : 0x02B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI23						RSV	RSV	PRI22					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	23						-	-	22					

Bits	Register Name	Description
13:8	PRI23[5:0]	Locate the interrupt factor number whose priority is high in the 24 th .
5:0	PRI22[5:0]	Locate the interrupt factor number whose priority is high in the 23 rd .

INTPRI12 (0003:0558) Offset : 0x02C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI25						RSV	RSV	PRI24					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	25						-	-	24					

Bits	Register Name	Description
13:8	PRI25[5:0]	Locate the interrupt factor number whose priority is high in the 26 th .
5:0	PRI24[5:0]	Locate the interrupt factor number whose priority is high in the 25 th .

INTPRI13 (0003:055A) Offset : 0x02D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI27						RSV	RSV	PRI26					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	27						-	-	26					

Bits	Register Name	Description
13:8	PRI27[5:0]	Locate the interrupt factor number whose priority is high in the 28 th .
5:0	PRI26[5:0]	Locate the interrupt factor number whose priority is high in the 27 th .

INTPRI14 (0003:055C) Offset : 0x02E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI29						RSV	RSV	PRI28					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	29						-	-	28					

Bits	Register Name	Description
13:8	PRI29[5:0]	Locate the interrupt factor number whose priority is high in the 30 th .
5:0	PRI28[5:0]	Locate the interrupt factor number whose priority is high in the 29 th .

INTPRI15 (0003:055E) Offset : 0x02F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI31						RSV	RSV	PRI30					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	31						-	-	30					

Bits	Register Name	Description
13:8	PRI31[5:0]	Locate the interrupt factor number whose priority is high in the 32 nd .
5:0	PRI30[5:0]	Locate the interrupt factor number whose priority is high in the 31 st .

INTPRI16 (0003:0560) Offset : 0x030

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI33						RSV	RSV	PRI32					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	33						-	-	32					

Bits	Register Name	Description
13:8	PRI29[5:0]	Locate the interrupt factor number whose priority is high in the 34 th .
5:0	PRI28[5:0]	Locate the interrupt factor number whose priority is high in the 33 rd .

INTPRI17 (0003:0562) Offset : 0x031

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI35						RSV	RSV	PRI34					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	35						-	-	34					

Bits	Register Name	Description
13:8	PRI35[5:0]	Locate the interrupt factor number whose priority is high in the 36 th .
5:0	PRI34[5:0]	Locate the interrupt factor number whose priority is high in the 35 th .

INTPRI18 (0003:0564) Offset : 0x032

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI37						RSV	RSV	PRI36					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	37						-	-	36					

Bits	Register Name	Description
13:8	PRI37[5:0]	Locate the interrupt factor number whose priority is high in the 38 th .
5:0	PRI36[5:0]	Locate the interrupt factor number whose priority is high in the 37 th .

INTPRI19 (0003:0566) Offset : 0x033

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PRI39						RSV	RSV	PRI38					
R/W	-	-	R/W						-	-	R/W					
Default	-	-	39						-	-	38					

Bits	Register Name	Description
13:8	PRI39[5:0]	Locate the interrupt factor number whose priority is the lowest.
5:0	PRI38[5:0]	Locate the interrupt factor number whose priority is high in the 39 th .

7. General I/O

The GIO module controls the I/O of external GIO port. The GIO module can control a total of 34 GIO ports. By setting of FSE register, the function of GIO26-GIO8 can be set to the original function of GIO or other functions. Through reading and writing of the register, the logic level of the I/O pin allocated outside can be read and controlled. It is available for control of various external devices including AD and DA, and the motor.

DIR0 (0003:0580) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DIR15	DIR14	DIR13	DIR12	DIR11	DIR10	DIR09	DIR08	DIR07	DIR06	DIR05	DIR04	DIR03	DIR02	DIR01	DIR00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	DIR15-DIR00	Port direction bit. Direction of each port I/O is set. DIR15 and DIR00 set GIO15 pin and GIO00 pin respectively. 0: Output 1: Input

DIR1 (0003:0582) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DIR31	DIR30	DIR29	DIR28	DIR27	DIR26	DIR25	DIR24	DIR23	DIR22	DIR21	DIR20	DIR19	DIR18	DIR17	DIR16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:0	DIR31-DIR16	Port direction bit. Direction of each port I/O is set. DIR31 and DIR16 set GIO31 pin and GIO16 pin respectively. 0: Output 1: Input

DIR2 (0003:0584) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DIR33	DIR32
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1:0	DIR33-DIR32	Port direction bit. Direction of each port I/O is set. DIR33 and DIR32 set GIO33 pin and GIO32 pin respectively. 0: Output 1: Input

INV0 (0003:0586) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV15	INV14	INV13	INV12	INV11	INV10	INV09	INV08	INV07	INV06	INV05	INV04	INV03	INV02	INV01	INV00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:0	INV15-INV00	Port inverse bit. The polarity of each input port is reversed. INV15 and INV00 set GIO15 pin and GIO0 pin respectively. While outputting, this bit is invalid. 0: Non inversion 1: Inversion

INV1 (0003:0588) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV31	INV30	INV29	INV28	INV27	INV26	INV25	INV24	INV23	INV22	INV21	INV20	INV19	INV18	INV17	INV16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	INV31-INV16	Port inverse bit. The polarity of each input port is reversed. INV31 and INV16 set GIO31 pin and GIO16 pin respectively. While outputting, this bit is invalid. 0: Non inversion 1: Inversion

INV2 (0003:058A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	INV33	INV32
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1:0	INV33-INV32	Port inverse bit. The polarity of each input port is reversed. INV33 and INV32 set GIO33 pin and GIO32 pin respectively. While outputting, this bit is invalid. 0: Non inversion 1: Inversion

BITSET0 (0003:058C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BS15	BS14	BS13	BS12	BS11	BS10	BS09	BS08	BS07	BS06	BS05	BS04	BS03	BS02	BS01	BS00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	BS15-BS00	GIO bit set 0. In case that GIO pin is set to output, when "1" is written in bit, corresponding GIO pin is set to "1". BS15 corresponds to GIO15 and BS00 corresponds to GIO0. In case that GIO pin is set to input, logical level of GIO pin can be read from corresponding bit. When GIO is set to output, current output value of port can be read. In case that it is set to inverse in INV register, the inverse value can be read if GIO is set to input. While setting as output, current output value can be read independently of INV register.

BITSET1 (0003:058E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BS31	BS30	BS29	BS28	BS27	BS26	BS25	BS24	BS23	BS22	BS21	BS20	BS19	BS18	BS17	BS16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	BS31-BS16	GIO bit set 1. This have same functions as GIO bit set 0. BS31 corresponds to GIO31 and BS16 corresponds to GIO16.

BITSET2 (0003:0590) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BS33	BS32
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1:0	BS33-BS32	GIO bit set 2. This has same functions as GIO bit set 0. BS33 corresponds to GIO33 and BS32 corresponds to GIO32.

BITCLR0 (0003:0592) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BC15	BC14	BC13	BC12	BC11	BC10	BC09	BC08	BC07	BC06	BC05	BC04	BC03	BC02	BC01	BC00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	BC15-BC00	GIO bit clear 0. In case that GIO is set to output, when "1" is written in corresponding bit, GIO pin is cleared to "0". BC15 corresponds to GIO15 and BS00 corresponds to GIO0. While in reading, current status of port can be read as well as BITSET0 register.

BITCLR1 (0003:0594) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BC31	BC30	BC29	BC28	BC27	BC26	BC25	BC24	BC23	BC22	BC21	BC20	BC19	BC18	BC17	BC16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	BC31-BC16	GIO bit clear 1. This has same functions as GIO bit clear 0. BC31 corresponds to GIO31 and BC16 corresponds to GIO16.

BITCLR2 (0003:0596) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BC33	BC32
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1:0	BC33-BC32	GIO bit clear 2. This has same functions as GIO bit clear 0. BC33 corresponds to GIO33 and BC32 corresponds to GIO32.

IRQPORT (0003:0598) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IRP15	IRP14	IRP13	IRP12	IRP11	IRP10	IRP9	IRP8	IRP7	IRP6	IRP5	IRP4	IRP3	IRP2	IRP1	IRP0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:0	IRP15-IRP0	Interrupt port. GIO15-GIO0 is used as external interrupt port. IRP15 and IRP0 set GIO15 and GIO0 respectively. 0: Normal GIO 1: External interrupt When it is used as the external interrupt, the I/O setting of DIR0 register should be set to input. Interruption detects more than 1 cycle of pulse of ARM clock that is 0 at the time of non-inverse input and 1 at inverse input by GIV register.

IRQEDGE (0003:059A) Offset : 0xD

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IRE15	IRE14	IRE13	IRE12	IRE11	IRE10	IRE9	IRE8	IRE7	IRE6	IRE5	IRE4	IRE3	IRE2	IRE1	IRE0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	IRE15-IRE0	Selection of interrupt detection edge 0: One-side (falling) edge detection (Except in input reverse setting. In this case, it comes to rising edge detection) 1: Both side edge detection Interruption occurs in both rising and falling edges.

CHAT0 (0003:059C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CHA15	CHA14	CHA13	CHA12	CHA11	CHA10	CHA9	CHA8	CHA7	CHA6	CHA5	CHA4	CHA3	CHA2	CHA1	CHA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	CHA15-CHA0	Chattering invalid setting 0. CHA15 and CHA0 set GIO15 and GIO0 respectively. 0: Normal 1: Chattering invalid mode When this is set to 1, input data is imported when certain level of interval longer than interval set in NCHAT is applied.

CHAT1 (0003:059E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CHA31	CHA30	CHA29	CHA28	CHA27	CHA26	CHA25	CHA24	CHA23	CHA22	CHA21	CHA20	CHA19	CHA18	CHA17	CHA16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	CHA31-CHA16	Chattering invalid setting 1. This has same functions as chattering invalid setting 0. CHA31 and CHA 16 set GIO31 and GIO16 respectively.

CHAT2 (0003:05A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CHA33	CHA32
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.
1:0	CHA33-CHA32	Chattering invalid setting 2. This has same functions as chattering invalid setting 0. CHA33 and CHA 32 set GIO33 and GIO32 respectively.

NCHAT (0003:05A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	NCH3	NCH2	NCH1	NCH0
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	1	0	0	0

Bits	Register Name	Description
15:4	RSV	Reserved bit. Not in use.
3:0	NCH3-NCH0	Chattering invalid width. The input data is imported when certain level of width more than width of (NCH[3:0]+1) x ARM cycle is applied. The data that changes at less than this width is invalid. The setting of chattering invalid of each port is made in CHAT register.

FSEL0 (0003:05A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FSEL15	FSEL14	FSEL13	FSEL12	FSEL11	FSEL10	FSEL09	FSEL08	FSEL07	FSEL06	FSEL05	FSEL04	FSEL03	FSEL02	FSEL01	FSEL00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	FSEL15	Function setting of GIO23 pin 0 : GIO23 1 : RXD1
14	FSEL14	Function setting of GIO22 pin 0 : GIO22 1 : SDEN1
13	FSEL13	Function setting of GIO21 pin 0 : GIO21 1 : SCLK1
12	FSEL12	Function setting of GIO20 pin 0 : GIO20 1 : SDI1
11	FSEL11	Function setting of GIO19 pin 0 : GIO19 1 : SDO1
10	FSEL10	Function setting of GIO18 pin 0 : GIO18 1 : CLKOUT2 CLKOUT can output clock generated by OSEL and O*DIV registers in general clock output.
9	FSEL9	Function setting of GIO17 pin 0 : GIO17 1 : CLKOUT1 CLKOUT can output clock generated by OSEL and O*DIV registers in general clock output.
8	FSEL8	Function setting of GIO16 pin 0 : GIO16 1 : CLKOUT0 CLKOUT can output clock generated by OSEL and O*DIV registers in general clock output.

Bits	Register Name	Description
7	FSEL7	Function setting of GIO15 pin 0 : GIO15 1 : DSP_XF
6	FSEL6	Function setting of GIO14 pin 0 : GIO14 1 : DSP_CLKS1
5	FSEL5	Function setting of GIO13 pin 0 : GIO13 1 : DSP_BDX1
4	FSEL4	Function setting of GIO12 pin 0 : GIO12 1 : DSP_BCLKX1
3	FSEL3	Function setting of GIO11 pin 0 : GIO11 1 : DSP_BFSX1
2	FSEL2	Function setting of GIO10 pin 0 : GIO10 1 : DSP_BDR1
1	FSEL1	Function setting of GIO9 pin 0 : GIO9 1 : DSP_BCLKR1
0	FSEL0	Function setting of GIO8 pin 0 : GIO8 1 : DSP_BFSR1

FSEL1 (0003:05A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	FSEL31	FSEL30	FSEL29	FSEL28	FSEL27	FSEL26	FSEL25	FSEL24	FSEL23	FSEL22	FSEL21	FSEL20	FSEL19	FSEL18	FSEL17	FSEL16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved(Set to 0 consistently)
10	FSEL26	Function setting of GIO33 pin 0 : GIO33 1 : EXT_LINE_ID
9	FSEL25	Function setting of GIO32 pin 0 : GIO32 1 : SDA
8	FSEL24	Function setting of GIO31 pin 0 : GIO31 1 : SCL
7	FSEL23	Function setting of GIO30 pin 0 : GIO30 1 : PWM1
6	FSEL22	Function setting of GIO29 pin 0 : GIO29 1 : PWM0
5	FSEL21	Function setting of GIO28 pin 0 : GIO28 1 : FIELD
4	FSEL20	Function setting of GIO27 pin 0 : GIO27 1 : CCDC_SHUT
3	FSEL19	Function setting of GIO26 pin 0 : GIO26 1 : C_WEN

Bits	Register Name	Description
2-1	FSEL18-FSEL17	Function setting of GIO25 pin FSEL18 FSEL17 0 0 GIO25 0 1 MIRQ 1 0 MRST 1 1 reserved MIRQ: Interrupt signal input in ARM core MRST: Reset signal of ARM core generated in WDT
0	FSEL16	Function setting of GIO24 pin 0 : GIO24 1 : TXD1

8. DSP Controller

HPIB_CTRL (0x30600) offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DBIO	DHOLD	DRST	DINT0	RSV	EXCHG	RSV	HPNMI	RSV	RSV	HPIEN
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	-	R/W	-	R/W	-	-	R/W
Default	-	-	-	-	-	1	1	1	1	-	0	-	1	-	-	1

Bits	Register Name	Description
15:11	RSV	Reserved bit
10	DBIO	DSP Branch Control. Connected to BIO terminal of DSP. When DSP is in BIO status, it can branch and control program. 0: DSP executes the conditional instruction 1: No operation
9	DHOLD	DSP hold request. Holds DSP. Connected to HOLDN of DSP. 0: Hold 1: Normal
8	DRST	Resets DSP. Connected to RSN terminal of DSP. 0: SP reset 1: Normal
7	DINT0	Signal input into INT0 port of DSP. Initial value is 1. INT0 is generated in DSP by the host setting this bit to 0. After interrupt is generated, the host must return this bit to 1. 0: Interrupt (INT0) 1: Normal
6	RSV	Reserved bit
5	EXCHG	Exchange Upper byte for Lower. This switches the upper byte and lower byte of the data in HPI16 mode. The initial value is 0. When set to 1, byte switching is performed. 0 : / HOST -> DSP: The MSB of 16-bit data passed from the host is written as is in correspondence with the MSB of the internal DARAM of the DSP. 0 : / DSP -> HOST: 16-bit data of the internal DARAM of DSP is passed as is to the host. 1 : / HOST -> DSP: The lower byte of 16-bit data passed from the host is written into the upper byte of the internal DARAM of DSP, and the upper byte is written into the lower byte of DARAM. 1 : / DSP -> HOST: The lower byte of the internal DARAM of DSP is written into the upper byte of the bus to the host, and the upper byte is written into the lower byte of the bus.
4	RSV	Reserved bit
3	HPNMI	HPI Non Mask able Interrupt. Host generates NMI in DSP. Connected to NMI port of DSP. After interrupt is generated, it must be reset to normal status. 0 : Interrupt (NMI) 1 : Normal
2:1	RSV	Reserved bit
0	HPIEN	Enables HPI. Connected to HPIEN of DSP. When disabled, HPI function is invalid. 0 : Disable 1 : Enable

HPIB_STAT (0x30602) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	XF	RSV	RSV	RSV	HOLDA	RSV	RSV	RSV	RSV	RSV	RSV	RSV	HINT
R/W	-	-	-	R	-	-	-	R	-	-	-	-	-	-	-	R
Default	-	-	-	1	-	-	-	1	-	-	-	-	-	-	-	1

Bits	Register Name	Description
15:13	RSV	Reserved bit
12	DXF	DSP XF indicator. Shows status o DSP XF signal output.
11:9	RSV	Reserved bit
8	HOLDA	DSP Hold Acknowledge. Indicates status of DSP HOLDA signal output. After a hold request is sent to DSP by the DHOLD bit of the HIB_CTRL register, acknowledgement can be checked by this bit. 0 : DSP hold status 1 : Normal
7:1	RSV	Reserved bit
0	HINT	Host port interrupt. Indicates status of DSP HINT signal output. 0 : DSP sends interrupt request to host 1 : Normal

9. OSD

OSD_MODE (0003:0680) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CS	OVRSZ	FSINV	EF	VVRSZ	VHRSZ	RSV	CPS	CABG7	CABG6	CABG5	CABG4	CABG3	CABG2	CABG1	CABG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	0
Trigger	#	VD	VD	VD	VD	VD	#	VD	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15	CS	Cb/Cr or Cr/Cb format. Select the order of output as Cb/Cr or Cr/Cb 0: Cb/Cr 1: Cr/Cb
14	OVRSZ	Vertical expansion of OSD window. When set, the OSD bitmap window images are stretched 6/5 times in the vertical direction. It is used when a VGA image is to be displayed on a PAL television. 0: x1 1: x 6/5
13	FSINV	Field signal reversing bit. 0: Unreversed/not Inversed 1: Reserved/Inversed
12	EF	Expand Filter ON/OFF. ON/OFF of the interpolation filter is specified. It is valid/effective, when VVRSZ or VHRSZ is ON or when the zoom of video window is set. Since the memory for the interpolation filter corresponds only up to the horizontal pixel number 720 pixels, extra care is necessary for using the filter. 0: OFF 1: ON
11	VVRSZ	Vertical expansion of video window. When set, the video window images are stretched 6/5 times in the vertical direction. It is used when a VGA image is to be displayed on a PAL television. 0: x1 1: x 6/5
10	VHRSZ	The horizontal expansion of video window. When set, the video window images are stretched 9/8 times in the vertical direction. It is used when a VGA image is to be displayed on a NTSC/PAL television. 0: x1 1: x 9/8
9	RSV	Please set 0.
8	CPS	Color Palette Selection for background color 0: ROM CLUT is used 1: RAM CLUT is used
7-0	CABG7-CABG0	The CLUT color value to be used for background color is specified. The part of the display area where image is not displayed, is filled with this background color.

VIDEOWIN_MODE (0003:0682) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	VHZ11	VHZ10	VVZ11	VVZ10	VFF1	ACT1	RSV	RSV	VHZ01	VHZ00	VVZ01	VVZ00	VFF0	ACT0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0
Trigger	-	-	VD	VD	VD	VD	VD	VD	-	-	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-14	RSV	Reserve Bit. Not in use.

Bits	Register Name	Description
13-12	VHZ11-VHZ10	Video Window1 horizontal zoom. VHZ11 VHZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
11-10	VVZ11-VVZ10	Video window 1 vertical zoom VVZ11 VVZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
9	VFF1	Video Window1 display mode. 0: Field mode. Reading every line. 1: Frame Mode. Reading every 1 line.
8	ACT1	Setting the ON/OFF of video window 1 image display. 0: OFF 1: ON
7-6	RSV	Reserved bit .Not in use.
5-4	VHZ01-VHZ00	Video Window 0 horizontal zoom. VHZ11 VHZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
3-2	VVZ01-VVZ00	Video Window 0 vertical zoom. VVZ11 VVZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
1	VFF0	Video Window 0 display mode. 0: Field Mode. Reading every line. 1: Frame Mode. Reading every 1 line.
0	ACT0	Setting the ON/OFF of Video Window 0 image display. 0: OFF 1: ON

OSDWIN0_MODE (0003:0684) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	CLUTS0	OVHZ01	OVHZ00	OVVZ01	OVVZ00	BW01	BW00	BF02	BF01	BF00	T0E	OFF0	OACT
R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-13	RSV	Reserve Bit. Not in use.
12	CLUTS0	CLUT selection, The look-up table which OSD window 0 uses is selected. 0: ROM-Lookup table. 1: RAM-Lookup table.
11-10	OVHZ01-OVHZ00	OSD/Window 0 horizontal zoom. OVHZ01 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')

Bits	Register Name	Description
9-8	OVVZ01-OVVZ00	OSD/Window 0 vertical zoom. OVVZ01 OVVZ00 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
7-6	BW01-BW00	The width of the bit of the bit map OSD/Window 0 is specified. BW11 BW10 Bit width 0 0 1 0 1 2 1 0 4 1 1 8
5-3	BF02-BF00	The mixture ratio of OSD window 0, video window 0, and 1 is set. BF02 BF01 BF00 OSD Window 0 Video Window 0 0 0 0 1 0 0 1 1/8 7/8 0 1 0 2/8 6/8 0 1 1 3/8 5/8 1 0 0 4/8 4/8 1 0 1 5/8 3/8 1 1 0 6/8 2/8 1 1 1 1 0
2	T0E	ON/OFF of transparency of OSD window 0 is set. When set to "1", for the pixel where the bit mapping value is "0" the mixing with colors specified in OSD007-OSD000 of W0BMP01(0003:06D0) and Video Window 0,1 is done according to blending ratio specified in BF02-BF00.
1	OFF0	OSD Window 0 display Mode. 0: Field Mode. Reading every line. 1: Frame Mode. Reading every 1 line.
0	OACT	Setting the ON/OFF of image display. 0: OFF 1: ON

OSDWIN1_MODE (0003:0686) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OASW	RSV	RSV	BIT12	BIT11	BIT10	BIT09	BIT08	BIT07	BIT06	BIT05	BIT04	BIT03	BIT02	BIT001	BIT00
R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15	OASW	OSD Window1, Attribute window conversion SW. 0: OSD Window 1 1: Attribute Window

OASW=0(In case of OSD window 1)

Bits	Register Name	Description
14-13	RSV	Reserve Bit .Not in use.
12	BIT12	CLUTSelection. The look-up table which OSD window 1 uses is selected. 0: ROM-Lookup Table 1: RAM-Lookup Table

Bits	Register Name	Description
11-10	BIT11-BIT10	OSD Window 1 horizontal zoom. BIT11 BIT10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
9-8	BIT09-BIT08	OSD Window 1 vertical zoom. BIT09 BIT08 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
7-6	BIT07-BIT06	The width of the bit of the bit map of OSD window 1 is specified. BIT07 BIT06 Bit Width 0 0 1 0 1 2 1 0 4 1 1 8
5-3	BIT05-BIT03	The mixture ratio of OSD window 1, video window 0, and 1 is set. BIT05 BIT04 BIT03 OSD Window1 Video Window 0 0 0 0 1 0 0 1 1/8 7/8 0 1 0 2/8 6/8 0 1 1 3/8 5/8 1 0 0 4/8 4/8 1 0 1 5/8 3/8 1 1 0 6/8 2/8 1 1 1 1 0
2	BIT02	ON/OFF of transparency of OSD window 1 is set. When set to "1", for the ipixel where the bit mapping value is "0" the mixing with colors specified in OSD007-OSD000 of W0BMP01(0003:06D0) and Video Window 0,1 is done according to blending ratio specified in BIT05-BIT03
1	BIT01	OSD window 1 Display mode. 0: Field mode. Reading every line. 1: Frame Mode. Reading every 1 line.
0	BIT00	Set the ON/OFF of Image display of OSD Window 1. 0: OFF 1: ON

OASW=1(in case of Attribute window)

Bits	Register Name	Description
14-13	RSV	Reserve Bit. Not in use.
12	BIT12	Enter 0.
11-10	BIT11-BIT10	Attribute Window horizontal zoom. BIT011 BIT10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
9-8	BIT09-BIT08	Attribute Window vertical zoom. BIT09 BIT08 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')

Bits	Register Name	Description
7-6	BIT07-BIT06	The blinking interval of the Attribute window is specified. Eight times of the VD signal becomes the basic intervals. BIT07 BIT06 0 0 Basic intervalx1 0 1 Basic intervalx2 1 0 Basic intervalx3 1 1 Basic intervalx4
5-2	BIT05-BIT03	Enter 0000.
1	BIT01	Attribute window display mode. 0: Field Mode. Reading every line. 1: Frame Mode. Reading every 1 line.
0	BIT00	Set the ON/OFF of blinker. 0: OFF 1: ON

The data format of Attribute window is as follows.

b3	B2	b1	B0
Blink	Blend factor		

Blink: Set the ON/OFF of Blinker for each pixel.

Blend factor: The mixture ratio of the video window for each pixel is set.

RECTANGLE_CURSOR (0003:0688) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CAC7	CAC6	CAC5	CAC4	CAC3	CAC2	CAC1	CAC0	CLUTS	CHW2	CHW1	CHW0	CVW2	CVW1	CVW0	CACT
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-8	CAC7-CAC0	The color of the cursor is specified in the address of the look-up table.
7	CLUTS	CLUT Selection. The look-up table used is selected. 0: ROM 1: RAM
6-4	CHW2-CHW0	The width of the line in the horizontal direction of the cursor is specified. 1,4,8,12,16,20,24,28 Pixel
3-1	CVW2-CVW0	The width of the line in the vertical direction of the cursor is specified. 1,2,4,6,8,10,12,14Line
0	CACT	Set the ON/OFF of cursor mode. 0: OFF 1: ON

VIDEOWIN0_OFFSET (0003:068C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0LO8	V0LO7	V0LO6	V0LO5	V0LO4	V0LO3	V0LO2	V0LO1	V0LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
8-0	V0LO8-V0LO0	1 line offset in video window 0 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example Horizontally there are 720 pixel(4:2:2) and the offset value of video window 0 is $(720 \times 2) / 32 = 45(0x02d)$

VIDEOWIN1_OFFSET (0003:069E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V1LO8	V1LO7	V1LO6	V1LO5	V1LO4	V1LO3	V1LO2	V1LO1	V1LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	V1LO8-V1LO0	1 line offset in video window 1 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example Horizontally there are 720 pixel(4:2:2) and the offset value of video window1 is $720 / 32 = 45(0x02d)$

OSDWIN0_OFFSET (0003:0690) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0LO8	O0LO7	O0LO6	O0LO5	O0LO4	O0LO3	O0LO2	O0LO1	O0LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	O0LO8-O0LO0	1 line offset in OSD window 0 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example The offset value of 64 pixel OSD window 0 with a width of eight bits per pixel, horizontally is $(64 \times 8) / 32 = 2(0x0010)$

OSDWIN1_ATTRIBUTE_OFFSET (0003:0692) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O1LO8	O1LO7	O1LO6	O1LO5	O1LO4	O1LO3	O1LO2	O1LO1	O1LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	O1LO8-O1LO0	Specify the 1 line offset of OSD Window 1 and Attribute Window 1. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example The offset value of 64 pixel OSD window 1 with a width of eight bits per pixel, horizontally is $(64 \times 8) / 32 = 2(0x0010)$

VIDEOWIN_AD_UPPER (0003:0694) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	V1A21	V1A20	V1A19	V1A18	V1A17	V1A16	RSV	RSV	V0A21	V0A20	V0A19	V0A18	V0A17	V0A16
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0
Trigger	-	-	VD	VD	VD	VD	VD	VD	-	-	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-8	V1A21-V1A16	The first address of SDRAM of the image data displayed in video window -1 is specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit..
7-6	RSV	Reserved bit. Not in use.
5-0	V0A21-V0A16	The first address of SDRAM of the image data displayed in video window 0 is specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

VIDEOWIN0_AD_LOWER (0003:0696) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V0A15	V0A14	V0A13	V0A12	V0A11	V0A10	V0A9	V0A8	V0A7	V0A6	V0A5	V0A4	V0A3	V0A2	V0A1	V0A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-0	V0A15-V0A0	The first address below 16 bits of SDRAM of image data (YCbCr) displayed in video window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

VIDEOWIN1_AD_LOWER (0003:0698) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V1A15	V1A14	V1A13	V1A12	V1A11	V1A10	V1A9	V1A8	V1A7	V1A6	V1A5	V1A4	V1A3	V1A2	V1A1	V1A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-0	V1A15-V1A0	The first address below 16 bits of SDRAM of image data (YCbCr) displayed in video window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN_AD_UPPER (0003:069A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	WA121	WA120	WA119	WA118	WA117	WA116	RSV	RSV	WA021	WA020	WA019	WA018	WA017	WA016
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0
Trigger	-	-	VD	VD	VD	VD	VD	VD	-	-	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-14	RSV	Reserved bit .Not in use.
13-8	WA121-WA116	OSD Window 1 address. The first address above 6 bits of SDRAM of image data (YCbCr) displayed in OSD window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.
7-6	RSV	Reserved bit .Not in use.

Bits	Register Name	Description
5-0	WA021-WA016	OSD Window 0 or address of Attribute window. The first address above 6 bits of SDRAM of image data (YCbCr) displayed in OSD window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN0_AD_LOWER (0003:069C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WA015	WA014	WA013	WA012	WA011	WA010	WA009	WA008	WA007	WA006	WA005	WA004	WA003	WA002	WA001	WA000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-0	WA015-WA000	OSD Window 0 address. The first address below 16 bits of SDRAM of image data (YCbCr) displayed in OSD window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN1_ATTRIBUTE_AD_LOWER (0003:068E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WA115	WA114	WA113	WA112	WA111	WA110	WA109	WA108	WA107	WA106	WA105	WA104	WA103	WA102	WA101	WA100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-0	WA115-WA100	OSD Window 1 or address of Attribute window. The first address below 16 bits of SDRAM of image data (YCbCr) displayed in OSD window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

BASEP_X (0003:06A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	BPX09	BPX08	BPX07	BPX06	BPX05	BPX04	BPX03	BPX02	BPX01	BPX00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit .Not in use.
9-0	BPX09-BPX00	The horizontal display of video window (standard position)0, 1, and OSD window 0 and 1 is set.Specify the number of pixels from the fall of HD. A minimum, set value is 24 pixels.

BASEP_Y (0003:06A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BPY08	BPY07	BPY06	BPY05	BPY04	BPY03	BPY02	BPY01	BPY00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit .Not in use.
8-0	BPY09-BPY00	The horizontal display of video window (standard position) 0, 1, and OSD window 0 and 1 is set. Specify the number of pixels from the fall of VD. A minimum, set value is 1 line. When it is "0", it becomes 1 line.

VIDEOWIN0_XP (0003:06A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	V0X09	V0X08	V0X07	V0X06	V0X05	V0X04	V0X03	V0X02	V0X01	V0X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-0	V0X09-V0X00	A display of beginning position in the horizontal direction of video window 0 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN0_YP (0003:06A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0Y08	V0Y07	V0Y06	V0Y05	V0Y04	V0Y03	V0Y02	V0Y01	V0Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	W0Y08-W0Y00	A display of beginning position in the vertical direction of video window 0 is set. The value of the line number corresponding to the field is set. Specify the number of lines from the displayed standard position.

VIDEOWIN0_XL (0003:06A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	V0W11	V0W10	V0W9	V0W8	V0W7	V0W6	V0W5	V0W4	V0W3	V0W2	V0W1	V0W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	V0W11-V0W0	Length of video window 0 in horizontal direction is specified by the number of pixels.

VIDEOWIN0_YL (0003:06AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	V0H10	V0H9	V0H8	V0H7	V0H6	V0H5	V0H4	V0H3	V0H2	V0H1	V0H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	V0H10-V0H0	The length of video window 0 in the vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For instance, if 240 lines are set in NTSC, it becomes the valid display range of vertical.

VIDEOWIN1_XP (0003:06AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	V1X09	V1X08	V1X07	V1X06	V1X05	V1X04	V1X03	V1X02	V1X01	V1X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-0	V1X09-V1X00	A display of beginning position in horizontal direction of video window 1 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN1_YP (0003:06AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V1Y08	V1Y07	V1Y06	V1Y05	V1Y04	V1Y03	V1Y02	V1Y01	V1Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	V1Y09-V1Y00	A display of beginning position in vertical direction of video window 1 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN1_XL (0003:06B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	V1W11	V1W10	V1W9	V1W8	V1W7	V1W6	V1W5	V1W4	V1W3	V1W2	V1W1	V1W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	V1W11-V1W0	The length of video window 1 in the horizontal direction is specified by the number of pixel.

VIDEOWIN1_YL (0003:06B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	V1H10	V1H9	V1H8	V1H7	V1H6	V1H5	V1H4	V1H3	V1H2	V1H1	V1H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	V1H10-V1H0	The length of video window 1 in the vertical direction is specified by the number of Lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

OSDWIN0_XP (0003:06B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	W0X09	W0X09	W0X07	W0X06	W0X05	W0X04	W0X03	W0X02	W0X01	W0X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-0	W0X09-W0X00	Set the display start position in horizontal direction of OSD window 0. Specify the number of pixels from the display standard position.

OSDWIN0_YP (0003:06B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	W0Y08	W0Y07	W0Y06	W0Y05	W0Y04	W0Y03	W0Y02	W0Y01	W0Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	W0Y08-W0Y00	Set the display start position in vertical direction of OSD window 0. The value of the line number corresponding to the field is set. Specify the number of lines from the display standard position.

OSDWIN0_XL (0003:06B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	W0W11	W0W10	W0W9	W0W8	W0W7	W0W6	W0W5	W0W4	W0W3	W0W2	W0W1	W0W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	W0W11-W0W0	Length of OSD window 0 in horizontal direction is specified by the number of pixels.

OSDWIN0_YL (0003:06BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	W0H10	W0H9	W0H8	W0H7	W0H6	W0H5	W0H4	W0H3	W0H2	W0H1	W0H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	W0H10-W0H0	The length of OSD window 0 in the vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

OSDWIN1_XP (0003:06BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	W0X09	W0X08	W0X07	W0X06	W0X05	W0X04	W0X03	W0X02	W0X01	W0X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-0	W1X09-W1X00	Set the display start position in horizontal direction of OSD window 1 or attribute window. Specify the number of lines from the display standard position.

OSDWIN1_YP (0003:06BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	W1Y08	W1Y07	W1Y06	W1Y05	W1Y04	W1Y03	W1Y02	W1Y01	W1Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	W1Y08-W1Y00	Set the display start position in vertical direction of OSD window 1 or attribute window. The value of the line number corresponding to the field is set. Specify the number of lines from the display standard position.

OSDWIN1_XL (0003:06C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	W1W11	W1W10	W1W9	W1W8	W1W7	W1W6	W1W5	W1W4	W1W3	W1W2	W1W1	W1W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	W1W11-W1W0	Length of OSD window 1 or Attribute window in horizontal direction is specified by the number of pixels.

OSDWIN1_YL (0003:06C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	W1H10	W1H9	W1H8	W1H7	W1H6	W1H5	W1H4	W1H3	W1H2	W1H1	W1H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	W1H10-W1H0	Length of OSD window 1 or Attribute window in vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

CUR_XP (0003:06C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	CSX09	CSX08	CSX07	CSX06	CSX05	CSX04	CSX03	CSX02	CSX01	CSX00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-0	CSX09-CSX00	The display start position in the horizontal direction of the cursor is set. Specify the number of lines from the display standard position.

CUR_YP (0003:06C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CSY08	CSY07	CSY06	CSY05	CSY04	CSY03	CSY02	CSY01	CSY00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	CSY08-CSY00	The display start position in the vertical direction of the cursor is set. Specify the number of lines from the display standard position.

CUR_XL (0003:06C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	CSW11	CSW10	CSW9	CSW8	CSW7	CSW6	CSW5	CSW4	CSW3	CSW2	CSW1	CSW0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

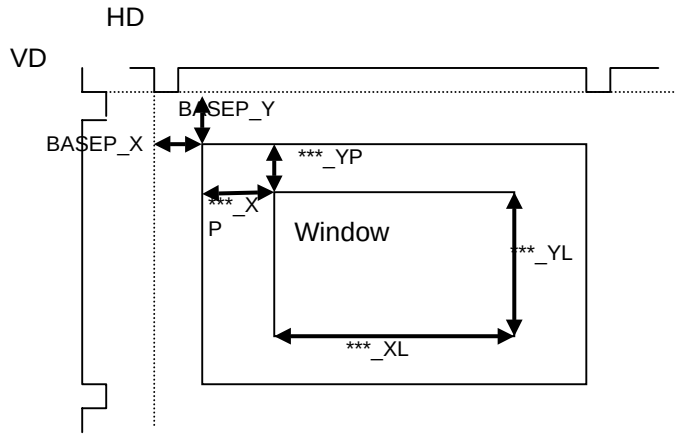
Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	CSW11-CSW0	Specify the length of the cursor in horizontal direction by number of pixels.

CUR_YL (0003:06CA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	CSH10	CSH9	CSH8	CSH7	CSH6	CSH5	CSH4	CSH3	CSH2	CSH1	CSH0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0
Trigger	-	-	-	-	-	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	CSH10-CSH0	Specify the length of the cursor in horizontal direction by number of lines. The value of the line number corresponding to the field is set. . For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

* The positions of windows as follows.



The display priority of each window is as follows.

Cursor > OSD window 1 > OSDWindow 0 > Video Window 1 > Video Window 0

W0BMP01 (0003:06D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD017	OSD016	OSD015	OSD014	OSD013	OSD012	OSD011	OSD010	OSD007	OSD006	OSD005	OSD004	OSD003	OSD002	OSD001	OSD000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD017-OSD010	The color of the bit map value "1" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD007-OSD000	The color of the bit map value "0" is specified in the CLUT (0-255) address. (width=1,2,4)

W0BMP23 (0003:06D2) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD037	OSD036	OSD035	OSD034	OSD033	OSD032	OSD031	OSD030	OSD027	OSD026	OSD025	OSD024	OSD023	OSD022	OSD021	OSD020
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD037-OSD030	The color of the bit map value "3" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD027-OSD020	The color of the bit map value "2" is specified in the CLUT (0-255) address. (width=4)

W0BMP45 (0003:06D4) Offset : 0x2A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD057	OSD056	OSD055	OSD054	OSD053	OSD052	OSD051	OSD050	OSD047	OSD046	OSD045	OSD044	OSD043	OSD042	OSD041	OSD040
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD057-OSD050	The color of the bit map value "5" is specified in the CLUT (0-255) address. (width=2,4)

Bits	Register Name	Description
7-0	OSD047-OSD040	The color of the bit map value "4" is specified in the CLUT (0-255) address. (width=4)

W0BMP67 (0003:06D6) Offset : 0x2B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD077	OSD076	OSD075	OSD074	OSD073	OSD072	OSD071	OSD070	OSD067	OSD066	OSD065	OSD064	OSD063	OSD062	OSD061	OSD060
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD077-OSD070	The color of the bit map value "7" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD067-OSD060	The color of the bit map value "6" is specified in the CLUT (0-255) address. (width=4)

W0BMP89 (0003:06D8) Offset : 0x2C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD097	OSD096	OSD095	OSD094	OSD093	OSD092	OSD091	OSD090	OSD087	OSD086	OSD085	OSD084	OSD083	OSD082	OSD081	OSD080
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD097-OSD090	The color of the bit map value "9" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD087-OSD080	The color of the bit map value "8" is specified in the CLUT (0-255) address. (width=4)

W0BMPAB (0003:06DA) Offset : 0x2D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0B7	OSD0B6	OSD0B5	OSD0B4	OSD0B3	OSD0B2	OSD0B1	OSD0B0	OSD0A7	OSD0A6	OSD0A5	OSD0A4	OSD0A3	OSD0A2	OSD0A1	OSD0A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD0B7-OSD0B0	The color of the bit map value "B" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD0A7-OSD0A0	The color of the bit map value "A" is specified in the CLUT (0-255) address. (width=2,4)

W0BMPCD (0003:06DC) Offset : 0x2E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0D7	OSD0D6	OSD0D5	OSD0D4	OSD0D3	OSD0D2	OSD0D1	OSD0D0	OSD0C7	OSD0C6	OSD0C5	OSD0C4	OSD0C3	OSD0C2	OSD0C1	OSD0C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD0D7-OSD0D0	The color of the bit map value "D" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD0C7-OSD0C0	The color of the bit map value "C" is specified in the CLUT (0-255) address. (width=4)

W0BMPEF (0003:06DE) Offset : 0x2F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0F7	OSD0F6	OSD0F5	OSD0F4	OSD0F3	OSD0F2	OSD0F1	OSD0F0	OSD0E7	OSD0E6	OSD0E5	OSD0E4	OSD0E3	OSD0E2	OSD0E1	OSD0E0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD0F7-OSD0F0	The color of the bit map value "F" is specified in the CLUT (0-255) address. (width=1,2,4)
7-0	OSD0E7-OSD0E0	The color of the bit map value "E" is specified in the CLUT (0-255) address. (width=4)

W1BMP01 (0003:06E0) Offset : 0x30

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD117	OSD116	OSD115	OSD114	OSD113	OSD112	OSD111	OSD110	OSD107	OSD106	OSD105	OSD104	OSD103	OSD102	OSD101	OSD100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD117-OSD110	The color of the bit map value "1" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD107-OSD100	The color of the bit map value "0" is specified in the CLUT (0-255) address. (width=1,2,4)

W1BMP23 (0003:06E2) Offset : 0x31

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD137	OSD136	OSD135	OSD134	OSD133	OSD132	OSD131	OSD130	OSD127	OSD126	OSD125	OSD124	OSD123	OSD122	OSD121	OSD120
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD137-OSD130	The color of the bit map value "3" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD127-OSD120	The color of the bit map value "2" is specified in the CLUT (0-255) address. (width=4)

W1BMP45 (0003:06E4) Offset : 0x32

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD157	OSD156	OSD155	OSD154	OSD153	OSD152	OSD151	OSD150	OSD147	OSD146	OSD145	OSD144	OSD143	OSD142	OSD141	OSD140
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD157-OSD150	The color of the bit map value "5" is specified in the CLUT (0-255) address. (width=2,4)
7-0	OSD147-OSD140	The color of the bit map value "4" is specified in the CLUT (0-255) address. (width=4)

W1BMP67 (0003:06E6) Offset : 0x33

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD177	OSD176	OSD175	OSD174	OSD173	OSD172	OSD171	OSD170	OSD167	OSD166	OSD165	OSD164	OSD163	OSD162	OSD161	OSD160
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD177-OSD170	The color of the bit map value "7" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD167-OSD160	The color of the bit map value "6" is specified in the CLUT (0-255) address. (width=4)

W1BMP89 (0003:06E8) Offset : 0x34

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD197	OSD196	OSD195	OSD194	OSD193	OSD192	OSD191	OSD190	OSD187	OSD186	OSD185	OSD184	OSD183	OSD182	OSD181	OSD180
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD197-OSD190	The color of the bit map value "9" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD187-OSD180	The color of the bit map value "8" is specified in the CLUT (0-255) address. (width=4)

W1BMPAB (0003:06EA) Offset : 0x35

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1B7	OSD1B6	OSD1B5	OSD1B4	OSD1B3	OSD1B2	OSD1B1	OSD1B0	OSD1A7	OSD1A6	OSD1A5	OSD1A4	OSD1A3	OSD1A2	OSD1A1	OSD1A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD1B7-OSD1B0	The color of the bit map value "B" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD1A7-OSD1A0	The color of the bit map value "A" is specified in the CLUT (0-255) address. (width=2,4)

W1BMPCD (0003:06EC) Offset : 0x36

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1D7	OSD1D6	OSD1D5	OSD1D4	OSD1D3	OSD1D2	OSD1D1	OSD1D0	OSD1C7	OSD1C6	OSD1C5	OSD1C4	OSD1C3	OSD1C2	OSD1C1	OSD1C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD1D7-OSD1D0	The color of the bit map value "D" is specified in the CLUT (0-255) address. (width=4)
7-0	OSD1C7-OSD1C0	The color of the bit map value "C" is specified in the CLUT (0-255) address. (width=4)

W1BMPEF (0003:06EE) Offset : 0x37

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1F7	OSD1F6	OSD1F5	OSD1F4	OSD1F3	OSD1F2	OSD1F1	OSD1F0	OSD1E7	OSD1E6	OSD1E5	OSD1E4	OSD1E3	OSD1E2	OSD1E1	OSD1E0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	OSD1F7-OSD1F0	The color of the bit map value "F" is specified in the CLUT (0-255) address. (width=1,2,4)
7-0	OSD1E7-OSD1E0	The color of the bit map value "E" is specified in the CLUT (0-255) address. (width=4)

CONTROL_MISC (0003:06F4) Offset : 0x3A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PPSW	PPRV	RSV
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0
Trigger	-	-	-	-	-	-	-	-	-	-	-	-	-	VD	#	-

Bits	Register Name	Description
15-3	RSV	Reserved bit
3	CPBSY	Writing status signal of color pallete data. 0: Ready 1: Busy
2	PPSW	Change/Switch signal of ping-pong buffer.Referance ()
1	PPRV	PPSW of the above-mentioned register is reversed.
0	RSV	Reserved bit. Set 0.

CLUT_RAM_YCB (0003:06F6) Offset : 0x3B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Y07	Y06	Y05	Y04	Y03	Y02	Y01	Y00	CB07	CB06	CB05	CB04	CB03	CB02	CB01	CB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	Y07- Y00	Writing data (Y) to internal CLUT-RAM. It is necessary to write prior to CLUT_RAM_CR.
7-0	CB07 – CB00	Writing data (Cb) to internal CLUT-RAM. It is necessary to write prior to CLUT_RAM_CR.

CLUT_RAM_CR (0003:06F8) Offset : 0x3C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CR07	CR06	CR05	CR04	CR03	CR02	CR01	CR00	AD07	AD06	AD05	AD04	AD03	AD02	AD01	AD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#	#

Bits	Register Name	Description
15-8	CR07- CR00	Writing data (Cb) to internal CLUT-RAM .It is necessary to write continuously in order of CLUT_RAM_YCB and CLUT_RAM_CR.
7-0	AD07 - AD00	CLUT writing RAM address.

VIDEOWIN0_AD0_PING_PONG_BUF (0003:06FC) Offset : 0x3E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0PA21	V0PA20	V0PA19	V0PA18	V0PA17	V0PA16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0
Trigger	-	-	-	-	-	-	-	-	-	-	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-6	RSV	Reserved bit. Not in used.
5-0	V0PA21-V0PA16	Video window 0 higher/above address of Ping_Pong buffer.

VIDEOWIN0_AD1_PING_PONG_BUF (0003:06FE) Offset : 0x3F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V0PA15	V0PA14	V0PA13	V0PA12	V0PA11	V0PA10	V0PA09	V0PA08	V0PA07	V0PA06	V0PA05	V0PA04	V0PA03	V0PA02	V0PA01	V0PA00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Trigger	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD	VD

Bits	Register Name	Description
15-0	V0PA15-V0PA00	Video window 0 Lower/Below address of Ping_Pong buffer.

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SYNCEN (0003:0700) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DWEN	SYEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	DWEN*	Data write enable. Storing of CCD RAW data to SDRAM is controlled. Moreover, this bit is taken by timing of internal VD. Immediately after writing 1 to this bit, it gets active by VD timing. 0 : Disable 1 : Enable
0	SYEN	VD/HD enable. The moment 1 is written to this bit, the internal timing generator becomes active and starts the VD/HD output. In input, taking of VD/HD is started. 0 : Disable 1 : Enable

MODESET (0003:0702) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MDFS	FICM	HLPF	SWEN	GAMC	GAMP	CCDW1	CCDW0	CCDMD	DPOL	EDGT	FIPOL	HDPOL	VDPOL	FIDD	HDVDD
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	MDFS	Field status. The field of CCD RAW data is shown. 0 : First field 1 : Second field
14	HLPF*	Horizontal low pass filter. ON/OFF of 3 tap LPF is set. When it is ON status, filter processing of $1/4 + 1/2 Z^{-2} + 1/4 Z^{-4}$ is conducted for CCD data. 0 : Off 1 : On
13-12	IMPMOD1-0	Setting of data input mode 0 : CCD RAW data 1 : YCbCr 16bit 2 : YCbCr 8bit 3 : Reserved
11	PACK8	Setting of SDRAM store mode 0 : 16 bit/pixel 1 : 8 bit/pixel
10-8	CCDW2-0	CCD data width. Bit width on taking capture data is set. 0 : 14bit 1 : 13bit 2 : 12bit 3 : 11bit 4 : 10bit 5 : 9bit 6 : 8bit 7 : Reserved
7	CCDMD	CCD mode. Setting of CCD reading method. 0 : Progressive 1 : Interlace

Bits	Register Name	Description
6	DPOL	Data polarity. CCD data is inverted. 0 : Non inverse 1 : Inverse
5	SWEN	Select write enable. External WEN signal can be used. When it is used, in case that WEN signal is 1 and bit 1 of SYNCEN register is 1, CCD data is written to SDRAM. 0 : Don't use 1 : Use external WEN
4	FIPOL	Field identification signal polarity. The polarity of field identification signal is selected. 0 : Positive polarity 1 : Negative polarity
3	HDPOL	HD polarity. The polarity of HD is selected. 0 : Positive polarity 1 : Negative polarity
2	VDPOL	VD polarity. The polarity of VD is selected. 0 : Positive polarity 1 : Negative polarity
1	FIDD	Field identification signal I/O setting. The input/output of field identification signal is set. 0 : Input 1 : Output
0	HDVDD	HD/VD I/O setting. The input/output of HD/VD is set. 0 : Input 1 : Output

HDW (0003 :0704) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HDW11	HDW10	HDW9	HDW8	HDW7	HDW6	HDW5	HDW4	HDW3	HDW2	HDW1	HDW0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	HDW11-0*	The width of HD is set by PCLK unit. It is setting value + 1.

VDW (0003 :0706) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VDW11	VDW10	VDW9	VDW8	VDW7	VDW6	VDW5	VDW4	VDW3	VDW2	VDW1	VDW0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	VDW15-0*	The width of VD is set by line unit. It is setting value + 1.

PPLN (0003 :0708) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	PPLN15	PPLN14	PPLN13	PPLN12	PPLN11	PPLN10	PPLN9	PPLN8	PPLN7	PPLN6	PPLN5	PPLN4	PPLN3	PPLN2	PPLN1	PPLN0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	PPLN15-0*	Number of clocks per line is set. The setting value +1 is the cycle of 1 line.

LPFR (0003 :070A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	LPFR15	LPFR14	LPFR13	LPFR12	LPFR11	LPFR10	LPFR9	LPFR8	LPFR7	LPFR6	LPFR5	LPFR4	LPFR3	LPFR2	LPFR1	LPFR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	LPFR15-0*	Number of lines per one frame or one field is set by 1/2 line unit. (Setting value + 1)/2 line is the circle of one frame or one field.

SPH (0003 :070C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	SPH11	SPH10	SPH9	SPH8	SPH7	SPH6	SPH5	SPH4	SPH3	SPH2	SPH1	SPH0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	SPH11-0*	Horizontal starting position. The position where capturing of data to SDRAM starts is set by number of pixel from HD.

LPH (0003 :070E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	LPH11	LPH10	LPH9	LPH8	LPH7	LPH6	LPH5	LPH4	LPH3	LPH2	LPH1	LPH0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	LPH11-0*	The size of data captured to SDRAM is set by number of pixels. Capturing size is setting value + 1 pixel.

SLV0 (0003 :0710) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	SLV011	SLV010	SLV09	SLV08	SLV07	SLV06	SLV05	SLV04	SLV03	SLV02	SLV01	SLV00
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	SLV011-0*	The position where capturing data to SDRAM starts is set by number of lines from VD when field identification signal is 0.

SLV1 (0003 :0712) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	SLV111	SLV110	SLV19	SLV18	SLV17	SLV16	SLV15	SLV14	SLV13	SLV12	SLV11	SLV 10
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	SLV111-0*	The position where capturing data to SDRAM starts is set by number of lines from VD when field identification signal is 1.

LNV (0003 :0714) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	LVN11	LVN10	LVN9	LVN8	LVN7	LVN6	LVN5	LVN4	LVN3	LVN2	LVN1	LVN0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	LVN11-0*	The vertical size capturing data to SDRAM is set by number of lines. Capturing size is setting value + 1 line.

CULH (0003 :0716) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CLHE7	CLHE6	CLHE5	CLHE4	CLHE3	CLHE2	CLHE1	CLHE0	CLHO7	CLHO6	CLHO5	CLHO4	CLHO3	CLHO2	CLHO1	CLHO0
R/W	R/W-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15-8	CLHE*	The horizontal decimation pattern when data of even line is captured to SDRAM is set. 0 : Decimation (Not writing to SDRAM) 1 : Writing to SDRAM Example) 0xAA: 1/2 decimation
7-0	CLHO*	The horizontal decimation pattern when data of odd line is captured to SDRAM is set. 0 : Decimation (Not writing to SDRAM) 1 : Writing to SDRAM

CULV (0003 :0718) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CULV7	CULV6	CULV5	CULV4	CULV3	CULV2	CULV1	CULV0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	CULV7-0*	The vertical decimation pattern when data is captured to SDRAM is set. 0 : Decimation (Not writing to SDRAM) 1 : Writing to SDRAM Example) 0x25: 3/8 decimation

HSIZE (0003 :071A) Offset : 0xD

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	HSIZE8	HSIZE7	HSIZE6	HSIZE5	HSIZE4	HSIZE3	HSIZE2	HSIZE1	HSIZE0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8-0	HSIZE8-0*	Number of pixels per 1 line that captures data to SDRAM is set by 16-pixel unit (burst unit).

SDOFST(0003 :071C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	FIINV	FOFST1	FOFST0	LOFST02	LOFST01	LOFST00	LOFST12	LOFST11	LOFST10	LOFST22	LOFST21	LOFST20	LOFST32	LOFST31	LOFST30
R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	FIINV*	Field identification signal inverse 0 : Non inverse 1 : Inverse
13-12	FOFST1-0*	Line offset value of field ID = 1 is specified. 0 : + 1 line 1 : + 2 line 2 : + 3 line 3 : + 4 line
11-9	LOFST02-00*	Line offset values of even line and field ID = 1 are specified. 0 : + 1 line 1 : + 2 line 2 : + 3 line 3 : + 4 line 4 : - 1 line 5 : - 2 line 6 : - 3 line 7 : - 4 line
8-6	LOFST12-10*	Line offset values of odd line and field ID = 1 are specified. 0 : + 1 line 1 : + 2 line 2 : + 3 line 3 : + 4 line 4 : - 1 line 5 : - 2 line 6 : - 3 line 7 : - 4 line
5-3	LOFST22-20*	Line offset values of even line and field ID = 1 are specified. 0 : + 1 line 1 : + 2 line 2 : + 3 line 3 : + 4 line 4 : - 1 line 5 : - 2 line 6 : - 3 line 7 : - 4 line
2-0	LOFST32-30*	Line offset values of odd line and field ID = 1 are specified. 0 : + 1 line 1 : + 2 line 2 : + 3 line 3 : + 4 line 4 : - 1 line 5 : - 2 line 6 : - 3 line 7 : - 4 line

CADU (0003 :071E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DWDS	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CADU5	CADU4	CADU3	CADU2	CADU1	CADU0
R/W	R/W	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-6	RSV	Reserved bit. Not in use.
5-0	CADU5-0*	Upper address of first address is specified when data is written to SDRAM.

CADL(0003 :0720) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CADL15	CADL14	CADL13	CADL12	CADL11	CADL10	CADL9	CADL8	CADL7	CADL6	CADL5	CADL4	CADL3	CADL2	CADL1	CADL0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	CADL15-0*	Lower address of first address is specified when data is written to SDRAM. Address is specified as offset from start of SDRAM in units of 32B.

CCLAMP (0003 :0722) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	CLEN	CLP	CLL	CLHS11	CLHS10	CLHS9	CLHS8	CLHS7	CLHS6	CLHS5	CLHS4	CLHS3	CLHS2	CLHS1	CLHS00
R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	CLEN*	Clamp enable 0 : Disable 1 : Enable
13	CLP	Clamp pixel. The number of pixels used for detection of the clamp level is set. 0: 8 pixels 1: 16 pixels
12	CLL	Clamp line. The number of lines used for detection of the clamp level is set. 0: 8 lines 1: 16 lines
11-0	CLHS11-0	Clamp start. The position where detection of clamp level is conducted is set by number of pixels from HD.

CDCSUB (0003 :0724) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	DCSB13	DCSB12	DCSB11	DCSB10	DCSB9	DCSB8	DCSB7	DCSB6	DCSB5	DCSB4	DCSB3	DCSB2	DCSB1	CLHS00
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-0	DCSB13-0	When in clamp disable (CLEN = 0), DC value subtracted from CCD data is set.

CCOLP (0003 :0726) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CPEE11	CPEE10	CPOE11	CPOE10	CPEO11	CPEO10	CPOO11	CPOO10	CPEE01	CPEE00	CPOE01	CPOE00	CPEO01	CPEO00	CPOO01	CPOO00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-14	CPEE11-0	Color patterns of even pixel, even line and field ID = 1 are specified. 0 : R / Ye 1 : Gr / Cy 2 : Gb / G 3 : B / Mg

Bits	Register Name	Description
13-12	CPOE11-0	Color patterns of odd pixel, even line and field ID = 1 are specified.
11-10	CPEO11-0	Color patterns of even pixel, odd line and field ID = 1 are specified.
9-8	CPOO11-0	Color patterns of odd pixel, odd line and field ID = 1 are specified.
7-6	CPEE01-0	Color patterns of even pixel, even line and field ID = 0 are specified. Only bit7-0 is referred in case of progressive CCD.
5-4	CPOE01-0	Color patterns of odd pixel, even line and field ID = 0 are specified.
3-2	CPEO01-0	Color patterns of even pixel, odd line and field ID = 0 are specified.
1-0	CPOO01-0	Color patterns of odd pixel, odd line and field ID = 0 are specified.

COFST0 (0003 :0728) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CO07	CO06	CO05	CO04	CO03	CO02	CO01	CO00	CO17	CO16	CO15	CO14	CO13	CO12	CO11	CO10
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	CO07-0	The offset adjustment of the R/Ye pixel is conducted. Setting value is signed 8 bit and CO07 becomes sign bit. The variable range is –128 thru 127.
7-0	CO17-0	The offset adjustment of the Gr/Cy pixel is conducted. Setting value is signed 8 bit and CO17 becomes sign bit. The variable range is –128 thru 127.

COFST1 (0003 :072A) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CO27	CO26	CO25	CO24	CO23	CO22	CO21	CO20	CO37	CO36	CO35	CO34	CO33	CO32	CO31	CO30
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	CO27-0	The offset adjustment of the Gb/G pixel is conducted. Setting value is signed 8 bit and CO27 becomes sign bit. The variable range is –128 thru 127.
7-0	CO37-0	The offset adjustment of the B/Mg pixel is conducted. Setting value is signed 8 bit and CO37 becomes sign bit. The variable range is –128 thru 127.

CMFIL (0003 :072C) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	CMFEN	CMTH13	CMTH12	CMTH11	CMTH10	CMTH9	CMTH8	CMTH7	CMTH6	CMTH5	CMTH4	CMTH3	CMTH2	CMTH1	CMTH0
R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	CMFEN	On/OFF of median filter is set. 0 : Off 1 : On
13-0	CMTH11-0	The threshold of median filter is set.

CGAIN (0003 :072E) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	CG10	CG9	CG8	CG7	CG6	CG5	CG4	CG3	CG2	CG1	CG0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	1	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	CG10-0	The gain of CCD data is adjusted. The decimal point is between bit8 and bit7, and the variable range is 0 thru x7.9921875.

COFSTA (0003 :0730) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	COFT9	COFT8	COFT7	COFT6	COFT5	COFT4	COFT3	COFT2	COFT1	COFT0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	COFT9-0	The offset of CCD data is adjusted. The variable range is 0 thru +1023.

OUTTH (0003 :0732) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	OTH13	OTH12	COF11	COF10	OTH9	OTH8	OTH7	OTH6	OTH5	OTH4	OTH3	OTH2	OTH1	OTH0
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-0	OTH13-0	The threshold clipped is set at the final stage of CCD signal processing.

OUTCLIP (0003 :0734) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	OCLIP13	OCLIP12	OCLIP11	OCLIP10	OCLIP9	OCLIP8	OCLIP7	OCLIP6	OCLIP5	OCLIP4	OCLIP3	OCLIP2	OCLIP1	OCLIP0
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-0	OCLIP13-0	The level clipped is set at the final stage of CCD signal processing.

VDINT0 (0003 :076) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	CVD011	CVD010	CVD09	CVD08	CVD07	CVD06	CVD05	CVD04	CVD03	CVD02	CVD01	CVD00
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	CVD011-0	CCD VD0 interrupt timing is set by line unit.

VDINT1 (0003 :0738) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	CVD111	CVD110	CVD19	CVD18	CVD17	CVD16	CVD15	CVD14	CVD13	CVD12	CVD11	CVD10
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
11-0	CVD111-0	CCD VD1 interrupt timing is set by line unit.

SHUTCTL (0003 :073A) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SHTCTL
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15-1	RSV	Reserved bit. Not in use.
0	SHTCTL	CCD SHUTTER signal output. When 1 is set to this bit, CCDSHUTTER signal (GIO21) becomes 1 by timing of CCD VD1, and when 0 is set to this bit, immediately it becomes to 0.

CGAMMAWD (0003 :073C) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	GWDI2	GWDI1	GWDI0	H3ATBL	CCDTBL
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0

Bits	Register Name	Description
15-5	RSV	Reserved bit. Not in use.
4-2	GWDI2-0*	Input data to A-LAW is selected. 0 : 13-4 bit 1 : 12-3 bit 2 : 11-2 bit 3 : 10-1bit 4 : 9-0 bit 5-7 : Reserved
1	H3ATBL*	ON/OFF setting of A-LAW table of 3A engine input data 0 : off 1 : on
0	CCDTBL*	ON/OFF setting of A-LAW table of data stored to SDRAM. 0 : off 1 : on

REC656IF (0003 :073E) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	GWDO1	GWDO0
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	ECCFVH	When 1 is set, 1 bit error correction of FVH code is conducted.
0	R656ON	CCIR Rec. 656 Interface 0: OFF 1: ON

CCDCFG (0003 :0740) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	VDLC	MSBINVO	MSBINVI	BSWD	Y8POS	EXTRG	TRGSEL	WENLOG	FIDMD1	FIDMD0	BW656	YCSWP	INPTRU	VDPOL	YCOSWP	YCIO
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	VDLC	ON/OFF of latching function of CPU register in internal VSYNC. 0 : Latched in VSYNC 1 : No latching function
14	MSBINVO	MSB of C signal of YOUT/COOUT terminal is inverted. 0 : OFF 1 : MSB inversion
13	MSBINVI	MSB of C signal of data stored to SDRAM is inverted. 0 : OFF 1 : MSB inversion
12	BSWD	ON/OFF of Byte SWAP of data stored to SDRAM is set. 0 : OFF 1 : ON (BYTE SWAP)
11	Y8POS	Location of Y signal is set when YCbCr 8bit data is input. 0 : Even pixel 1 : Odd pixel
10	EXTRG	When 1 is set, the address of SDRAM is initialized at rising of external input terminal or at rising of EWEN bit (SYNCEN register).
9	TRGSEL	The signal that initializes SDRAM address is selected when EXTRG is 1. 0 : DWEN bit (SYNCEN register) 1 : FID input terminal
8	WENLOG	Setting of data capturing when WEN input terminal is used. 0 : Data is stored to SDRAM by signal that makes internal data capturing valid area signal and WEN signal are OR. 1 : Data is stored to SDRAM by signal that makes internal data capturing valid area signal and WEN signal are AND.
7-6	FIDMD1-0	Setting of FID detection function 0 : FID input signal is captured at the head of VD. 1 : FID input signal is used directly. 2 : FID is 0 at rising of VD input signal and 1 at falling. 3 : FID is 0 when VD input signal and HD input signal are in phase, and 1 when they are out-of-phase.
5	BW656	The data width in CCIR656 input mode is set. 0 : 8 bit 1 : 10 bit
4	YCINSWP	Y input (YIN7-0) and C input (CIN7-0) are swapped. 0 : YIN7-0 = Y signal / CIN7-0 = C signal 1 : YIN7-0 = C signal / CIN7-0 = Y signal
3	INPTRU	Y output and C output are controlled. 0 : Video Encoder output 1 : Input through mode YOUT7-0 = YIN7-0 / COOUT7-0 = CIN7-0
2	YCPUTSWP	Y output (YOUT7-0) and C output (COOUT7-0) are swapped. 0 : YOUT7-0 = Y signal / COOUT7-0 = C signal 1 : YOUT7-0 = C signal / COOUT7-0 = Y signal
1	IDS	Selection of Y signal / C signal capturing port 0 : YIN7-0/CIN7-0 1 : YOUT7-0/COOUT7-0
0	YCIO	I / O setting of YOUT/COOUT output terminal 0 : Output 1 : Input

- Registers with an *(asterisk) are latched by VD.

11.Preview

TMS320DM270 has the hardware preview engine and it can vary various setting by register.

PVEN (0003:0780) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PVEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	PVEN	Preview Engine enable. 0: Disable 1: Enable

PVSET1 (0003:0782) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	8BITS	ALAW	PVOS	RSOLY	INMOD	BSTAL	WEN
R/W	-	-	-	-	-	-	-	-	-	-R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0

Bits	Register Name	Description
15:7	RSV	Reserved bit. Not in use.
6	8BITS	When set, the data from CCDC is 8 bits width for each color. 0: 10 bits mode 1: 8 bits mode
5	ALAW	Inverse A-LAW mode 0: Normal mode 1: Inverse A-LAW mode
4	PVOS	One shot preview 0: Normal (Continuous process) 1: One shot (Outputs only one screen)
3	RSOLY	Resize only. Selecting the process contents. 0: Normal process. The input data is Raw data 1: Resize process only. The input data is YcbCr 4:2:2
2	INMOD	Input Mode. Select the Input signal. 0: Input from CCDC 1: Input from SDRAM
1	BSTAL	Burst align 0: Normal forwarding/Transferring 1: Forwarding /Transferring by burst boundary size.
0	WEN	Write enable. Set if the input write enable from CCDC is Valid or Invalid. 0: Invalid 1: Valid

RADDU (0003:0784) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RADD21	RADD20	RADD19	RADD18	RADD17	RADD16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
5:0	RADD5-0	When data from SDRAM is read, specify the upper/above address of first address specified as offset from start of SDRAM in units of 32 bytes.

RADDL (0003:0786) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RADD15	RADD14	RADD13	RADD12	RADD11	RADD10	RADD9	RADD8	RADD7	RADD6	RADD5	RADD4	RADD3	RADD2	RADD1	RADD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	RADD15-0	When data from SDRAM is read, specify the lower/below address of first address specified as offset from start of SDRAM in units of 32 bytes.

WADU (0003:0788) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	WAD21	WAD20	WAD19	WAD18	WAD17	WAD16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved bit. Not in use.
5:0	WADD5-0	When data from SDRAM is read, specify the upper/above address of first address specified as offset from start of SDRAM in units of 32 bytes.

WADL (0003:078A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WAD15	WAD14	WAD13	WAD12	WAD11	WAD10	WAD9	WAD8	WAD7	WAD6	WAD5	WAD4	WAD3	WAD2	WAD1	WAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	RADD15-0	When data from SDRAM is read, specify the lower/below address of first address specified as offset from start of SDRAM in units of 32 bytes.

HSRT (0003:078C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HSRT11	HSRT10	HSRT9	HSRT8	HSRT7	HSRT6	HSRT5	HSRT4	HSRT3	HSRT2	HSRT1	HSRT0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit. Not in use.
11:0	HRST11-0	Horizontal start position. When INMOD=0, the beginning position of taking data to SDRAM is set by number of pixels from HD. When INMOD=1, the horizontal size of the data read from SDRAM is set by the number of pixels.

HSIZE (0003:078E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HSIZE11	HSIZE10	HSIZE9	HSIZE8	HSIZE7	HSIZE6	HSIZE5	HSIZE4	HSIZE3	HSIZE2	HSIZE1	HSIZE0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit. Not in use.
11:0	HSIZE11-0	Horizontal Size The horizontal size of capturing data to SDRAM is set by the number of pixels. When INMOD=0, It is necessary to set more than 32 pixels from the number of horizontal total pixels by a small value. In other words, the Blanking period should be 32 or more.

VSRT (0003:0790) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VSRT11	VSRT10	VSRT9	VSRT8	VSRT7	VSRT6	VSRT5	VSRT4	VSRT3	VSRT2	VSRT1	VSRT0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit. Not in use.
11:0	VSRT11-0	Vertical start position. The beginning position of capturing data to SDRAM is set by number of lines from VD.

VSIZ (0003:0792) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VSIZ11	VSIZ10	VSIZ9	VSIZ8	VSIZ7	VSIZ6	VSIZ5	VSIZ4	VSIZ3	VSIZ2	VSIZ1	VSIZ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit. Not in use.
11:0	VSIZ11-0	Vertical Size A vertical size of capturing data to SDRAM is set by the number of lines.

PVSET2 (0003:0794) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	H_INTP	V_INTP	V_NFIL	CCDM	RSV	GBLD2	GBLD1	GBLD0	CFA7	CFA6	CFA5	CFA4	CFA3	CFA2	CFA1	CFA0
R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	H_INTP	Selection of horizontal interpolation process. 0: Other peripheral color information is used (5 tap). 1: Only the same color is used (2 tap).
14	V_INTP	Selection of vertical interpolation process. 0: Other peripheral color information is used (5 tap). 1: Only the same color is used (2 tap).
13	V_NFIL	Vertical noise filter 0: Off 1: On

Bits	Register Name	Description
12	CCDMD	CCDMode 0: RGB 1: Complementary color
11	RSV	Reserved bit. Not in use.
10:8	GBLD2-0	Selection of G process. When complementary color CCD is used, the G signal Process is selected. In $G_{out} = rG_{org} + (1-r)(Y_e + C_y - M_g)$ GBLD2 GBLD1 GBLD0 0 0 0 $r=1$ 0 0 1 $r=3/4$ 0 1 0 $r=1/2$ 0 1 1 $r=1/4$ 1 0 0 $r=0$
7:0	CFA7-0	Specification of colors. The color of the pixel is set according to the color filter of CCD. CFA7-6: Even pixel/Even lines CFA5-4: Odd Pixel/Even lines CFA3-2: Even pixel/Odd lines CFA1-0: Odd Pixel/Odd lines CFA7,5,3,1 CFA6,4,2,0 0 0: G/Ye 0 1: B/Cy 1 0: R/Mg 1 1: -/G

N_FIL (0003:0796) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	NF_EN	RSV	RSV	NF_RT1	NF_RT0
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	0	0

Bits	Register Name	Description
15:5 3:2	RSV	Reserved bit. Not in use.
4	NF_EN	The horizontal noise filter 0: Off 1: On
1:0	NF_RT1-0	Setting the horizontal noise filter level NF_RT1 NF_RT0 Noise removal level 0 0 Less 0 1 1 0 1 1 More

DGAIN (0003:0798) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	DGAIN9	DGAIN8	DGAIN7	DGAIN6	DGAIN5	DGAIN4	DGAIN3	DGAIN2	DGAIN1	DGAIN0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
9:0	DGAIN9-0	Digital gain. It is possible to set within the range of x0 thru x3.99609375 (3+255 / 256). The decimal point is located between DGAIN8 and DGAIN7. DGAIN9 - DGAIN0 11 1111 1111: x3.99609375 (3 + 255 / 256) 01 0000 0000: x1 00 1000 0000: x0.5 00 0000 0001: x0.00390625 (1/256) 00 0000 0000: x0

WB_GAIN0 (0003:079A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WG007	WG006	WG005	WG004	WG003	WG002	WG001	WG000	WG017	WG016	WG015	WG014	WG013	WG012	WG011	WG010
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	WG007-0	White balance gain setting. It is possible to set within the range of area 0 and Gb/Ye pixel x0 thru x3.984375 (3+63 / 64). The decimal point is located between WG006 and WG005.
7:0	WG017-0	White balance gain setting. B/Cy pixel

WB_GAIN1 (0003:079C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WG027	WG026	WG025	WG024	WG023	WG022	WG021	WG020	WG037	WG036	WG035	WG034	WG033	WG032	WG031	WG030
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	WG027-0	White balance gain setting. Gr/G pixel
7:0	WG037-0	White balance gain setting. R/Mg pixel

WGxx7 - WGxx0	Gain
1111 1111	x3.984375 (3 + 63 / 64)
0100 0000	x1
0010 0000	x0.5
0000 0001	x0.03125 (1/64)
0000 0000	x0

SMTH (0003:079E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SMTEN	SMTLV
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
1	SMTEN	Smoothing enable 0: Disable 1: Enable
0	SMTLV	Smoothing level 0: Less 1: More

H_RSZ (0003:07A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	HRSZ9	HRSZ8	HRSZ7	HRSZ6	HRSZ5	HRSZ4	HRSZ3	HRSZ2	HRSZ1	HRSZ0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	HRSZ9-0	Horizontal Resizing. When horizontal resizing is done, the denominator of the below-mentioned formula is set. RH shows the ratio of resizing.

$$R_H = \frac{16}{HRS}$$

V_RSZ (0003:07A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	VRSZ9	VRSZ8	VRSZ7	VRSZ6	VRSZ5	VRSZ4	VRSZ3	VRSZ2	VRSZ1	VRSZ0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	VRSZ9-0	Vertical resizing. When vertical resizing is done, the denominator of the below mentioned formula is set. RV shows the ratio of resizing.

$$R_V = \frac{16}{VRS}$$

BL_OFST0 (0003:07A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BL_R7	BL_R6	BL_R5	BL_R4	BL_R3	BL_R2	BL_R1	BL_R0	BL_G7	BL_G6	BL_G5	BL_G4	BL_G3	BL_G2	BL_G1	BL_G0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	BL_R7-0	Black level / adjustment. A black level of R signal after insert process is adjusted. (2's compl.)
7:0	BL_G7-0	Black level / adjustment. A black level of G signal after insert process is adjusted. (2's compl.)

BL_OFST1 (0003:07A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BL_B7	BL_B6	BL_B5	BL_B4	BL_B3	BL_B2	BL_B1	BL_B0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	BL_B7-0	Black level / adjustment. A black level of B signal after insert process is adjusted. (2's compl.)
7:0	RSV	Reserved bit. Not in use.

MTX_GAIN0 (0003:07A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXRR7	MXRR6	MXRR5	MXRR4	MXRR3	MXRR2	MXRR1	MXRR0	MXGR7	MXGR6	MXGR5	MXGR4	MXGR3	MXGR2	MXGR1	MXGR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXRR7-0	Matrix coefficient setting. The MTX_RR value of RGB2RGB conversion matrix is set.
7:0	MXGR7-0	Matrix coefficient setting. The MTX_GR value of RGB2RGB conversion matrix is set.

$$\begin{bmatrix} R_{out} \\ G_{out} \\ B_{out} \end{bmatrix} = \begin{bmatrix} MTX_RR & MTX_GR & MTX_BR \\ MTX_RG & MTX_GG & MTX_BG \\ MTX_RB & MTX_GB & MTX_BB \end{bmatrix} \begin{bmatrix} R_{in} \\ G_{in} \\ B_{in} \end{bmatrix} + \begin{bmatrix} R_offset \\ G_offset \\ B_offset \end{bmatrix}$$

MXxx7 - MXxx0	Gain
0111 1111	x 1.984375 (1 + 63/64)
0100 0000	x 1.0
0000 0001	x 0.015625(1/64)
0000 0000	x 0.0
1111 1111	x -0.015625(-1/64)
1100 0000	x -1.0
1000 0000	x -2.0

MTX_GAIN1 (0003:07AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXBR7	MXBR6	MXBR5	MXBR4	MXBR3	MXBR2	MXBR1	MXBR0	MXRG7	MXRG6	MXRG5	MXRG4	MXRG3	MXRG2	MXRG1	MXRG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXBR7-0	Matrix coefficient setting. The MTX_BR value of RGB2RGB conversion matrix is set.
7:0	MXRG7-0	Matrix coefficient setting. The MTX_RG value of RGB2RGB conversion matrix is set.

MTX_GAIN2 (0003:07AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXGG7	MXGG6	MXGG5	MXGG4	MXGG3	MXGG2	MXGG1	MXGG0	MXBG7	MXBG6	MXBG5	MXBG4	MXBG3	MXBG2	MXBG1	MXBG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXGG7-0	Matrix coefficient setting. The MTX_GG value of RGB2RGB conversion matrix is set.
7:0	MXBG7-0	Matrix coefficient setting. The MTX_BG value of RGB2RGB conversion matrix is set.

MTX_GAIN3 (0003:07AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXRB7	MXRB6	MXRB5	MXRB4	MXRB3	MXRB2	MXRB1	MXRB0	MXGB7	MXGB6	MXGB5	MXGB4	MXGB3	MXGB2	MXGB1	MXGB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXRB7-0	Matrix coefficient setting. The MTX_RB value of RGB2RGB conversion matrix is set.
7:0	MXGB7-0	Matrix coefficient setting. The MTX_GB value of RGB2RGB conversion matrix is set.

MTX_GAIN4 (0003:07B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXBB7	MXBB6	MXBB5	MXBB4	MXBB3	MXBB2	MXBB1	MXBB0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	MXBB7-0	Matrix coefficient setting. The MTX_BB value of RGB2RGB conversion matrix is set.
7:0	RSV	Reserved bit. Not in use.

MTX_OFST0 (0003:07B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXR9	MXR8	MXR7	MXR6	MXR5	MXR4	MXR3	MXR2	MXR1	MXR0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	MXR9-0	Hosu set value setting. The R_offset value of the RGB2RGB conversion matrix is set.

MTX_OFST1 (0003:07B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXG9	MXG8	MXG7	MXG6	MXG5	MXG4	MXG3	MXG2	MXG1	MXG0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:0	MXG9-0	Hosu set value setting. The G_offset value of the RGB2RGB conversion matrix is set.

MTX_OFST2 (0003:07B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXB9	MXB8	MXB7	MXB6	MXB5	MXB4	MXB3	MXB2	MXB1	MXB0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:0	MXB9-0	Hosu set value setting. The B_offset value of the RGB2RGB conversion matrix is set.

BYP_GAM (0003:07B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TBSEL	BYPR	BYPG	BYPB
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:4	RSV	Reserved bit. Not in use.
3	TBSEL	Gamma table selection. Select the gamma table. 0: Table 0 1: Table 1
2	BYP_R	Gamma table by-pass. In R signal gamma table is not used and it is bypassed. 0: Use Table 1: By-pass Table
1	BYP_G	Gamma table by-pass. In G signal gamma table is not used and it is bypassed. 0: Use Table 1: By-pass Table
0	BYP_B	Gamma table by-pass. In B signal gamma table is not used and it is bypassed. 0: Use Table 1: By-pass Table

CSC0 (0003:07BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CRY7	CRY6	CRY5	CRY4	CRY3	CRY2	CRY1	CRY0	CGY7	CGY6	CGY5	CGY4	CGY3	CGY2	CGY1	CGY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CRY7-0	Matrix coefficient setting. The CSC_RY value of RGB2RGB conversion matrix is set. It is possible to set within the range x-2 thru x1.984375 (1+63 / 64). The decimal point is located between CRY06 and CRY05.
7:0	CGY7-0	Matrix coefficient setting. The CSC_GY value of RGB2RGB conversion matrix is set.

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} CSC_RY & CSC_GY & CSC_BY \\ CSC_RCB & CSC_GCB & CSC_BCB \\ CSC_RCR & CSC_GCR & CSC_BCR \end{bmatrix} \begin{bmatrix} Rin \\ Gin \\ Bin \end{bmatrix} + \begin{bmatrix} y_offset \\ Cb_offset \\ Cr_offset \end{bmatrix}$$

Cxx7 - Cxx0	Parameter value
0111 1111	1.984375(1+63/64)
: :	:
0100 0000	1.0
: :	:
0000 0001	0.015625 (1/64)
0000 0000	0
1111 1111	-0.015625 (-1/64)
: :	:
1100 0000	-1.0
: :	:

1000	0000	-2.0
------	------	------

CSC1 (0003:07BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CBY7	CBY6	CBY5	CBY4	CBY3	CBY2	CBY1	CBY0	CRCB7	CRCB6	CRCB5	CRCB4	CRCB3	CRCB2	CRCB1	CRCB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CBY7-0	Matrix coefficient setting. The CSC_BY value of RGB2RGB conversion matrix is set.
7:0	CRCB7-0	Matrix coefficient setting. The CSC_RCB value of RGB2RGB conversion matrix is set.

CSC2 (0003:07BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CGCB7	CGCB6	CGCB5	CGCB4	CGCB3	CGCB2	CGCB1	CGCB0	CBCB7	CBCB6	CBCB5	CBCB4	CBCB3	CBCB2	CBCB1	CBCB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CGCB7-0	Matrix coefficient setting. The CSC_GCB value of RGB2RGB conversion matrix is set.
7:0	CBCB7-0	Matrix coefficient setting. The CSC_BCB value of RGB2RGB conversion matrix is set.

CSC3 (0003:07C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CRCR7	CRCR6	CRCR5	CRCR4	CRCR3	CRCR2	CRCR1	CRCR0	CGCR7	CGCR6	CGCR5	CGCR4	CGCR3	CGCR2	CGCR1	CGCR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CRCR7-0	Matrix coefficient setting. The CSC_RCR value of RGB2RGB conversion matrix is set.
7:0	CGCR7-0	Matrix coefficient setting. The CSC_GCR value of RGB2RGB conversion matrix is set.

CSC4 (0003:07C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CBCR7	CBCR6	CBCR5	CBCR4	CBCR3	CBCR2	CBCR1	CBCR0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	CBCR7-0	Matrix coefficient setting. The CSC_BCR value of RGB2RGB conversion matrix is set.
7:0	RSV	Reserved bit. Not in use.

Y_OFST (0003:07C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	OTY9	OTY8	OTY7	OTY6	OTY5	OTY4	OTY3	OTY2	OTY1	OTY0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit. Not in use.
9:0	OTY7-0	Offset Y. After YCbCr conversion, the Y signal Offset is set. Yout = Yin – setting value (0,1,....., 1023)

C_OFST (0003:07C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OTCB7	OTCB6	OTCB5	OTCB4	OTCB3	OTCB2	OTCB1	OTCB0	OTCR7	OTCR6	OTCR5	OTCR4	OTCR3	OTCR2	OTCR1	OTCR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	OTCB7-0	Offset CB. After YCbCr conversion, the Cb signal Offset is set. Cout = Cin – setting value (-128,....., 0,.....127)
7:0	OTCR7-0	Offset CR. After YCbCr conversion, the Cr signal Offset is set. Cout = Cin – setting value (-128,....., 0,.....127)

CNT_BRT (0003:07C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CNTR7	CNTR6	CNTR5	CNTR4	CNTR3	CNTR2	CNTR1	CNTR0	BRT7	BRT6	BRT5	BRT4	BRT3	BRT2	BRT1	BRT0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CBTR7-0	Contrast. After offset adjustments, the contrast of Y signal is set.(x0 thru x15.9375)
7:0	BRT7-0	Bright. After contrast adjustment, bright of Y signal is set.

C_SUP0 (0003:07CA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SUPEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	SUPEN	Color signal pseudocolor suppression. 0: Off 1: On

C_SUP1 (0003:07CC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SUPTH7	SUPTH6	SUPTH5	SUPTH4	SUPTH3	SUPTH2	SUPTH1	SUPTH0	SUPG7	SUPG6	SUPG5	SUPG4	SUPG3	SUPG2	SUPG1	SUPG0	SUPTH7
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

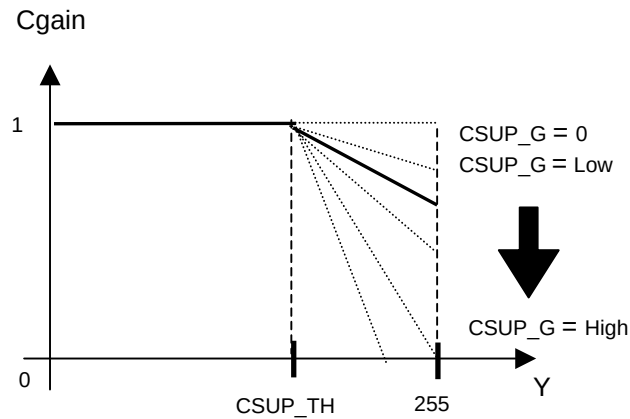
Bits	Register Name	Description
15:8	SUPTH7-0	Various signal pseudocolor suppression threshold. Set the CSUP_TH.
7:0	SUPG7-0	Various signal pseudocolor suppression gain. Set the CSUP_G.

C gain is suppressed smaller than 1 while luminance signal level is higher than CSUP_TH.

```

IF (Y > CSUP_TH)
    Cgain = 1 - (Y - CSUP_TH) / 256 * CSUP_G
ELSE
    Cgain = 1

```



SUPG		CSUP_G
1111	1111	0.99609375
1111	1110	0.99218750
...		...
0000	0001	0.00390625
0000	0000	0.00000000

SETUP_Y (0003:07CE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MAXY7	MAXY6	MAXY5	MAXY4	MAXY3	MAXY2	MAXY1	MAXY0	MINY7	MINY6	MINY5	MINY4	MINY3	MINY2	MINY1	MINY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MAXY7-0	Y signal maximum value. Data more than set value is clipped to the set value.
7:0	MINY7-0	Y signal minimum value. Data less than set value is clipped to set value.

SETUP_C (0003:07D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MAXC7	MAXC6	MAXC5	MAXC4	MAXC3	MAXC2	MAXC1	MAXC0	MINC7	MINC6	MINC5	MINC4	MINC3	MINC2	MINC1	MINC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MAXC7-0	C signal maximum value. Data more than set value is clipped to the set value.
7:0	MINC7-0	C signal minimum value. Data less than set value is clipped to the set value.

12. NTSC/PAL Encoder

VID01 (0003:0800) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	NTPLS	FFSW	CRCUT	CBAE	CBTY	STUP	DFLT1	DFLT0	YFLT	DAPD	DAOE	RSV	RSV	VIE	SYEXT	VIEN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	NTPLS	Selection switch for NTSC/PAL 0: NTSC 1: PAL
14	FFSW	Selection switch for Interlace/non-Interlace 0: interlace 1: non-interlace
13	CRCUT	Low pass filter of color-difference signal is set. 0: 1.5MHz cutoff 1: 3MHz cutoff
12	CBAE	Switching of color bar and normal output is set. 0: Normal output 1: Color bar output
11	CBTY	Type of color bar is set. 0: 75% 1: 100%
10	STUP	Setting up for Composite and Y/C signal is set. 0: 0% 1: 7.5%
9-8	DFLT1-DFLT0	Low pass filter for digital output of RGB is set. 0: No filter 1: 3-tap filter 2: 5-tap filter 3: No filter
7	YFLT	Low pass filter for luminance signal is set. 0: ON 1: OFF
6	DAPD	DAC power-down 0: Normal mode 1: Power-down mode
5	DAOE	DAC output enable 0: OFF (Output level is L) 1: ON
4-3	RSV	Reserved bit. Not in use.
2	VIE	Video output enable 0: OFF 1: ON
1	SYEXT	It is set whether the image data is put or not on the composite signal and Y signal. 0: Normal output 1: Only Sync signal output
0	VIEN	Active or non active of video encoder is set. 0: OFF 1: ON

VID02 (0003:0802) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	VIMD1	VIMD0	INLVL	SYSW	VSSW	SYNE	BREN	BRPL	BCNW7	BCNW6	BCNW5	BCNW4	BCNW3	BCNW2	BCNW1	BCNW0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-14	VIMD1-VIMD0	Video mode is selected. 00: Composite/YC 01: Reserved. 10: Digital RGB 11: Reserved.
13	INLVL	Amplitude of the input signal is controlled. 0: 16-235 amplitude 1: 0-255 amplitude
12	SYSW	Output Sync signal is selected. 0: Analog Sync 1: Digital Sync
11	VSSW	Output signal of VSYNC port is selected. 0: Normal V Sync 1: Composite Sync
10	SYNE	Sync signal enable 0: OFF 1: ON
9	BREN	Bright control enable 0: OFF 1: ON
8	BRPL	Polarity of bright control signal is set. 0: Normal 1: Inverse
7-0	BCNW7-BCNW0	Pulse width of bright control signal is set. Units of (CLK-VE NC/2)

DLCD_MODE0 (0003:0804) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DRGBE	VPL	HPL	RSV	LRCE	EXLID	EXIDP	IONM	RGBF	LINID	EVNF2	EVNF1	EVNF0	ODDF2	ODDF1	ODDF0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	DRGBE	Digital RGB enable 0: OFF 1: ON
14	VPL	Polarity of V Sync is set. 0: low active 1: high active
13	HPL	Polarity of H Sync is set. 0: low active 1: high active
12	RSV	Reserved bit.
11	LRCE	Setting of RGB data output line inversion function 0: OFF 1: ON
10	EXLID	ON/OFF of line selection signal input (EXTLINE_ID) is set. 0: non-ID setting 1: ID setting
9	EXIDP	Polarity of line selection signal input (EXTLINE_ID) is set.

Bits	Register Name	Description
8	IRONM	ON/OFF of the interface of Iron Man device in RGB565 mode is set. 0: OFF 1: ON
7	RGBF	ON/OFF of 5/6 resize filter at the time of Digital RGB output is set. 0: OFF 1: ON
6	LINID	ID setting for odd/even line in even field 0: Odd/Odd 1: Odd/Even
5-3	EVNF	RGB format of even line is set. 0: R0-G1-B2 1: R0-B1-G2 2: G0-R1-B2 3: G0-B1-R2 4: B0-R1-G2 5: B0-G1-R2
2-0	ODDF	RGB format of odd line is set. 0: R0-G1-B2 1: R0-B1-G2 2: G0-R1-B2 3: G0-B1-R2 4: B0-R1-G2 5: B0-G1-R2

DLCD_MODE1 (0003:0806) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DRGBI	DOE	DOEP	DACI2	DACI1	DACI0	ACE	HFDCK	DCKME	DCKP	PATW5	PATW4	PATW3	PATW2	PATW1	PATW0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	DRGBI	Output of Digital RGB is controlled. 0: Normal output 1: Fixed to 1
14	DOE	LCD_OE output enable 0: OFF 1: ON
13	DOEP	Polarity of LCD_OE output is set. 0: Normal output 1: Inverse output
12-10	DACI2-DACI0	Switching interval of LCD_AC output is set. Switching is done by field rate specified.
9	ACE	LCD_AC output enable 0: OFF 1: ON
8	HFDCK	Fundamental frequency of DCLK is set. 0: Same as picture rate 1: Twice as picture rate
7	DCKME	It is set whether DCLK is masked or not by the register which consists of Offset 0xC thru 0x10. 0: Mask OFF 1: Mask ON
6	DCKP	Polarity of DCLK is set. 0: Normal output 1: Inverse output
5-0	PATW5-0	Valid width of register (Offset 0x04-0x0B) which controls the pattern of DCLK is specified.

DCLK_PATTERN0E (0003:0808) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPE015	DPE014	DPE013	DPE012	DPE011	DPE010	DPE009	DPE008	DPE007	DPE006	DPE005	DPE004	DPE003	DPE002	DPE001	DPE000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPE015-DPE000	Pattern of DCLK in even line is set.

DCLK_PATTERN1E (0003:080A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPE115	DPE114	DPE113	DPE112	DPE111	DPE110	DPE109	DPE108	DPE107	DPE106	DPE105	DPE104	DPE103	DPE102	DPE101	DPE100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPE115-DPE100	Pattern of DCLK in even line is set.

DCLK_PATTERN2E (0003:080C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPE215	DPE214	DPE213	DPE212	DPE211	DPE210	DPE209	DPE208	DPE207	DPE206	DPE205	DPE204	DPE203	DPE202	DPE201	DPE200
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPE215-DPE200	Pattern of DCLK in even line is set.

DCLK_PATTERN3E (0003:080E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPE315	DPE314	DPE313	DPE312	DPE311	DPE310	DPE309	DPE308	DPE307	DPE306	DPE305	DPE304	DPE303	DPE302	DPE301	DPE300
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPE315-DPE300	Pattern of DCLK in even line is set.

DCLK_PATTERN0O (0003:0810) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPO015	DPO014	DPO013	DPO012	DPO011	DPO010	DPO009	DPO008	DPO007	DPO006	DPO005	DPO004	DPO003	DPO002	DPO001	DPO000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPO015-DPO000	Pattern of DCLK in odd line is set.

DCLK_PATTERN1O (0003:0812) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPO115	DPO114	DPO113	DPO112	DPO111	DPO110	DPO109	DPO108	DPO107	DPO106	DPO105	DPO104	DPO103	DPO102	DPO101	DPO100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-0	DPO115-DPO100	Pattern of DCLK in odd line is set.

DCLK_PATTERN2O (0003:0814) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPO215	DPO214	DPO213	DPO212	DPO211	DPO210	DPO209	DPO208	DPO207	DPO206	DPO205	DPO204	DPO203	DPO202	DPO201	DPO200
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPO215-DPO200	Pattern of DCLK in odd line is set.

DCLK_PATTERN3O (0003:0816) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DPO315	DPO314	DPO313	DPO312	DPO311	DPO310	DPO309	DPO308	DPO307	DPO306	DPO305	DPO304	DPO303	DPO302	DPO301	DPO300
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DPO315-DPO300	Pattern of DCLK in odd line is set.

DCLK_START_HE (0003:0818) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DSHE10	DSHE09	DSHE08	DSHE07	DSHE06	DSHE05	DSHE04	DSHE03	DSHE02	DSHE01	DSHE00
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	DSHE10-DSHE0	Horizontal starting position of DCLK enable in even line is set.

DCLK_START_HO (0003:081A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DSHO10	DSHO09	DSHO08	DSHO07	DSHO06	DSHO05	DSHO04	DSHO03	DSHO02	DSHO01	DSHO00
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	DSHO10-DSHO0	Horizontal starting position of DCLK enable in odd line is set.

DCLK_VALID_H (0003:081C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DVH10	DVH09	DVH08	DVH07	DVH06	DVH05	DVH04	DVH03	DVH02	DVH01	DVH00
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	DVH10-DVH0	Horizontal valid range of DCLK enable is set.

DCLK_START_V (0003:081E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DSV10	DSV09	DSV08	DSV07	DSV06	DSV05	DSV04	DSV03	DSV02	DSV01	DSV00
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	DSV10-DSV0	Vertical starting position of DCLK enable is set.

DCLK_VALID_V (0003:0820) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DVV10	DVV09	DVV08	DVV07	DVV06	DVV05	DVV04	DVV03	DVV02	DVV01	DVV00
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit. Not in use.
10-0	DVV10-DVV0	Vertical valid range of DCLK enable is set.

PULSE_WIDTH_H_V (0003:0822) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	LHW7	LHW6	LHW5	LHW4	LHW3	LHW2	LHW1	LHW0	DVLD07	DVLD06	DVLD05	DVLD04	DVLD03	DVLD02	DVLD01	DVLD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	LHW7-LHW0	Pulse width of H Sync is set.
7-0	DVLD7-DVLD0	Pulse width of V Sync is set.

H_INTERVAL (0003:0824) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HITV11	HITV10	HITV9	HITV8	HITV7	HITV6	HITV5	HITV4	HITV3	HITV2	HITV1	HITV0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	HITV11-HITV0	H Sync interval is set.

H_START (0003:0826) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HSTP11	HSTP10	HSTP9	HSTP8	HSTP7	HSTP6	HSTP5	HSTP4	HSTP3	HSTP2	HSTP1	HSTP0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	HSTP11-HSTP0	Starting position of H Sync is set.

H_VALID (0003:0828) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HVLD11	HVLD10	HVLD9	HVLD8	HVLD7	HVLD6	HVLD5	HVLD4	HVLD3	HVLD2	HVLD1	HVLD0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	HVLD11-HVLD0	Valid period of H Sync is set.

V_INTERVAL (0003:082A) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VITV11	VITV10	VITV9	VITV8	VITV7	VITV6	VITV5	VITV4	VITV3	VITV2	VITV1	VITV0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	VITV11-VITV0	V Sync interval is set.

V_START (0003:082C) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VSTP11	VSTP10	VSTP9	VSTP8	VSTP7	VSTP6	VSTP5	VSTP4	VSTP3	VSTP2	VSTP1	VSTP0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	VSTP11-VSTP0	Starting position of V Sync is set.

V_VALID (0003:082E) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VVLD11	VVLD10	VVLD9	VVLD8	VVLD7	VVLD6	VVLD5	VVLD4	VVLD3	VVLD2	VVLD1	VVLD0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	VVLD11-VVLD0	Valid period of V Sync is set.

HSYNC_DELAY (0003:0830) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HDLY11	HDLY10	HDLY9	HDLY8	HDLY7	HDLY6	HDLY5	HDLY4	HDLY3	HDLY2	HDLY1	HDLY0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	HDLY11-HDLY0	Delay of H sync is set.

VSYSN_DELAY (0003:0832) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VDLY11	VDLY10	VDLY9	VDLY8	VDLY7	VDLY6	VDLY5	VDLY4	VDLY3	VDLY2	VDLY1	VDLY0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	VDLY11-VDLY0	Delay of V sync is set.

CULL_LINE (0003:0834) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ODDF3	ODDF2	ODDF1	ODDF0	EVNF3	EVNF2	EVNF1	EVNF0	RSV	RSV	RSV	RSV	CULI3	CULI2	CULI1	CULI0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15-12	ODDF3-ODDF0	Odd line/even line output inverse line number in odd field is specified. The order of RGB data output is reversed from the line set up within the cycle set by CULI3-CULI0 register. It is valid when LRCE register is 1.
11-8	EVNF3-EVNF0	Odd line/even line output inverse line number in even field is specified. The order of RGB data output is reversed from the line set up within the cycle set by CULI3-CULI0 register. It is valid when LRCE register is 1.
7-4	RSV	Reserved bit. Not in use.
3-0	CULI3-CULI0	Cycle (line unit) of line inverse function is set. It is valid when LRCE register is 1.

PWM_CONTROL (0003:0836) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	PWME	PWMP	RSV	RSV	PRS11	PRS10	PRS9	PRS8	PRS7	PRS6	PRS5	PRS4	PRS3	PRS2	PRS1	PRS0
R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	PWME	PWM enable 0: OFF 1: ON
14	PWMP	PWM output polarity is set. 0: Normal output 1: Inverse
13-12	RSV	Reserved bit. Not in use.
11-0	PRS11-PRES0	Counts of PWM1 cycle at the time of 27MHz count lock is set. PWM cycle = (setting value of PRS[11:0]+1)/ 27MHz

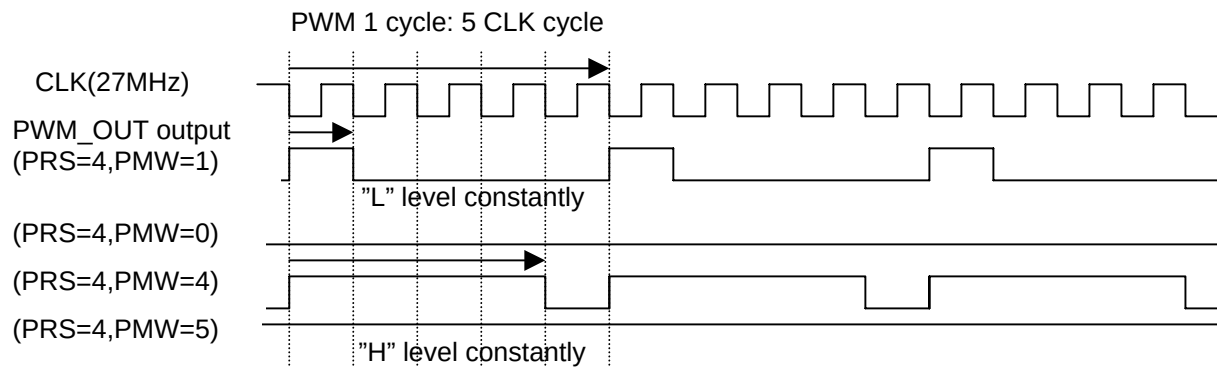
PWM_COUNT (0003:0838) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	PMW11	PMW10	PMW09	PMW08	PMW07	PMW06	PMW05	PMW04	PMW03	PMW02	PMW01	PMW00
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11-0	PMW11-PMW00	Pulse width of PWM_OUT output is set. (setting value 0-4095) When 0 is set, PWM_OUT is constantly "L" level, and when the value higher than PRS [11:0] is set, it is constantly set to "H" level output.

PRS[11:0] determines the number of counts in PWM1 cycle and 27MHz is used as the count clock. Moreover, PMW [11:0] determines "H" pulse (in normal output) width of PWM_OUT, and sets up the number of counts at the time of a 27MHz count clock.

Example) In case of PRS=4, (PWMP=0)



RGB_LEVEL (0003:083A) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CLIP7	CLIP6	CLIP5	CLIP4	CLIP3	CLIP2	CLIP1	CLIP0	RSV	RSV	BLK5	BLK4	BLK3	BLK2	BLK1	BLK0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	0	1	0	1	1	-	-	1	0	0	0	0	0

Bits	Register Name	Description
15-8	CLIP7-CLIP0	Clip level for upper side of RGB output is set.
7-6	RSV	Reserved bit. Not in use.
5-0	BLK5-BLK0	Black level of RGB output is set.

ATTRIBUTE0 (0003:083C) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ATR7	ATR6	ATR5	ATR4	ATR3	ATR2	ATR1	ATR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	ATR7-ATR0	Video attribute data register0 is set.

ATTRIBUTE1 (0003:083E) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ATR7	ATR6	ATR5	ATR4	ATR3	ATR2	ATR1	ATR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	ATR7-ATR0	Video attribute data register1 is set.

ATTRIBUTE2 (0003:0840) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ATR7	ATR6	ATR5	ATR4	ATR3	ATR2	ATR1	ATR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	ATR7-ATR0	Video attribute data register 2 is set.

REC656 (0003:0842) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	FIDPOL	RSV	VOUTHLD	CBCR1	CBCR0	SYNCE	RECE
R/W	-	-	-	-	-	-	-	-	-	R/W	-	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	-	0	0	0	0	0

Bits	Register Name	Description
15-7	RSV	Reserved bit. Not in use.
6	FIDPOL	Polarity of FIELD output (selection output signal from GIO28) is set. 0: Normal rotation 1: Inversion
5	RSV	Reserved bit. Not in use.
4	VOUTHLD	HOLD function of digital output When 1 is set, video output is stopped upon end of output of current line. Meanwhile, SYNC and video data output are stopped. When 0 is set, the output is started from the next line.
3-2	CBCR1-CBCR0	Output order of Y/Cb/Cr is specified. 0: Cb-Y-Cr-Y 1: Y-Cr-Y-Cb 2: Cr-Y-Cb-Y 3: Y-Cb-Y-Cr
1	SYNCE	SYNC processing enable 0: No SYNC processing 1: SYNC processing
0	RECE	REC656 mode is selected. 0: RGB output 1: REC656 (YCbCr) output

EPSON_LCD (0003:0844) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	EPVM	EPSYSC	EPGRBR	RSV	RSV	FIDPOL	FIDSEL	VOUTHLD	CBCR1	CBCR0	GCPSL1	GCPSL0
R/W	-	-	-	-	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	-	-	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserved bit. Not in use.
11	EPVM	Vertical signal selection. Generation circuit of vertical signal provided to LCD panel manufactured by EPSON is selected. 0: Normal mode 1: For test

Bits	Register Name	Description
10	EPSYSC	SYS control bit. SYS of LCD panel manufactured by EPSON is controlled. 0: Normal operation 1: B-field up
9	EPRGBR	RGB filter operation rate 0: 13.5MHz 1: 27MHz
8-7	RSV	Reserved bit. Not in use.
6-5	EPYFS	Prefilter selection 0: No filter 1: 1+1 2: 1+2+1
4	EP18DBT	EPSON 18DBT mode. In normal mode, pattern of DCLK is toggled per line, however DCLK is altered per 3-line in 18DBT mode. 0: Normal mode 1: 18DBT type
3	GCPRD	GCP RAM access prohibition bit. Read/write to GCPDATA register is controlled. 0: Access permit 1: Access prohibit
2	GCPRU	RAM is selected to GCP pulse.
1-0	GCPSL1-GCPSL0	ROM output pattern of GCP pulse is selected.

GCPDATA (0003:0846) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GCPD15	GCPD14	GCPD13	GCPD12	GCPD11	GCPD10	GCPD09	GCPD08	GCPD07	GCPD06	GCPD05	GCPD04	GCPD03	GCPD02	GCPD01	GCPD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	GCPD15-GCPD00	GCP RAM port register. When it is GCPRD=0, it is accessible to RAM table. When GCP table is set, firstly GCPRD register is set to 1 and is returned to 0. According to this operation, internal counter is cleared to 0. Then, writing pulse data required one after another to GCPDATA register can make setting. When GCP table is read out, firstly GCPRD register is set to 1 and is returned to 0. According to this operation, internal counter is cleared to 0. Then, setting contents can be read out by reading GCPDATA register required one after another. The data written at the beginning of GCP output is output in MSB first.

CASIO (0003:0848) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CGRES	CDM	CRIT	CSTB	CLS1	CLS0
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-6	RSV	Reserved bit. Not in use.
5	CGRES	GRES port control 0: Normal operation 1: Fixed at LOW level
4	CDM	Display mode 0: Normal mode 1: Vertical inverse mode
3	CRIT	RIT port control
2	CSTB	STBYB port control

Bits	Register Name	Description
1-0	CLS1- CLS0	CASIO LCD panel selection 00: Standard 01: NTSC 10: PAL

DOUTCTL (0003:084A) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	DOF1	DOF0	VCLKS	ESGEN	EFIV	EFEN	ESYIV	ESYEN	Y16O1	Y16O0	DOIV	DOMD1	DOMD0	DODIS
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-12	DOF1-DOF0	DCLK offset 0: 0 1: -0.5 2: +0.5
11	VCLKS	VCLK select 0: DCLK 1: External clock input
10	ESGEN	External SYNC analog synchronous mode 0: OFF 1: ON
9	EFIV	External field input inversion 0: Non-inversion 1: Inversion
8	EFEN	External field input mode 0: Use internal generation field 1: Use external field input
7	ESYIV	External SYNC input inversion 0: Non-inversion 1: Inversion
6	ESYEN	External SYNC input mode 0: Normal mode 1: External SYNC mode
5-4	Y16O[1.0]	Output order of C output is set. 00: Cb-Cr (based on 13.5MHz) 01: Cb-Cr (based on 27MHz) 10: Cr-Cb (based on 13.5MHz) 11: Cr-Cb (based on 27MHz)
3	DOIV	Digital output inversion 0: Normal output 1: Inverse output
2-1	DOMD1-DOMD0	Digital output mode 00: Digital RGB 01: YUV16 bit 10: EPSON LCD 11: CASIO LCD
0	DODIS	Digital output disable 0: Normal output 1: Fixed at LOW level

SGHSYNP (0003:084C) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SGVP3	SGVP2	SGVP1	SGVP0	RSV	RSV	RSV	SGHP8	SGHP7	SGHP6	SGHP5	SGHP4	SGHP3	SGHP2	SGHP1	SGHP0
R/W	R/W	R/W	R/W	R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	0	0	0	-	-	-	0	0	0	0	0	0	0	1	0

Bits	Register Name	Description
15-12	SGVP3-SGVP0	Vertical counter preset value at the time of external SYNC analog synchronous mode. It is 8 by default.
11-9	RSV	Reserved bit. Not in use.
8-0	SGHP8-SGHP0	Horizontal counter preset value at the time of external SYNC analog synchronous mode. It is 2 by default.

13.Clock Controller

PLLA (0003:0880) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	PTA	PBA	PPDA	RSV	PODA2	PODA1	PODA0	PMA3	PMA2	PMA1	PMA0	RSV	PNA2	PNA1	PNA0
R/W	R	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W
Default	0	0	0	0	-	0	0	0	1	0	0	0	-	0	0	1

Bits	Register Name	Description
15	RSV	Reserved bit
14	PTA	TEST for PLLA. It is for test. Set to 0 constantly.
13	PBA	BYPASS for PLLA. It is for test. Set to 0 constantly.
12	PPDA	PWRDN for PLLA. PLLA is powered down by setting to 1. It is 0 by default.
11	RSV	Reserved bit. Not in use.
10:8	PODA2-PODA0	OD for PLLA. It is for test. Set to 0 constantly.
7:4	PMA3-PMA0	Dividing ratio of M divider for PLLA. $M = PMA[3:0] + 1$. The output frequency of PLL is determined by $F_{in} * (M/N)$. N is dividing ratio of N divider given in PNA register. The input clock (F_{in}) is SYSCLK or MXI. Values of settable M and N should meet the following requirements. $200MHz < F_{in} * (M/N) < 351MHz$ It is 8 by default so that it can use 60MHz-leveled clock at the time of external host boot. (In case of 27MHz clock input)
3	RSV	Reserved bit. Not in use.
2:0	PNA2-PNA0	Dividing ratio of N divider of PLLA. $N = PNA[2:0] + 1$. It is 1 by default.

PLLB (0003:0882) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	PTB	PBB	PPDB	RSV	PODB2	PODB1	PODB0	PMB3	PMB2	PMB1	PMB0	RSV	RNB2	PNB1	PNB0
R/W	R	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W
Default	0	0	0	1	-	0	0	0	0	0	0	0	-	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit
14	PTB	TEST for PLLB. It is for test. Set to 0 constantly.
13	PBB	BYPASS for PLLB. It is for test. Set to 0 constantly.
12	PPDB	PWRDN for PLLB. PLLB is powered down by setting to 1. It is 1 by default.
11	RSV	Reserved bit. Not in use.
10:8	PODB2-PODB0	OD for PLLB. It is for test. Set to 0 constantly.
8:4	PMB3-PMB0	Dividing ratio of M divider for PLLB. $M = PMB[3:0] + 1$. Setting method is same as PMA bit of PLLA register. It is 0 by default.
3:2	RSV	Reserved bit. Not in use.
1:0	PNB2-PNB0	Dividing ratio of N divider for PLLB. $N = PNB[2:0] + 1$. Setting method is same as PNA bit of PLLB register. It is 0 by default.

CLKC (0003:0884) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	CMCIV	CCDIV	CENIV	CMTST0	CUA1S	CUA0S	CTM3S	CTM2S	CTM1S	CTM0S	COSDS	CCDS	CENS1	CENS0	CFINS
R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
14	CMCIV	MMC clock inverse bit. Switch after turning MMC clock off. 0 : Non-inverse 1 : Inverse
13	CCDIV	CCD clock inverse bit. Switch after turning OSD clock off. 0 : Non-inverse 1 : Inverse
12	CENIV	ENC clock inverse bit. Switch after turning ENC clock off. 0 : Non-inverse 1 : Inverse
11	CMTST0	Test bit. Fixed "0". No not use.
10	CUA1S	UART1 clock select. Switch after turning UART1 clock off. 0 : ARM clock 1 : PLLIN
9	CUA0S	UART0 clock select. Switch after turning UART0 clock off. 0 : ARM clock 1 : PLLIN
8	CTM3S	TMR3 clock select. Switch after turning TMR3 clock off. 0 : ARM clock 1 : PLLIN
7	CTM2S	TMR2 clock select. Switch after turning TMR2 clock off. 0 : ARM clock 1 : PLLIN
6	CTM1S	TMR1 clock select. Switch after turning TMR1 clock off. 0 : ARM clock 1 : PLLIN
5	CTM0S	TMR0 clock select. Switch after turning TMR0 clock off. 0 : ARM clock 1 : PLLIN
4	COSDS	OSD clock select. Switch after turning OSD clock off. 0 : SYSCLK 1 : PCLK
3	CCCDS	CCD clock select. Switch after turning CCD clock off. 0 : PCLK 1 : SYSCLK
2	CENS1	ENC clock select 1. Switch after turning ENC clock off. 0 : MXI/SYSCLK 1 : PCLK
1	CENS0	ENC clock select 0. Switch after turning ENC clock off. 0 : MXI 1 : SYSCLK
0	CFINS	PLL input clock select. Setting of PLL should be conducted again after setting alteration. In case of TEST2 = L 0 : MXI 1 : SYSCLK In case of TEST2 = H 0 : SYSCLK 1 : MXI

SEL (0003:0886) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	CAXLS	RSV	RSV	RSV	CSDRS	RSV	RSV	RSV	CDSPS	RSV	RSV	RSV	CARMS
R/W	-	-	-	R/W	-	-	-	R/W	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	0	-	-	-	1	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:13	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
12	CAXLS	AXL clock select 0 : PLL A 1 : PLL B
11:9	RSV	Reserved bit. Not in use.
8	CSDRS	SDRC clock select. PLLB is used by default. 0 : PLL A 1 : PLL B
7:5	RSV	Reserved bit. Not in use.
4	CDSPS	DSP clock select 0 : PLL A 1 : PLL B
3:1	RSV	Reserved bit. Not in use.
0	CARMS	ARM clock select 0 : PLL A 1 : PLL B

DIV (0003:0888) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	CMDIV1	CMDIV0	RSV	RSV	CSDIV1	CSDIV0	RSV	RSV	CDDIV1	CDDIV0	RSV	RSV	CADIV1	CADIV0
R/W	-	-	R/W	R/W	-	-	R/W	R/W	-	-	R/W	R/W	-	-	R/W	R/W
Default	-	-	0	0	-	-	0	0	-	-	0	0	-	-	0	1

Bits	Register Name	Description
15:14	RSV	Reserved bit. Not in use.
13:12	CAXDIV1-CAXDIV0	Dividing of accelerator clock. Dividing ratio = CAXDIV [1:0]+1.
11:10	RSV	Reserved bit. Not in use.
9:8	CSDIV1-CSDIV0	Dividing of SDRAM clock. Dividing ratio = CSDIV [1:0]+1.
7:6	RSV	Reserved bit. Not in use.
5:4	CDDIV1-CDDIV0	Dividing of DSP clock. Dividing ratio = CDDIV [1:0]+1.
3:2	RSV	Reserved bit. Not in use.
1:0	CADIV1-CADIV0	Dividing of ARM clock. Dividing ratio = CADIV [1:0]+1. It is 2 dividing by default.

BYP (0003:088A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	CMB	RSV	RSV	RSV	CSB	RSV	RSV	RSV	CDB	RSV	RSV	RSV	CAB
R/W	-	-	-	R/W	-	-	-	R/W	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	0	-	-	-	0	-	-	-	0	-	-	-	0/1

Bits	Register Name	Description
15:13	RSV	Reserved bit. Not in use.
12	CMB	Bypass PLL. Accelerator clock is bypassed. 0 : Does not bypass 1 : Bypass
11:9	RSV	Reserved bit. Not in use.
8	CSB	Bypass PLL. SDRAM clock is bypassed. 0 : Does not bypass 1 : Bypass
7:5	RSV	Reserved bit. Not in use.
4	CDB	Bypass PLL. DSP clock is bypassed. 0 : Does not bypass 1 : Bypass
3:1	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
0	CAB	Bypass PLL. ARM clock is bypassed. 0 : Does not bypass 1 : Bypass It is 0 by default in the external host boot mode (EXTHOST=H and BTSEL=H) and 1 by default in other status. Therefore, when 27 MHz is set to input frequency of PLL in the external host boot mode, the clock having (9/2)/2 times frequency 60.75 MHz by default is given as ARM clock after PLL locking.

MMCLK (0003:088C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CMDV7	CMDV6	CMDV5	CMDV4	CMDV3	CMDV2	CMDV1	CMDV0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Register Name	Description
15:8	RSV	Reserved bit. Not in use.
7:0	CMDV7-CMDV0	Setting of dividing ratio of MMC clock. ARM clock is divided by (setting value + 1). It is 255 by default.

CTST (0003:088E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:0	RSV	Reserved bit. Not in use. Fixed "0".

MOD0 (0003:0890) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CTST	RSV	RSV	RSV	RSV	CHPIB	C5409	CEHIF	CSDRC	CEMIF	CINTC	CATM	CADMAC	CICA	CBUSC	CACOR
R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	-	-	-	-	1	1	1	1	1	1	0	0	1	1	1

Bits	Register Name	Description
15	CTST	Test bit. Fixed "0". Do not use.
14:11	RSV	Reserved bit. Not in use.
10	CHPIB	HPIB clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
9	C5409	C5409 clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
8	CEHIF	External host interface clock 0 : Disable 1 : Enable
7	CSDRC	SDRAM controller clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.

Bits	Register Name	Description
6	CEMIF	External memory interface clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
5	CINTC	Interrupt controller clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
4	CATM	ATM clock 0 : Disable 1 : Enable
3	CADMAC	AIM DMA controller clock 0 : Disable 1 : Enable
2	CICA	i-CACHE clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
1	CBUSC	Bus controller clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.
0	CACORE	ARM core clock 0 : Disable 1 : Enable It is set to 1 automatically at the time of wake-up.

MOD1 (0003:0892) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CIMG	CIMX	CVLC	CDAC	CVENC	COSD	CPRV	CH3A	CCDC
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit. Not in use.
8	CIMG	Image buffer clock 0 : Disable 1 : Enable
7	CIMX	IMX clock 0 : Disable 1 : Enable
6	CVLC	VLC clock 0 : Disable 1 : Enable
5	CDAC	DAC clock 0 : Disable 1 : Enable
4	CVENC	NTSC/PAL encoder clock 0 : Disable 1 : Enable
3	COSD	OSD clock 0 : Disable 1 : Enable
2	CPRV	Preview clock 0 : Disable 1 : Enable

Bits	Register Name	Description
1	CH3A	H3A clock 0 : Disable 1 : Enable
0	CCDC	CCD controller clock 0 : Disable 1 : Enable

MOD2 (0003:0894) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	CI2C	CMTST1	CMMC	CSIF	CSIFD	CUAT1	CUAT0	CUSB	CGIO	CTMR3	CTMR2	CTMR1	CTMR0	CWDT
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:14	RSV	Reserved bit. Not in use.
13	CI2C	I2C clock 0 : Disable 1 : Enable
12	CMTST1	Test bit. Fixed "0". Do not use.
11	CMMC	MMC interface clock 0 : Disable 1 : Enable
10	CSIF1	Serial interface 1 clock 0 : Disable 1 : Enable
9	CSIF0	Serial interface 0 clock 0 : Disable 1 : Enable
8	CUAT1	UART1 clock 0 : Disable 1 : Enable
7	CUAT0	UART0 clock 0 : Disable 1 : Enable
6	CUSB	USB clock 0 : Disable 1 : Enable
5	CGIO	GIO clock 0 : Disable 1 : Enable
4	CTMR3	TIMER3 clock 0 : Disable 1 : Enable
3	CTMR2	TIMER2 clock 0 : Disable 1 : Enable
2	CTMR1	TIMER1 clock 0 : Disable 1 : Enable
1	CTMR0	TIMER0 clock 0 : Disable 1 : Enable
0	CWDT	Watch dog timer clock 0 : Disable 1 : Enable

LPCTL0 (0003:0896) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SLPMD
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
0	SLPMD	Sleep mode 0 : Normal 1 : Sleep When 1 is written, PLL is powered down and all clocks are turned off, and then it moves into sleep status. Recovery is done by external interrupt signal from GPIO0.

LPCTL1 (0003:0898) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PDMD	OSC48
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:1	RSV	Reserved bit. Not in use.
1	PDMD	Power down mode 0 : Normal 1 : Auto power down mode When it is 0, internal crystal resonators of MXI and M48XI can be stopped by making PWDWN pin H. When it is 1, moving from normal operation status into sleep status and stopping of crystal resonator are conducted automatically by only making PWDWN pin H. Moreover, bringing PWDWN pin back to L, recovery of crystal resonator and recovery from sleep are done automatically.
0	OSC48	M48XI crystal resonator enable 0 : Enable 1 : Disable

OSEL (0003:089A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O2SEL2	O2SEL1	O2SEL0	RSV	O1SEL2	O1SEL1	O1SEL0	RSV	O0SEL2	O0SEL1	O0SEL0
R/W	-	-	-	-	-	R/W	R/W	R/W	-	R/W	R/W	R/W	-	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	-	0	0	0	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit. Not in use.
10:8	O2SEL2-O2SEL0	General-purpose clock output 2 (GPIO18) source select 0 : ARM clock 1 : DSP clock 2 : SDR clock 3 : AXL clock 4 : PLL input clock 5 : M48XI
7	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
6:4	O1SEL2-O1SEL0	General-purpose clock output 1(GIO17) source select 0 : ARM clock 1 : DSP clock 2 : SDR clock 3 : AXL clock 4 : PLL input clock 5 : M48XI
3	RSV	Reserved bit. Not in use.
2:0	O0SEL2-O0SEL0	General-purpose clock output 0(GIO16) source select 0 : ARM clock 1 : DSP clock 2 : SDR clock 3 : AXL clock 4 : PLL input clock 5 : M48XI

O0DIV (0003:089C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O0DV15	O0DV14	O0DV13	O0DV12	O0DV11	O0DV10	O0DV9	O0DV8	O0DV7	O0DV6	O0DV5	O0DV4	O0DV3	O0DV2	O0DV1	O0DV0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O0DV15-O0DV0	Dividing ratio setting of general-purpose clock output 0. Clock source selected by O0SEL is divided into 2*(setting value +1).

O1DIV (0003:089E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O1DV15	O1DV14	O1DV13	O1DV12	O1DV11	O1DV10	O1DV9	O1DV8	O1DV7	O1DV6	O1DV5	O1DV4	O1DV3	O1DV2	O1DV1	O1DV0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O1DV15-O1DV0	Dividing ratio setting of general-purpose clock output 1. Clock source selected by O1SEL is divided into 2*(setting value +1).

O2DIV (0003:08A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O2DV3	O2DV2	O2DV1	O2DV0
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:4	RSV	Reserved bit. Not in use.
3:0	O2DV3-O2DV0	Dividing ratio setting of general-purpose clock output 2. Clock source selected by O2SEL is divided into setting value +1.

PWM0C (0003:08A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	P0C15	P0C14	P0C13	P0C12	P0C11	P0C10	P0C9	P0C8	P0C7	P0C6	P0C5	P0C4	P0C3	P0C2	P0C1	P0C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:0	P0C15-P0C0	Cycle of PWM output 0. It is number of counts on PLL input clock base. PWM output 0 is output selecting from GIO29.

PWM0H (0003:08A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	P0H15	P0H14	P0H13	P0H12	P0H11	P0H10	P0H9	P0H8	P0H7	P0H6	P0H5	P0H4	P0H3	P0H2	P0H1	P0H0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	P0H15-P0H0	High level period of PWM output 0. It is number of counts on PLL input clock basis.

PWM1C (0003:08A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	P1C15	P1C14	P1C13	P1C12	P1C11	P1C10	P1C9	P1C8	P1C7	P1C6	P1C5	P1C4	P1C3	P1C2	P1C1	P1C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	P1C15-P1C0	Cycle of PWM output 1. It is number of counts on PLL input clock base. PWM output 1 is output selecting from GIO30.

PWM1H (0003:08A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	P1H15	P1H14	P1H13	P1H12	P1H11	P1H10	P1H9	P1H8	P1H7	P1H6	P1H5	P1H4	P1H3	P1H2	P1H1	P1H0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	P1H15-P1H0	High level period of PWM output 1. It is number of counts on PLL input clock basis.

14. Bus Controller

ARM accesses to all RAM, ROM, and memory mapped registers through BUS controller.

ECR (0003:0900) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ECR15	ECR14	ECR13	ECR12	ECR11	ECR10	ECR09	ECR08	ECR07	ECR06	ECR05	ECR04	ECR03	ECR02	ECR01	ECR00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	ECR15-ECR00	The endian of written data is converted. The converted results can be read through EBYTER and EBITR registers.

EBYTER (0003:0902) Offset : 0x01

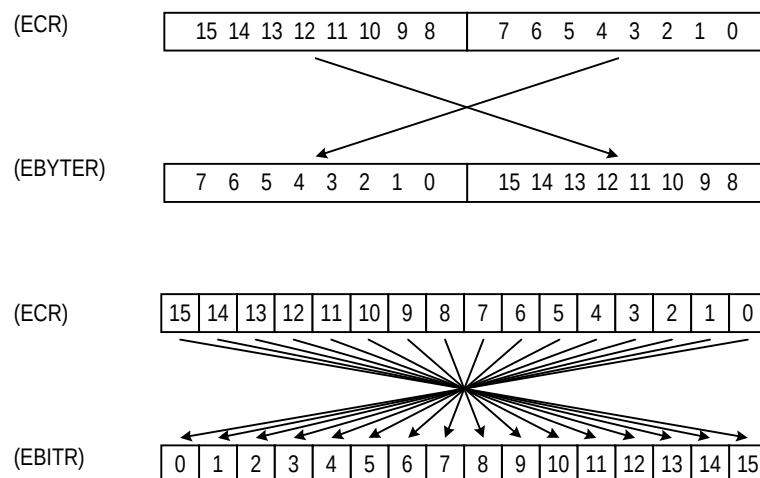
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EBYT15	EBYT14	EBYT13	EBYT12	EBYT11	EBYT10	EBYT09	EBYT08	EBYT07	EBYT06	EBYT05	EBYT04	EBYT03	EBYT02	EBYT01	EBYT00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	EBYT15-EBYT00	The data that is swapped the upper byte for the lower byte can be read when the data to be converted the endian is written to ECR register.

EBITR (0003:0904) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EBIT15	EBIT14	EBIT13	EBIT12	EBIT11	EBIT10	EBIT09	EBIT08	EBIT07	EBIT06	EBIT05	EBIT04	EBIT03	EBIT02	EBIT01	EBIT00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	EBIT15-EBIT00	The data that is swapped each bit can be read when the data to be converted the endian is written to ECR register.



REVR (0003:0906) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	REV7	REV6	REV5	REV4	REV3	REV2	REV1	REV0
R/W	-	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	REV7-REV0	Device revision code. eg.) ES1.0 => 0x10

15.SDRAM Controller

TMS320DM270 has two kinds of access with SDRAM. It is a burst access through the buffer and a normal memory access. SDRAM controller has a register to do this burst-access including 16 units of 16-bit data register for burst transfer of 16 words. In case of writing, when writing command is written after the data is set to these data register and sets to SDRAM address of writing destination, the writing is done for SDRAM. In case of reading, when SDRAM address of reading destination is set and read command is generated, the data is read from the data register. Initialization should operate for SDRAM when the power supply is turned on.

Power ON -> PREA -> REF x 8 -> MRS (Refer to the SDRAM Specification document.)

SDBUF_D0L (0003:0980) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD015	SD014	SD013	SD012	SD011	SD010	SD009	SD008	SD007	SD006	SD005	SD004	SD003	SD002	SD001	SD000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD015-SD000	SDRAM data register 0L

SDBUF_D0 H (0003:0982) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD031	SD030	SD029	SD028	SD027	SD026	SD025	SD024	SD023	SD022	SD021	SD020	SD019	SD018	SD017	SD016
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD031-SD016	SDRAM data register 0H

SDBUF_D1L (0003:0984) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD115	SD114	SD113	SD112	SD111	SD110	SD109	SD108	SD107	SD106	SD105	SD104	SD103	SD102	SD101	SD100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD115-SD100	SDRAM data register 1L

SDBUF_D1 H (0003:0986) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD131	SD130	SD129	SD128	SD127	SD126	SD125	SD124	SD123	SD122	SD121	SD120	SD119	SD118	SD117	SD116
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD131-SD116	SDRAM data register 1H

SDBUF_D2L (0003:0988) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD215	SD214	SD213	SD212	SD211	SD210	SD209	SD208	SD207	SD206	SD205	SD204	SD203	SD202	SD201	SD200
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD215-SD200	SDRAM data register 2L

SDBUF_D2 H (0003:098A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD231	SD230	SD229	SD228	SD227	SD226	SD225	SD224	SD223	SD222	SD221	SD220	SD219	SD218	SD217	SD216
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD231-SD216	SDRAM data register 2H

SDBUF_D3L (0003:098C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD315	SD314	SD313	SD312	SD311	SD310	SD309	SD308	SD307	SD306	SD305	SD304	SD303	SD302	SD301	SD300
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD315-SD300	SDRAM data register 3L

SDBUF_D3 H (0003:098E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD331	SD330	SD329	SD328	SD327	SD326	SD325	SD324	SD323	SD322	SD321	SD320	SD319	SD318	SD317	SD316
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD331-SD316	SDRAM data register 3H

SDBUF_D4L (0003:0990) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD415	SD414	SD413	SD412	SD411	SD410	SD409	SD408	SD407	SD406	SD405	SD404	SD403	SD402	SD401	SD400
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD415-SD400	SDRAM data register 4L

SDBUF_D4 H (0003:0992) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD431	SD430	SD429	SD428	SD427	SD426	SD425	SD424	SD423	SD422	SD421	SD420	SD419	SD418	SD417	SD416
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD431-SD416	SDRAM data register 4H

SDBUF_D5L (0003:0994) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD515	SD514	SD513	SD512	SD511	SD510	SD509	SD508	SD507	SD506	SD505	SD504	SD503	SD502	SD501	SD500
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD515-SD500	SDRAM data register 5L

SDBUF_D5H (0003:0996) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD531	SD530	SD529	SD528	SD527	SD526	SD525	SD524	SD523	SD522	SD521	SD520	SD519	SD518	SD517	SD516
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD531-SD516	SDRAM data register 5H

SDBUF_D6L (0003:0998) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD615	SD614	SD613	SD612	SD611	SD610	SD609	SD608	SD607	SD606	SD605	SD604	SD603	SD602	SD601	SD600
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD615-SD600	SDRAM data register 6L

SDBUF_D6H (0003:099A) Offset : 0xD

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD615	SD614	SD613	SD612	SD611	SD610	SD609	SD608	SD607	SD606	SD605	SD604	SD603	SD602	SD601	SD600
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD615-SD600	SDRAM data register 6H

SDBUF_D7L (0003:099C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD715	SD714	SD713	SD712	SD711	SD710	SD709	SD708	SD707	SD706	SD705	SD704	SD703	SD702	SD701	SD700
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD715-SD700	SDRAM data register 7L

SDBUF_D7H (0003:099E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SD731	SD730	SD729	SD728	SD727	SD726	SD725	SD724	SD723	SD722	SD721	SD720	SD719	SD718	SD717	SD716
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SD731-SD716	SDRAM data register 7H

SDBUF_AD1 (0003:09A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AAI	RSV	RSV	SDA21	SDA20	SDA19	SDA18	SDA17	SDA16
R/W	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	-	-	0	0	0	0	0	0

Bit	Register Name	Description
15-9	RSV	Reserved bit. Not in use.
8	AAI	Address auto increment. When "1" is set, the address is automatically incremented for the address set in SDBUF_AD0 and SDBUF_AD1.
7-6	RSV	Reserved bit. Not in use.
5-0	SDA21-SDA16	SDRAM address 1. The higher 6-bit of SDRAM address that reads and writes the data is set by this register. 1 address here means 32- byte unit, and it becomes an offset from SDRAM area header.

SDBUF_AD2 (0003:09A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDA15	SDA14	SDA13	SDA12	SDA11	SDA10	SDA9	SDA8	SDA7	SDA6	SDA5	SDA4	SDA3	SDA2	SDA1	SDA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	SDA15-SDA0	SDRAM address 2. The lower 16 bit of SDRAM address that reads and writes the data is set by this register. 1 address here means 32-bytes unit, and it becomes an offset of SDRAM area header.

SDBUF_CTL (0003:09A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PTSEL1	PTSEL0	BUFC	WM	WA	RSD
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bit	Register Name	Description
15-6	RSV	Reserved bit. Not in use.
5-4	PTSEL1-0	SDRAM request port selection in FIFO access 0: FIFO access function OFF 1: Preview Engine port SDRAM access of Preview Engine is prohibited and FIFO access is available. 2: 3A Engine port SDRAM access of 3A Engine is prohibited and FIFO access is available. 3: External CPUI/F port SDRAM access of external CPUI/F is prohibited and FIFO access is available.
3	BUFC	Buffer clear. When "1" is set, all data in the data register is cleared.
2	WM	Modified data write. When "1" is set, Only the data of register written in data register is written to SDRAM. Moreover, when transfer to SDRAM ends, this register is cleared automatically.
1	WA	All data write. When "1" is set, all data in data register is written to SDRAM. When transfer to SDRAM ends, this register is cleared automatically.
0	RSD	SDRAM data read. When "1" is set, the data from SDRAM is captured in data register. When transfer from SDRAM ends, this register is cleared automatically.

Transfer with SDRAM is conducted by 16 words unit.

SDMODE (0003:09A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TRDL	SDBW	TRCD	BNS	RSV	CASL	MEMT1	MEMT0	DQMC	APO	SDCTL5	SDCTL4	SDCTL3	SDCTL2	SDCTL1	SDCTL0
R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	-	0	0	0	1	0	0	0	0	0	0	0

Bit	Register Name	Description
15	TRDL	tRDL select. 1 or 2 cycle selection is possible by SDRAM clock unit. 0: 1 cycle 1: 2 cycle
14	SDBW	SDRAM bus width select 0: 32bit 1: 16bit
13	TRCD	tRCD select. 2 or 3 cycle selection is possible by SDRAM clock unit. 0: 2 cycle 1: 3 cycle
12	BNS	Bank select 0: 2 bank 1: 4 bank
11	RSV	Reserved bit. Not in use.
10	CASL	CAS latency. CAS latency selection is possible by SDRAM clock unit. 0: 2 cycle 1: 3 cycle
9-8	MEMT1-MEMT0	Memory type MEMT1 MEMT0 0 0 2k x 256 words 0 1 4k x 256 words 1 0 4k x 512 words 1 1 8k x 512 words
7	DQMC	DQM control 0: Normal 1: DQM is set to "1" compulsorily
6	APO	Auto power down mode 0: Disable 1: Enable
5-0	SDCTL5-SDCTL0	SDRAM control bit SDCTL[5:0] 000000: No operation (NOP) 000001: SDRAM mode setting (MSR) 000010: All banks pre charge (PREA) 000100: Auto refresh command (REF) 001000: Self refresh mode (REF) 010000: Self refresh mode and power-down mode are cleared. (SELFC) 100000: Power-down mode (PDN)

REFCTL (0003:09A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDMA12	SDMA11	SDMA10	SDMA22	SDMA21	SDMA20	RSV	REFEN	REFC7	REFC6	REFC5	REFC4	REFC3	REFC2	REFC1	REFC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	1	-	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-13	SDMA12-SDMA10	DMA select 1

		0: External memory interface 1: Serial interface 2: USB 3: MMC 4: DSP(HPIB)
12-10	SDMA22-SDMA20	DMA select 2 0: External memory interface 1: Serial interface 2: USB 3: MMC 4: DSP(HPIB) Note) It is not possible to set the same value at the same in bit15-13 and bit12-10. Example: 0x00XX prohibited
9	RSV	Reserved bit. Not in use.
8	REFEN	Auto refresh enable 0: Disable 1: Enable
7-0	REFC7-REFC0	Refresh cycle. The interval of auto refresh is set. (Setting value + 1) x 8 SDRAM clock comes to the interval.

SDPRTY1 (0003:09AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PRTY19	PRTY18	PRTY17	PRTY16	PRTY15	PRTY14	PRTY13	PRTY12	PRTY11	PRTY10
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	1	0	0	0	0	0	0	0	0	0

SDPRTY2 (0003:09AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PRTY29	PRTY28	PRTY27	PRTY26	PRTY25	PRTY24	PRTY23	PRTY22	PRTY21	PRTY20
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	1	0	0	0	0	0	0	0	0

SDPRTY3 (0003:09AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PRTY39	PRTY38	PRTY37	PRTY36	PRTY35	PRTY34	PRTY33	PRTY32	PRTY31	PRTY30
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	1	0	0	0	0	0	0	0

SDPRTY4 (0003:09B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PRTY49	PRTY48	PRTY47	PRTY46	PRTY45	PRTY44	PRTY43	PRTY42	PRTY41	PRTY40
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	1	0	0	0	0	0	0

SDPRTY5 (0003:09B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PRTY59	PRTY58	PRTY57	PRTY56	PRTY55	PRTY54	PRTY53	PRTY52	PRTY51	PRTY50
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	1	0	0	0	0	0

SDPRTY6 (0003:09B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PTY69	PTY68	PTY67	PTY66	PTY65	PTY64	PTY63	PTY62	PTY61	PTY60
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	1	0	0	0	0

SDPRTY7 (0003:09B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PTY79	PTY78	PTY77	PTY76	PTY75	PTY74	PTY73	PTY72	PTY71	PTY70
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	1	0	0	0

SDPRTY8 (0003:09B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PTY89	PTY88	PTY87	PTY86	PTY85	PTY84	PTY83	PTY82	PTY81	PTY80
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	1	0	0

SDPRTY9 (0003:09BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PTY99	PTY98	PTY97	PTY96	PTY95	PTY94	PTY93	PTY92	PTY91	PTY90
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	1	0

SDPRTY10 (0003:09BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PTY109	PTY108	PTY107	PTY106	PTY105	PTY104	PTY103	PTY102	PTY101	PTY100
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	1

The priority of SDRAM controller is determined.

It is set by writing "1" into any of the bits from bit0 to bit9. Writing "1" more than two in one register is inhibited. Bit9 has the highest priority and bit0 has the lowest. When "1" is written in the same bit position in a different register, the request of lower address register is more prioritized and the request of a higher address register is ignored.

The contents of the request that is selected by each register are as follows.

SDPRTY1: CCD Controller DMA request
SDPRTY2: Preview Engine DMA request
SDPRTY3: Hardware 3A Engine DMA request
SDPRTY4: OSD DMA request
SDPRTY5: Ext HOST request
SDPRTY6: ARM CPU request
SDPRTY7: External Memory I/F, Serial I/F MMC DMA request
SDPRTY8: External Memory I/F, Serial I/F MMC DMA request
SDPRTY9: ImageBuffer DMA request
SDPRTY10: SDRAM auto refresh request

PRTYON (0003:09BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PTYON
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bit	Register Name	Description
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15-1	RSV	Reserved bit. Not in use.
0	PTYON	According to writing "1" to this register, the priority written in register of SDPRTY1 to 10 is valid. The user need not worry about the update timing because the priority changing within SDRAM controller is automatically done when the access to SDRAM becomes IDLE.

SDRCTEST (0003:09C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TEST
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bit	Register Name	Description
15-1	RSV	Reserved bit. Not in use.
0	TEST	It is for test. Set to 0 in normal use.

16.External Memory Interface

- External memory space

The external memory area is divided into 5 areas by ARM bus controller. The respective area size and start address can be set freely. In DM270, various memories, devices, etc. can be connected within the five areas, and the chip select signals output respectively. Area 1(EM_CS1) can connect to CompactFlash/CF+ card, and when this area is accessed, CFE1/CFE2 is asserted. Area 2(EM_CS2) can connect to SmartMedia card, and when this area is accessed, SCE is asserted.

External memory area	Chip select terminal name
EM_CS0	FLASH_CE
EM_CS1	CFE1/CFE2
EM_CS2	SCE
EM_CS3	CE0
EM_CS4	CE1

Mapping of area1/area2

External memory area	Address in area	Memory
EM_CS1	00:0000 – 00:07FF	CF memory space - attribute memory area - common memory area
	40:0000 – 40:07FF	CF I/O space
EM_CS2	00:0000	SmartMedia Command Write
	00:0004	SmartMedia Address Write
	00:0008	SmartMedia Data Read/Write

- WAIT control

Description for setting parameter

The below parameter setting is available for each CS (CS0 - CS4).

CS0/CS3/CS4

Parameter	Description	Clock variables	Variable unit	Initial value after RESET
CYCLE	Number of cycles	2-16 clock.	1CLK unit (ARM_CLK)	11
CS_SU	Number of Setup of CS	0-15 clock	1CLK unit	0
WE_SU	Number of Setup of WE	same as above	same as above	1
OE_SU	Number of Setup of OE	same as above	same as above	1
CS_WAIT	Number of Wait of CS	1-16 clock	1CLK unit	10
WE_WAIT	Number of Wait of WE	same as above	same as above	9
OE_WAIT	Number of Wait of OE	same as above	same as above	9
IDLE	Number of IDLEs	0-3 clock	1CLK unit	1

CS1

Parameter	Description	Clock variables	Variable unit	Initial value after RESET
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CYCLE	Number of cycles	2-32 clock	1CLK unit (ARM_CLK)	22
CS_SU	Number of Setup of CS	0-15 clock	1 CLK unit	0
WE_SU	Number of Setup of WE	same as above	same as above	2
OE_SU	Number of Setup of OE	same as above	same as above	2
CS_WAIT	Number of Wait of CS	1-32 clock	1CLK unit	21
WE_WAIT	Number of Wait of WE	same as above	same as above	10
OE_WAIT	Number of Wait of OE	same as above	same as above	18
IDLE	Number of IDLEs	0-3 clock	1CLK unit	1

CS2

Parameter	Description	Clock variables	Variable unit	Initial value after RESET
CYCLE	Number of cycles	2-16 clock	1CLK unit (ARM_CLK)	8
WE_SU	Number of Setup of WE	0-15 clock	1CLK unit	2
OE_SU	Number of Setup of OE	same as above	same as above	1
WE_WAIT	Number of Wait of WE	1-16 clock	1CLK unit	3
OE_WAIT	Number of Wait of OE	same as above	same as above	4
READ_SU	Number of Setup of READ	3-18 clock	1CLK unit	12
IDLE	Number of IDLEs	0-3 clock	1CLK unit	1

It is possible to transfer from ARM internal memory, SDRAM and external memory to ARM internal memory, SDRAM and external memory using external bus DMA controller without ARM intervention.

Transfer source device	Receive device
External memory CS0	External memory CS0
External memory CS1	External memory CS1
External memory CS2	External memory CS2
External memory CS3	External memory CS3
External memory CS4	External memory CS4
SDRAM	SDRAM
AIM	AIM

The data of Image Buffer and DSP is transferred to not only SDRAM but also external memory.

CS0CTRL1 (0003:0A00) Offset : 0x0

CS3CTRL1 (0003:0A0E) Offset : 0x07

CS4CTRL1 (0003:0A12) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OECx3	OECx2	OECx1	OECx0	WECx3	WECx2	WECx1	WECx0	CECx3	CECx2	CECx1	CECx0	CYCLx3	CYCLx2	CYCLx1	CYCLx0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	0	0	0	1	0	0	0	1	0	0	1	1	0	1	0

Bits	Register Name	Description
15-12	OECx3-OECx0	Width of output enable OECx[3:0] Width (clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It is setting value + 1 clock. Consequently, set as the value below (cycle number – 1).
11-8	WECx3-WECx0	Width of write enable WECx[3:0] Width (clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It is setting value + 1 clock. Consequently, set as the value below (cycle number – 1).
7-4	CECx3-CECx0	Width of chip enable CECx[3:0] Width (clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It is setting value + 1 clock. Consequently, set as the value below (cycle number – 1).

Bits	Register Name	Description
3-0	CYCLx3-CYCLx0	Width of cycle CYCLx[3:0] Width (clock) 0 0 0 0 2 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It is setting value + 1 clock. However, in case of "0000", it is 2 clocks.

However, the update of CS0CTRL1 contents is executed at the same time as CS0CTRL2 ((0003:0A02) Offset : 0x01) is updated.

CS1CTRL1A (0003:0A04) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	CEC14	CEC13	CEC12	CEC11	CEC10	RSV	RSV	RSV	CYCL14	CYCL13	CYCL12	CYCL11	CYCL10
R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	-	-	-	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	1	0	1	0	0	-	-	-	1	0	1	0	1

Bits	Register Name	Description
15-13	RSV	Reserved bit. Not in use.
12-8	CEC13-CEC10	Width of chip enable CEC1[4:0] Width(clock) 0 0 0 0 0 1 0 0 0 0 1 2 0 0 0 1 0 3 0 0 0 1 1 4 0 0 1 0 0 5 : 1 1 1 1 0 31 1 1 1 1 1 32 It is setting value + 1 clock. Consequently, set as the value below (cycle number – 1).
7-5	RSV	Reserved bit. Not in use.
4-0	CYCL13-CYCL10	Width of cycle CYCL1[4:0] Width(clock) 0 0 0 0 0 2 0 0 0 0 1 2 0 0 0 1 0 3 0 0 0 1 1 4 0 0 1 0 0 5 : 1 1 1 1 0 31 1 1 1 1 1 32 It is setting value + 1 clock. However, in case of "0000", it is 2 clocks.

CS1CTRL1B (0003:0A06) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	OEC14	OEC13	OEC12	OEC11	OEC10	RSV	RSV	RSV	WEC14	WEC13	WEC12	WEC11	WEC10
R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	-	-	-	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	1	0	0	0	1	-	-	-	0	1	0	0	1

Bits	Register Name	Description
15-13	RSV	Reserved bit. Not in use.
12-8	OEC13-OEC10	Width of output enable OEC1[4:0] Width(clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 0 1 0 0 5 : 1 1 1 1 0 31 1 1 1 1 1 32 It becomes the setting value +1 clock. Please set the value less than (cycle number – 1).
7-5	RSV	Reserved bit. Not in use.
4-0	WEC13-WEC10	Width of write enable WEC1[4:0] Width(clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 0 1 0 0 5 : 1 1 1 1 0 31 1 1 1 1 1 32 It becomes the setting value +1 clock. Please set the value less than (cycle number – 1).

CS2CTRL1 (0003:0A0A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OEC23	OEC22	OEC21	OEC20	WEC23	WEC22	WEC21	WEC20	RSV	RSV	RSV	RSV	CYCL23	CYCL22	CYCL21	CYCL20
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W
Default	0	0	1	1	0	0	1	0	-	-	-	-	0	1	1	1

Bits	Register Name	Description
15-12	OEC23-OEC20	Width of output enable OEC2[3:0] Width(clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It becomes the setting value +1 clock. Please set the value less than (cycle number – 1).

Bits	Register Name	Description
11-8	WEC23-WEC20	Width of write enable WEC2[3:0] Width(clock) 0 0 0 0 1 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It becomes the setting value +1 clock. Please set the value less than (cycle number – 1).
7-4	RSV	Reserved bit. Not in use.
3-0	CYCL23-CYCL20	Width of cycle CYCL2[3:0] Width(clock) 0 0 0 0 2 0 0 0 1 2 0 0 1 0 3 0 0 1 1 4 0 1 0 0 5 : 1 1 1 0 15 1 1 1 1 16 It is setting value + 1 clock. However, in case of "0000", it is 2 clocks.

CS0CTRL2 (0003:0A02) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	IDLE01	IDLE00	OESU03	OESU02	OESU01	OESU00	WESU03	WESU02	WESU01	WESU00	CESU03	CESU02	CESU01	CESU00
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	1	0	0	0	1	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-12	IDLE01-IDLE00	Number of clocks of IDLE state IDLE1 IDLE0 Number of clocks 0 0 0 0 1 1 1 0 2 1 1 3
10-8	OESU03-OESU00	Number of clocks of output enable setup OESU0[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15

Bits	Register Name	Description
7-4	WESU03-WESU00	Number of write enable setup WESU0[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15
3-0	CESU03-CESU00	Number of clocks of chip enable setup CESU0[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15

CS1CTRL2 (0003:0A08) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	IDLE11	IDLE10	OESU13	OESU12	OESU11	OESU10	WESU13	WESU12	WESU11	WESU10	CESU13	CESU12	CESU11	CESU10
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	1	0	0	1	0	0	0	1	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-12	IDLE11-IDLE10	Number of clocks of IDLE state IDLE11 IDLE10 Number of clocks 0 0 0 0 1 1 1 0 2 1 1 3
11-8	OESU13-OESU10	Number of clocks of output enable setup OESU1[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15
7-4	WESU13-WESU10	Number of write enable setup WESU1[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15

Bits	Register Name	Description
3-0	CESU13-CESU10	Number of clocks of chip enable setup CESU1[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15

CS2CTRL2 (0003:0A0C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	IDLE21	IDLE20	OESU23	OESU22	OESU21	OESU20	WESU23	WESU22	WESU21	WESU20	RESU23	RESU22	RESU21	RESU20
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	1	0	0	0	1	0	0	1	0	1	0	0	1

Bits	Register Name	Description
15-14	RSV	Reserved bit. Not in use.
13-12	IDLE21-IDLE20	Number of clocks of IDLE state IDLE21 IDLE20 Number of clocks 0 0 0 0 1 1 1 0 2 1 1 3
11-8	OESU23-OESU20	Number of clocks of output enable setup OESU1[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15
7-4	WESU23-WESU20	Number of write enable setup WESU1[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : 1 1 1 0 14 1 1 1 1 15
3-0	RESU23-RESU20	Number of clocks of read cycle setup Number of setup of read cycle following after address input cycle during access to SmartMedia is set. After changed ALE terminal of SmartMedia to Low Level, commencement of read cycle is hold during the period set in READ_SU. RESU2[3:0] Number of clocks 0 0 0 0 3 0 0 0 1 4 0 0 1 0 5 : 1 1 1 0 17 1 1 1 1 18 It is setting value + 3 clock.

CS3CTRL2 (0003:0A10) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	BUSW3	IDLE31	IDLE30	OESU33	OESU32	OESU31	OESU30	WESU33	WESU32	WESU31	WESU30	CESU33	CESU32	CESU31	CESU30
R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved bit. Not in use.
14	BUSW3	Bus width of external device BUSW3 Bus width 0 8 bit 1 16 bit
13-12	IDLE31-IDLE30	Number of clocks of IDLE state IDLE31 IDLE30 Number of clocks 0 0 0 0 1 1 1 0 2 1 1 3
11-8	OESU33-OESU30	Number of clocks of output enable setup OESU3[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15
7-4	WESU33-WESU30	Number of write enable setup WESU3[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15
3-0	CESU33-CESU30	Number of clocks of chip enable setup CESU3[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15

CS4CTRL2 (0003:0A14) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CESL4	BUSW4	IDLE41	IDLE40	OESU43	OESU42	OESU41	OESU40	WESU43	WESU42	WESU41	WESU40	CESU43	CESU41	CESU41	CESU40
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15	CESL4	Polarity of chip enable of memory area 4 is selected. 0: Chip enable (CE1 pin) is Low Active. 1: Chip enable (CE1 pin) is High Active.

Bits	Register Name	Description
14	BUSW4	Bus width of external device BUSW4 Bus width 0 8 bit 1 16 bit
13-12	IDLE41-IDLE40	Number of clocks of IDLE state IDLE41 IDLE40 Number of clocks 0 0 0 0 1 1 1 0 2 1 1 3
11-8	OESU43-OESU40	Number of clocks of output enable setup OESU4[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15
7-4	WESU43-WESU40	Number of write enable setup WESU4[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15
3-0	CESU43-CESU40	Number of clocks of chip enable setup CESU4[3:0] Number of clocks 0 0 0 0 0 0 0 0 1 1 0 0 1 0 2 : : 1 1 1 0 14 1 1 1 1 15

BUSCTRL (0003:0A16) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	EMWE4	EMWE3
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15-2	RSV	Reserved bit. Not in use.
1	EMWE4	CS4 external wait (EM_WAIT) enable 0: When CS4 is accessed, external wait is invalid. 1: When CS4 is accessed, external wait is valid.
0	EMWE3	CS3 external wait(EM_WAIT) enable 0: When CS3 is accessed, external wait is invalid. 1: When CS3 is accessed, external wait is valid.

BUSRLS (0003:0A18) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BUSSEL	SDREL	EMREL
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15-3	RSV	Reserved bit. Not in use.
2	BUSSEL	SDRAM bus sharing mode 0: Normal mode 1: SDRAM bus is shared for use (CS0-CS4).
1	SDREL	SDRAM bus open control bit 0: SDRAM bus is not opened. According to writing "0", it is cleared. 1: SDRAM bus is opened.
0	EMREL	External memory bus open control bit 0: External memory bus is not opened. According to writing "0", it is cleared. 1: External memory bus is opened.

CFCTRL1 (0003:0A1A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CFSEL
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Bit Name	Description
15:1	RSV	Reserved bit. Not in use.
0	CFSEL	Selection of CF card IF(CFC) or Smart Media card IF(SSFDC) 0: CFC 1: SSFDC

CFCTRL2 (0003:0A1C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BUSSZ	RSV	RSV	RSV	CFMOD
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	-	0

Bits	Bit Name	Description
15:5	RSV	Reserved bit. Not in use.
4	BUSSZ	CF card dynamic bus sizing enable 0: The access to CF card is done by 16-bit bus width. 1: When CF card is in IO mode, dynamic bus sizing by IOIS16 terminal is executed.
3:1	RSV	Reserved bit. Not in use.
0	CFMOD	The memory space of CF card is accessed, REG output (ARM_A22 terminal) is controlled. When the I/O space is accessed, Low Level is output in REG regardless of this setting. 0: When memory space of CF is accessed, Low Level is output to REG. The access to attribute memory area is available. 1: When memory space of CF is accessed, High Level is output to REG. The access to common memory area is available.

SMCTRL (0003:0A1E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SCEH
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1

Bits	Bit Name	Description
15:1	RSV	Reserved bit. Not in use.
0	SCEH	Chip enable terminal (SCE) of SmartMedia is controlled. 0: SCE terminal is set to Low Level. 1: SCE terminal is set to High Level.

BUSINTEN (0003:0A20) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CBCND	CBIEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Bit Name	Description
14:2	RSV	Reserved bit. Not in use.
1	CBCND	CFRDY terminal interrupt condition 0: When CFRDY terminal transits from Low Level to High level, interrupt is generated. 1: When CFRDY terminal transits from High Level to Low Level, interrupt is generated.
0	CBIEN	Ready/busy interrupt enable of card connected to CFRDY terminal. When the state of CFRDY terminal changed, according to the above CFRDY terminal interrupt condition, interrupt is generated. 0: Disable 1: Enable

BUSSTS (0003:0A24) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	EMWT	IO16	CFWT	CFRDY
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	X	X	X	X

Bits	Bit Name	Description
15:4	RSV	Reserved bit. Not in use.
3	EMWT	The state of EM_WAIT terminal is shown.
2	IO16	The state of IOIS16 terminal is shown.
1	CFWT	The state of CFWAIT terminal is shown.
0	CFRDY	The state of CFRDY terminal is shown.

BUSWAITMD (0003:0A26) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CFRSL	CFRW
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Bit Name	Description
15:2	RSV	Reserved bit. Not in use.
1	CFRSL	The card type connected to CFRDY terminal is specified. 0: RDY/-BSY pin of CF card is connected to CFRDY terminal. 1: R/-B pin of SmartMedia card is connected to CFRDY terminal.

Bits	Bit Name	Description
0	CFRW	CFRDY wait. When the card specified by the above CFRSL accesses, it is set whether the start of bus cycle is controlled or not by CFRDY terminal. 0: Regardless of the state of CFRDY terminal, access for the card is begun. 1: While CFRDY terminal is low level, access for the card is not begun.

ECC1CP (0003:0A28) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	1CP5	1CP4	1CP3	1CP2	1CP1	1CP0	RD11	RD10
R/W	-	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Bit Name	Description
15:8	RSV	Reserved bit. Not in use.
7:2	1CP[5:0]	SmartMedia ECC CP. The result of the ECC calculation of Storing Area-1 of SmartMedia can be read.
1:0	RD1[1:0]	"1" can be read anytime.

ECC1LP (0003:0A2A) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	1LP15	1LP14	1LP13	1LP12	1LP11	1LP10	1LP09	1LP08	1LP07	1LP06	1LP05	1LP04	1LP03	1LP02	1LP01	1LP00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Bit Name	Description
15:0	1LP[15:0]	SmartMedia ECC LP. The result of ECC calculation of Storing Area-1 of SmartMedia can be read.

ECC2CP (0003:0A2C) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	2CP5	2CP4	2CP3	2CP2	2CP1	2CP0	RD11	RD10
R/W	-	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bits	Bit Name	Description
15:8	RSV	Reserved bit. Not in use.
7:2	2CP[5:0]	SmartMedia ECC CP. The result of ECC calculation of Storing Area-2 of SmartMedia can be read.
1:0	RD1[1:0]	"1" can be read anytime.

ECC2LP (0003:0A2E) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	2LP15	2LP14	2LP13	2LP12	2LP11	2LP10	2LP09	2LP08	2LP07	2LP06	2LP05	2LP04	2LP03	2LP02	2LP01	2LP00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Bit Name	Description
15:0	2LP[15:0]	SmartMedia ECC LP. The result of ECC calculation of Storing Area-2 of SmartMedia can be read.

ECCCLR (0003:0A30) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ECLR
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Bit Name	Description
15:1	RSV	Reserved bit. Not in use.
0	ECLR	ECC clear. When "1" is written, ECC is cleared. This bit is automatically cleared to "0".

PAGESZ (0003:0A32) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PAGESZ
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1

Bits	Bit Name	Description
15:1	RSV	Reserved bit. Not in use.
0	PAGESZ	Page size of used SmartMedia is set. 0 : 256+8byte/page 1 : 512+16byte/page

IMGDSPDEST (0003:0A34) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	IDST2	IDST1	IDST0	RSV	DDST2	DDST1	DDST0
R/W	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	-	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	0	0	-	0	0	0

Bits	Register Name	Description
15-7	RSV	Reserved bit. Not in use.
6-4	IDST2-IDST0	The data transfer source and transfer destination of image buffer are specified. The addresses of transfer source and transfer destination of data, etc. are set by register (internal DSP) of image buffer for CS0, CS3 and CS4 and set by register IMGDSPADD1/2 of EMIF for CS1 and CS2. CDST[2:0] Transfer destination 0 0 0 CS0 0 0 1 CS1 0 1 0 CS2 0 1 1 CS3 1 0 0 CS4
3	RSV	Reserved bit. Not in use.
2-0	DDST2-DDST0	The transfer source and transfer destination of data DSP (HPIB) are specified. The addresses of transfer source and transfer destination of data, etc. are set by register of HPIB for CS0, CS3 and CS4 and set by register IMGDSPADD1/2 of EMIF for CS1 and CS2. DDST[2:0] transfer source/transfer destination 0 0 0 CS0 0 0 1 CS1 0 1 0 CS2 0 1 1 CS3 1 0 0 CS4

PRIORCTL (0003:0A36) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	1PRY1	1PRY0	2PRY1	2PRY0	3PRY1	3PRY0	4PRY1	4PRY0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	1	1	0	1	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-6	1PRY1-1PRY0	1 st priority is specified. 1PRY1 1PRY0 1 st priority 0 0 ARM 0 1 DMA 1 0 Image Buffer 1 1 DSP(HPIB)
5-4	2PRY1-2PRY0	2 nd Priority is specified. 2PRY1 2PRY0 2 nd Priority 0 0 ARM 0 1 DMA 1 0 Image Buffer 1 1 DSP(HPIB)
3-2	3PRY1-3PRY0	3 rd Priority is specified. 3PRY1 3PRY0 3 rd Priority 0 0 ARM 0 1 DMA 1 0 Image Buffer 1 1 DSP(HPIB)
1-0	4PRY1-4PRY0	4 th Priority is specified. 4PRY1 4PRY0 4 th priority 0 0 ARM 0 1 DMA 1 0 Image Buffer 1 1 DSP(HPIB)

SOURCEADD1 (0003:0A38) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	SAFIX	RSV	RSV	SA25	SA24	SA23	SA22	SA21	SA20	SA19	SA18	SA17	SA16
R/W	-	-	-	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-13	RSV	Reserved bit. Not in use.
12	SAFIX	DMA transfer source byte address is fixed. 0: Address is incremented at the time of each transfer. 1: Fixed address.
11-10	RSV	Reserved bit. Not in use.
9-0	SA25-SA16	The upper 10 bits address of DMA transfer source byte address is described. Specify the relative address from the header address of each bank.

SOURCEADD2 (0003:0A3A) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SA15	SA15	SA15	SA15	SA15	SA10	SA9	SA8	SA7	SA6	SA5	SA4	SA3	SA2	SA1	SA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-0	SA15-SA0	The lower 16-bit address of DMA transfer source byte address is described. Since the transfer address is specified by multiples of 4, even though the lower 2 bits DA1/DA0 are described, it is ignored. Specify the relative address from the header address of each bank.

DESTADD1 (0003:0A3C) Offset 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	DAFIX	RSV	RSV	DA25	DA24	DA23	DA22	DA21	DA20	DA19	DA18	DA17	DA16
R/W	-	-	-	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-13	RSV	Reserved bit. Not in use.
12	DAFIX	DMA transfer destination address is fixed. 0: Address is incremented at the time of each transfer. 1: Fixed address.
11-10	RSV	Reserved bit. Not in use.
9-0	DA25-DA16	Upper 10-bit address of DMA transfer destination byte address is described. Specify the relative address from the header address of each bank.

DESTADD2 (0003:0A3E) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DA15	DA15	DA15	DA15	DA15	DA10	DA9	DA8	DA7	DA6	DA5	DA4	DA3	DA2	DA1	DA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	DA15-DA0	The lower 16-bit address of DMA transfer destination byte address is described. Since the transfer address is specified by multiples of 4, even though the lower 2 bits DA1/DA0 are described, it is ignored. Specify the relative address from the header address of each bank.

DMASIZE (0003:0A40) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TXB15	TXB15	TXB15	TXB15	TXB15	TXB10	TXB9	TXB8	TXB7	TXB6	TXB5	TXB4	TXB3	TXB2	TXB1	TXB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	TXB15-TXB0	Total transfer size is specified by byte. The transfer size must be specified consistently by multiples of 4 bytes. Therefore, even though "1" is written to TXB1 or TXB0, it is ignored.

DMADEVSEL (0003:0A42) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SDV2	SDV1	SDV0	RSV	DDV2	DDV1	DDV0
R/W	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	-	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	0	0	-	0	0	0

Bits	Register Name	Description
15-7	RSV	Reserved bit. Not in use.

Bits	Register Name	Description
6-4	SDV2-SDV0	DMA transfer source device is specified. SDV[6:4] Transfer source device 0 0 0 CS0 0 0 1 CS1 0 1 0 CS2 0 1 1 CS3 1 0 0 CS4 1 0 1 SDRAM 1 1 0 AIM
2-0	DDV2-DDV0	DMA transfer destination device is specified. DDV[2:0] Transfer destination device 0 0 0 CS0 0 0 1 CS1 0 1 0 CS2 0 1 1 CS3 1 0 0 CS4 1 0 1 SDRAM 1 1 0 AIM

DMACTL (0003:0A44) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	ENDI1	ENDI0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	GO
R/W	-	-	-	-	-	-	R/W	R/W	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	0	0	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15-10	RSV	Reserved bit. Not in use.
9-8	ENDI1-ENDI0	Switching of endian is controlled. (each 8 bit data swap) ENDI1 ENDI0 Switching of endian 0 0 D3 D2 D1 D0 -> D3 D2 D1 D0 0 1 D3 D2 D1 D0 -> D2 D3 D0 D1 1 0 D3 D2 D1 D0 -> D0 D1 D2 D3 1 1 D3 D2 D1 D0 -> D1 D0 D3 D2
7-1	RSV	Reserved bit. Not in use.
0	GO	DMA transfer commencement bit. When "1" is set, DMA transfer is started. After transfer completed, it is automatically cleared to "0".

IMGDSPADD1 (0003:0A46) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CDA22	CDA21	CDA20	CDA19	CDA18	CDA17	CDA16
R/W	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0

Bits	Register Name	Description
15-7	RSV	Reserved bit. Not in use.
6-0	CDA25-CDA16	Upper 7 bits address of byte address of external memory when data of image buffer or DSP (HPIB) and external memory CS1 or CS2 is described. Specify the relative address from the header address of each bank. During data transfer between Image buffer or DSP (HPIB) and CS1 or CS2, the CS1 or CS2 address specified by IMGDSPADD1/2 acts as a fixed address, i.e., address is not incremented after transfer of each unit of data.

IMGDSPADD2 (0003:0A48) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CDA15	CDA15	CDA15	CDA15	CDA15	CDA10	CDA9	CDA8	CDA7	CDA6	CDA5	CDA4	CDA3	CDA2	CDA1	CDA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	CDA15-CDA0	Lower 16 bits address of byte address of external memory when data of image buffer or DSP (HPIB) and external memory CS1 or CS2 is described. Specify the relative address from the header address of each bank. When it is transferred, it becomes a fixed address.

DPSTR0 (0003:0A4A) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST07	DST06	DST05	DST04	DST03	DST02	DST01	DST00
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	1	0	0	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DST07-DST00	Register for setting CS0 area maximum address. CS0 area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

DPSTR1 (0003:0A4C) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST17	DST16	DST15	DST14	DST13	DST12	DST11	DST10
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	1	0	0	0	0	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DST17-DST10	Register for setting SDRAM area maximum address. SDRAM area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

DPSTR2 (0003:0A4E) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST27	DST26	DST25	DST24	DST23	DST22	DST21	DST20
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	1	1	1	0	0	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DST27-DST20	Register for setting CS1 area maximum address. CS1 area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

DPSTR3 (0003:0A50) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST37	DST36	DST35	DST34	DST33	DST32	DST31	DST30
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	1	0	1	0	0	0	1

Bits	Register Name	Description
15-7	RSV	Reserved bit. Not in use.
6-0	DST37-DST30	Register for setting CS2 area maximum address. CS2 area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

DPSTR4 (0003:0A52) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST47	DST46	DST45	DST44	DST43	DST42	DST41	DST40
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	1	1	0	1	0	0	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DST47-DST40	Register for setting CS3 (Smart Media) area maximum address. CS3 area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

DPSTR5 (0003:0A54) Offset : 0x2A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DST57	DST56	DST55	DST54	DST53	DST52	DST51	DST50
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	0	0	0	0	0	0	1

Bits	Register Name	Description
15-8	RSV	Reserved bit. Not in use.
7-0	DST57-DST50	Register for setting CS4 area maximum address. CS4 area is set by specifying absolute address in external memory area Setting unit is 1M byte. Minimum value: 0 Maximum value: 255

TEST (0003:0A56) Offset : 0x2B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TEST
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15-1	RSV	Reserved bit. Not in use.
0	TEST	TEST mode 0: Normal mode 1: Test mode Internal ARM address is output to address bus when external memory access is IDLE.

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EP0OAD (0003:0A80) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O0AD10	O0AD9	O0AD8	O0AD7	O0AD6	O0AD5	O0AD4	O0AD3	O0AD2	O0AD1	O0AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	O0AD10-O0AD0	EP0 OUT RAM Base Address. Specify the base address of RAM allocated for EP0 OUT FIFO. Setting bit0/1 has no effect.

EP0OSZ (0003:0A82) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0SZ7	O0SZ6	O0SZ5	O0SZ4	O0SZ3	O0SZ2	O0SZ1	O0SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserve.
7:0	O0SZ7-O0SZ0	EP0 OUT Buffer Size. Specify the buffer size of EP0 OUT FIFO. The actual buffer size equals to this register value added 1. The maximum size of EP0 OUT is 256.

EP0OCTL (0003:0A84) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O0EMDE	O0BYTE	O0ENDI	RSV	RSV	RSV	RSV	RSV	O0STL	O0RST	O0ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	O0EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	O0BYTE	Byte access. When '1', data can be read with one byte.
8	O0ENDI	Endian. '0' if little endian, '1' if big endian when reading EP0ORDT.
7:3	RSV	Reserve.
2	O0STL	STALL bit for EP0 OUT. USB specifies the protocol stall in addition to the function stall of Endpoint by SetFeature command. The protocol stall is that when the data packet of the received setup transaction is illegal or unsupported command, the IN/OUT transaction of the successive data or status stage is stalled by the device. When set to one, EP0 OUT is stalled. The device returns a STALL handshake to the HOST after an OUT token is received. Since this bit is automatically cleared when the device receives the next SETUP token, users do not have to clear this bit. Note that do not set O0STL and O0ENA simultaneously.
1	O0RST	RESET bit for EP0 OUT FIFO. When set to one, the status and the data stored in the EP0 OUT FIFO are cleared. This bit is automatically cleared and write only. EP0OAD, EP0OSZ and EP0OIRQSZ are not cleared by this register. When a SETUP token is received, this bit is set automatically and the FIFO is reset to receive the DATA packet without a failure.

Bits	Register Name	Description
0	O0ENA	ENABLE bit for EP0 OUT. When this bit is set, the device returns an ACK handshake after receiving a DATA packet from HOST in OUT transactions to EP0. When this bit is cleared, the device receives but ignores the DATA packet, and returns a NAK handshake. This bit is cleared by hardware when □ the DEVICE sends an ACK handshake to the HOST (O0EMDE = 0) □ the number of data sent from the HOST in the transaction is less than MaxPktSize when the DEVICE sends an ACK handshake to the HOST (O0EMDE = 1) The enable bit auto clearing scheme can be selected by setting O0EMDE register (bit10 of EP0OCTL register). As for the SETUP transaction, after receiving a SETUP token the device always receives a DATA0 packet and returns an ACK handshake regardless of this bit.

EP0OIRQSZ (0003:0A86) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0IRQ8	O0IRQ7	O0IRQ6	O0IRQ5	O0IRQ4	O0IRQ3	O0IRQ2	O0IRQ1	O0IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8:0	O0IRQ8-O0IRQ0	EP0 OUT Threshold for Interrupt. Specify the threshold value for the interrupt assertion. (See the description of USBINTST register)

EP0ORDT (0003:0A8A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O0RDT15	O0RDT14	O0RDT13	O0RDT12	O0RDT11	O0RDT10	O0RDT9	O0RDT8	O0RDT7	O0RDT6	O0RDT5	O0RDT4	O0RDT3	O0RDT2	O0RDT1	O0RDT0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O0RDT15-O0RDT0	EP0 OUT Read Port. Read port of EP0 OUT FIFO. The data field of the received data packet is stored in the FIFO. Read only.

EP0OREST (0003:0A8C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0REST8	O0REST7	O0REST6	O0REST5	O0REST4	O0REST3	O0REST2	O0REST1	O0REST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8:0	O0REST8-O0REST0	EP0 OUT Available Byte Size. Indicate the available byte size for read from EP0 OUT FIFO.

EP0OST (0003:0A8E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0FUL	O0EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserved bit.

Bits	Register Name	Description
1	O0FUL	EP0 OUT Full. Set when the FIFO is full. Cleared when the FIFO is not full.
0	O0EMP	EP0 OUT Empty. Set when the FIFO is empty. Cleared when the FIFO is not empty.

EP0IAD (0003:0A90) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I0AD10	I0AD9	I0AD8	I0AD7	I0AD6	I0AD5	I0AD4	I0AD3	I0AD2	I0AD1	I0AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	I0AD10-I0AD0	EP0 IN RAM Base Address. Same as EP0OAD.

EP0ISZ (0003:0A92) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0SZ7	I0SZ6	I0SZ5	I0SZ4	I0SZ3	I0SZ2	I0SZ1	I0SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	I0SZ7- I0SZ0	EP0 IN Buffer Size. Same as EP0OSZ. The maximum FIFO size is 256.

EP0ICTL (0003:0A94) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I0EMDE	I0BYTE	I0ENDI	RSV	RSV	RSV	RSV	RSV	I0STL	I0RST	I0ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10	I0EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	I0BYTE	Byte access. When '1', data can be written with one byte.
8	I0ENDI	Endian. '0' if little endian, '1' if big endian when writing EP0IWDI.
7:3	RSV	Reserved bit.
2	I0STL	STALL bit for EP0 IN. When set, the protocol stall condition for EP0 IN is activated. When an IN token to EP0 is received, the device returns a STALL handshake. This bit is automatically cleared after the device receives the next SETUP packet. Note that this stall bit and the enable bit, bit0 of this EP0ICTL, must not be set simultaneously.
1	I0RST	RESET bit for EP0 IN FIFO. When set to one, the status and the data stored in EP0 OUT FIFO are cleared. This register is automatically cleared and write only. EP0IAD, EP0ISZ and EP0IIRQSZ are not cleared by this register.

Bits	Register Name	Description
0	I0ENA	ENABLE bit for EP0 IN. When this bit is set, the DEVICE returns a DATA0/DATA1 packet with data in the FIFO after receiving an IN token to EP0. When this bit is cleared, the DEVICE returns a NAK handshake for IN token. This bit is automatically cleared by hardware when □ the DEVICE receives an ACK handshake from the HOST (I0EMDE = 0) □ the rest of bytes in FIFO when the DEVICE receives an ACK handshake from the HOST is less than MaxPktSize (I0EMDE = 1) The ENABLE bit control scheme can be selected by setting I0EMDE. If a NAK handshake is received after sending the DATA packet, the data sent to the HOST is recovered. The FIFO status (EP0IST) and this bit are not changed from the status before the transaction.

EP0IIRQSZ (0003:0A96) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0IRQ8	I0IRQ7	I0IRQ6	I0IRQ5	I0IRQ4	I0IRQ3	I0IRQ2	I0IRQ1	I0IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8:0	I0IRQ8- I0IRQ0	EP0 IN Threshold for Interrupt. Same as EP0IIRQSZ.

EP0IWDT (0003:0A98) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I0WDT15	I0WDT14	I0WDT13	I0WDT12	I0WDT11	I0WDT10	I0WDT9	I0WDT8	I0WDT7	I0WDT6	I0WDT5	I0WDT4	I0WDT3	I0WDT2	I0WDT1	I0WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I0WDT15- I0WDT0	EP0 IN Write Port. Write port of EP0 IN FIFO. The data field of the data packet to be sent to HOST should be stored in the FIFO by writing this register. Write Only.

EP0IREST (0003:0A9C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0REST8	I0REST7	I0REST6	I0REST5	I0REST4	I0REST3	I0REST2	I0REST1	I0REST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8:0	I0REST8- I0REST0	EP0 IN Available Byte Size. The available byte size for write to EP0 IN FIFO.

EP0IST (0003:0A9E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0FUL	I0EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserved bit.
1	I0FUL	EP0 IN Full.
0	I0EMP	EP0 IN Empty.

EP1OAD (0003:0AA0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1AD10	O1AD9	O1AD8	O1AD7	O1AD6	O1AD5	O1AD4	O1AD3	O1AD2	O1AD1	O1AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserved bit.
10-0	O1AD10- O1AD0	EP1 OUT RAM Base Address.

EP1OSZ (0003:0AA2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1SZ10	O1SZ9	O1SZ8	O1SZ7	O1SZ6	O1SZ5	O1SZ4	O1SZ3	O1SZ2	O1SZ1	O1SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	O1SZ10- O1SZ0	EP1 OUT Buffer Size. The maximum FIFO size of EP1 OUT is 2048.

EP1OCTL (0003:0AA4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1EMDE	O1BYTE	O1ENDI	RSV	RSV	RSV	RSV	RSV	O1STL	O1RST	O1ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10	O1EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	O1BYTE	Byte access. When '1', data can be read with one byte.
8	O1ENDI	Endian. '0' if little endian, '1' if big endian when reading EP1ORDT.
7:3	RSV	Reserved bit.
2	O1STL	STALL bit for EP1 OUT. When set, the function stall condition for EP1 OUT can be applied. On receiving an OUT token with setting O1STL, the internal state of EP1 OUT changes to HALT and the endpoint is stalled. The device receives and ignores the DATA packet, and sends a STALL handshake. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting O1ENA when ClearFeature(HALT) is received, both bits are automatically cleared at that time. Setting STALL bit without setting O1ENA does not clear them automatically.
1	O1RST	RESET bit for EP1 OUT FIFO. When set to one, the status and the data stored in EP1 OUT FIFO are cleared. This register is automatically clear and write only. EP1OAD, EP1OSZ and EP1OIRQSZ are not cleared by this register.
0	O1ENA	ENABLE bit for EP1 OUT. When this bit is set, the device returns an ACK handshake after receiving the DATA packet from the HOST in the OUT transactions to EP1. When this bit is cleared, the device receives but ignores the DATA packet, and returns a NAK handshake. This bit is cleared by hardware when □ the DEVICE sends an ACK handshake to the HOST (O1EMDE = 0) □ the number of data sent from the HOST in the transaction is less than MaxPktSize when the DEVICE sends an ACK handshake to the HOST (O1EMDE = 1) The ENABLE bit control scheme can be selected by setting O1EMDE.

EP1OIRQSZ (0003:0AA6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	O1IRQ11	O1IRQ10	O1IRQ9	O1IRQ8	O1IRQ7	O1IRQ6	O1IRQ5	O1IRQ4	O1IRQ3	O1IRQ2	O1IRQ1	O1IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:0	O1IRQ11- O1IRQ0	EP1 OUT Threshold for Interrupt.

EP1ORDT (0003:0AAA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O1RDT15	O1RDT14	O1RDT13	O1RDT12	O1RDT11	O1RDT10	O1RDT9	O1RDT8	O1RDT7	O1RDT6	O1RDT5	O1RDT4	O1RDT3	O1RDT2	O1RDT1	O1RDT0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O1RDT15- O1RDT0	EP1 OUT Read Port.

EP1OREST (0003:0AAC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	O1REST11	O1REST10	O1REST9	O1REST8	O1REST7	O1REST6	O1REST5	O1REST4	O1REST3	O1REST2	O1REST1	O1REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:0	O1REST11- O1REST0	EP1 OUT Available Byte Size for Read.

EP1OST (0003:0AAE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O1FUL	O1EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserved bit.
1	O1FUL	EP1 OUT Full. Same as EP0OST.
0	O1EMP	EP1 OUT Empty. Same as EP0OST.

EP1IAD (0003:0AB0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I1AD10	I1AD9	I1AD8	I1AD7	I1AD6	I1AD5	I1AD4	I1AD3	I1AD2	I1AD1	I1AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	I1AD10- I1AD0	EP1 IN RAM Base Address.

EP1ISZ (0003:0AB2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I1SZ7	I1SZ6	I1SZ5	I1SZ4	I1SZ3	I1SZ2	I1SZ1	I1SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	I1SZ7- I1SZ0	EP1 IN Buffer Size. The maximum FIFO size of EP1 IN is 2048.

EP1ICTL (0003:0AB4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I1EMDE	I1BYTE	I1ENDI	RSV	RSV	RSV	RSV	RSV	I1STL	I1RST	I1ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10	I1EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	I1BYTE	Byte access. When '1', data can be written with one byte.
8	I1ENDI	Endian. '0' if little endian, '1' if big endian when writing EP1IWDT.
7:3	RSV	Reserved bit.
2	I1STL	EP1 OUT STALL. STALL bit for EP1 IN. When set, the function stall condition for EP1 IN can be applied. On receiving an IN token with setting I1STL, the internal state of the EP1 IN changes to HALT and the EP1 IN is stalled. The device returns a STALL handshake. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting I1ENA when ClearFeature (HALT) is received, both bits are automatically cleared at that time. Setting I1STL without setting I1ENA does not clear them automatically.
1	I1RST	EP1 OUT FIFO RESET.
0	I1ENA	EP1 OUT ENABLE. ENABLE bit for EP1 IN. When this bit is set, the device returns a DATA0/DATA1 packet with the data in the FIFO after receiving an IN token to EP1. When this bit is cleared, the device returns a NAK handshake for the IN token. This bit is cleared by hardware when □ the DEVICE receives an ACK handshake from the HOST (I1EMDE = 0) □ the rest of bytes in the FIFO when the DEVICE receives an ACK handshake from the HOST is less than MaxPktSize (I1EMDE = 1) The enable bit control scheme can be selected by setting I1EMDE register (bit10 of EP1ICTL register). If a NAK handshake is received after sending a DATA packet, the data sent to the HOST is recovered, and the FIFO status (EP1IST) and this bit are not changed from the status before the transaction.

EP1IRQSZ (0003:0AB6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I1IRQ11	I1IRQ10	I1IRQ9	I1IRQ8	I1IRQ7	I1IRQ6	I1IRQ5	I1IRQ4	I1IRQ3	I1IRQ2	I1IRQ1	I1IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I1IRQ11-IRQ0	EP1 IN Threshold for Interrupt. Same as EP0IRQSZ except of the bitwidth.

EP1IWDT (0003:0AB8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I1WDT15	I1WDT14	I1WDT13	I1WDT12	I1WDT11	I1WDT10	I1WDT9	I1WDT8	I1WDT7	I1WDT6	I1WDT5	I1WDT4	I1WDT3	I1WDT2	I1WDT1	I1WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I1WDT15- I1WDT0	EP1 IN Write Port. Same as EP0IWDT.

EP1IREST (0003:0ABC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I1REST11	I1REST10	I1REST9	I1REST8	I1REST7	I1REST6	I1REST5	I1REST4	I1REST3	I1REST2	I1REST1	I1REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I1REST8- I1REST0	EP1 IN Available Byte Size for Write. Same as EP0IREST except of the bitwidth.

EP1IST (0003:0ABE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I1FUL	I1EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserved bit
1	I1FUL	EP1 IN Full.
0	I1EMP	EP1 IN Empty.

EP2IAD (0003:0AC0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I2AD10	I2AD9	I2AD8	I2AD7	I2AD6	I2AD5	I2AD4	I2AD3	I2AD2	I2AD1	I2AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	I2AD10- I2AD0	EP2 IN RAM Base Address. Same as EP0IAD.

EP2ISZ (0003:0AC2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2SZ7	I2SZ6	I2SZ5	I2SZ4	I2SZ3	I2SZ2	I2SZ1	I2SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit
7:0	I2SZ7- I2SZ0	EP2 IN Buffer Size. Same as EP0ISZ. The maximum FIFO size of EP2 IN is 256.

EP2ICTL (0003:0AC4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I2EMDE	I2BYTE	I2ENDI	RSV	RSV	RSV	RSV	RSV	I2STL	I2RST	I2ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10	I2EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	I2BYTE	Byte access. When '1', data can be written with one byte.
8	I2ENDI	Endian. '0' if little endian, '1' if big endian when writing EP2IWDT.
7:3	RSV	Reserved bit.
2	I2STL	EP2 IN STALL.
1	I2RST	EP2 IN FIFO RESET.
0	I2ENA	EP2 IN ENABLE.

EP2IIRQSZ (0003:0AC6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2IRQ8	I2IRQ7	I2IRQ6	I2IRQ5	I2IRQ4	I2IRQ3	I2IRQ2	I2IRQ1	I2IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit
8:0	I2IRQ8- I2IRQ0	EP2 IN Threshold for Interrupt.

EP2IWDT (0003:0AC8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I2WDT15	I2WDT14	I2WDT13	I2WDT12	I2WDT11	I2WDT10	I2WDT9	I2WDT8	I2WDT7	I2WDT6	I2WDT5	I2WDT4	I2WDT3	I2WDT2	I2WDT1	I2WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I2WDT15- I2WDT0	EP2 IN Write Port.

EP2IREST (0003:0ACC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2REST8	I2REST7	I2REST6	I2REST5	I2REST4	I2REST3	I2REST2	I2REST1	I2REST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:9	RSV	Reserved bit
8:0	I2REST8-REST0	EP2 IN Available Byte Size for Write.

EP2IST (0003:0ACE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2FUL	I2EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserved bit
1	I2FUL	EP2 IN Full.

Bits	Register Name	Description
0	I2EMP	EP2 IN Empty.

EP3IAD (0003:0AD0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I3AD10	I3AD9	I3AD8	I3AD7	I3AD6	I3AD5	I3AD4	I3AD3	I3AD2	I3AD1	I3AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	I3AD10- I3AD0	EP3 IN RAM Base Address.

EP3ISZ (0003:0AD2) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I3SZ10	I3SZ9	I3SZ8	I3SZ7	I3SZ6	I3SZ5	I3SZ4	I3SZ3	I3SZ2	I3SZ1	I3SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	I3SZ10- I3SZ0	EP3 IN Buffer Size. The maximum FIFO size of EP3 IN is 2048.

EP3ICTL (0003:0AD4) Offset : 0x2A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	I3BYTE	I3ENDI	RSV	RSV	RSV	RSV	RSV	I3STL	I3RST	I3ENA
R/W	-	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	-	0	0	0	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10	RSV	Reserved bit. Writing '1' has no effect.
9	I3BYTE	Byte access. When '1', data can be written with one byte.
8	I3ENDI	Endian. '0' if little endian, '1' if big endian when writing EP3IWDI.
7:3	RSV	Reserved bit
2	I3STL	EP3 IN STALL. STALL bit for EP3 IN. On receiving an IN token with setting I3STL, the internal state of the EP3 IN changes to HALT and the endpoint is stalled. The device does not send any data packets to the HOST for IN tokens. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting I3ENA when ClearFeature(HALT) is received, both bits are automatically cleared at that time. Setting I3STL without setting I3ENA does not clear them automatically.
1	I3RST	EP3 IN FIFO RESET. Same as the bit1 of EP0ICTL.
0	I3ENA	EP3 IN ENABLE. ENABLE bit for EP3 IN. When this bit is set, the device returns a DATA0 packet with the data in the FIFO after receiving an IN token to EP3. When this bit is cleared, the device returns a null DATA0 packet. This bit is not automatically cleared by hardware.

EP3IIRQSZ (0003:0AD6) Offset : 0x2B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I3IRQ11	I3IRQ10	I3IRQ9	I3IRQ8	I3IRQ7	I3IRQ6	I3IRQ5	I3IRQ4	I3IRQ3	I3IRQ2	I3IRQ1	I3IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I3IRQ11- I3IRQ0	EP3 IN Threshold for Interrupt.

EP3IWDT (0003:0AD8) Offset : 0x2C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I3WDT15	I3WDT14	I3WDT13	I3WDT12	I3WDT11	I3WDT10	I3WDT9	I3WDT8	I3WDT7	I3WDT6	I3WDT5	I3WDT4	I3WDT3	I3WDT2	I3WDT1	I3WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I3WDT15- I3WDT0	EP3 IN Write Port.

EP3IREST (0003:0ADC) Offset : 0x2E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I3REST11	I3REST10	I3REST9	I3REST8	I3REST7	I3REST6	I3REST5	I3REST4	I3REST3	I3REST2	I3REST1	I3REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I3REST11- I3REST0	EP3 IN Available Byte Size for Write.

EP3IST (0003:0ADE) Offset : 0x2F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I3FUL	I3EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserved bit
1	I3FUL	EP3 IN Full.
0	I3EMP	EP3 IN Empty.

EP4IAD (0003:0AE0) Offset : 0x30

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I4AD10	I4AD9	I4AD8	I4AD7	I4AD6	I4AD5	I4AD4	I4AD3	I4AD2	I4AD1	I4AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	I4AD10- I4AD0	EP4 IN RAM Base Address.

EP4ISZ (0003:0AE2) Offset : 0x31

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I4SZ10	I4SZ9	I4SZ8	I4SZ7	I4SZ6	I4SZ5	I4SZ4	I4SZ3	I4SZ2	I4SZ1	I4SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	I4SZ10- I4SZ0	EP4 IN Buffer Size. The maximum FIFO size of EP4 IN is 2048.

EP4ICTL (0003:0AE4) Offset : 0x32

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	I4BYTE	I4ENDI	RSV	RSV	RSV	RSV	RSV	I4STL	I4RST	I4ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10	RSV	Reserved bit. Writing '1' has no effect.
9	I4BYTE	Byte access. When '1', data can be written with one byte.
8	I4ENDI	Endian. '0' if little endian, '1' if big endian when writing EP4IWDT.
7:3	RSV	Reserved bit
2	I4STL	EP4 IN STALL. Same as the bit2 of EP3ICTL.
1	I4RST	EP4 IN FIFO RESET. Same as the bit1 of EP0ICTL.
0	I4ENA	EP4 IN ENABLE. Same as the bit0 of EP3ICTL.

EP4IRQSZ (0003:0AE6) Offset : 0x33

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I4IRQ11	I4IRQ10	I4IRQ9	I4IRQ8	I4IRQ7	I4IRQ6	I4IRQ5	I4IRQ4	I4IRQ3	I4IRQ2	I4IRQ1	I4IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I4IRQ11- I4IRQ0	EP4 IN Threshold for Interrupt.

EP4IWDT (0003:0AE8) Offset : 0x34

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I4WDT15	I4WDT14	I4WDT13	I4WDT12	I4WDT11	I4WDT10	I4WDT9	I4WDT8	I4WDT7	I4WDT6	I4WDT5	I4WDT4	I4WDT3	I4WDT2	I4WDT1	I4WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I4WDT15- I4WDT0	EP4 IN Write Port.

EP4IREST (0003:0AEC) Offset : 0x36

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I4REST11	I4REST10	I4REST9	I4REST8	I4REST7	I4REST6	I4REST5	I4REST4	I4REST3	I4REST2	I4REST1	I4REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	I4REST11- I4REST0	EP4 IN Available Byte Size for Write.

EP4IST (0003:0AEE) Offset : 0x37

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I4FUL	I4EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserved bit
1	I4FUL	EP4 IN Full.
0	I4EMP	EP4 IN Empty.

USBST (0003:0B00) Offset : 0x40

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MIRST
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
14:1	RSV	Reserved bit.
0	MIRST	Soft Reset. When set, the protocol engine (UDC) is reset. USBST and USBINTST registers are also reset. This bit is automatically cleared. When the cable is attached, this bit must be set by user.

USBRS (0003:0B02) Offset : 0x41

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MICKOF	MIRSM
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
14:2	RSV	Reserved bit.
1	MICKOF	Clock Off. Internal clock off control. When set to one, the internal clock is disabled. To enable the clock, clear this bit. This can be used for power saving when USB is in the suspend state. Note that when the device is resumed (including the remote wakeup), this bit is automatically cleared, then the clocks are enabled.
0	MIRSM	Resume. Remote wakeup control register. When USB bus is in the suspend condition, the device can resume the bus by setting this register to one. It is not until DEVICE_REMOTE_WAKEUP feature is enabled by SetFeature command from the HOST that remote wakeup function can be available. At power on reset, this function is disabled and setting this bit has no effects. When set this register, do not set the bit1 of USBRS to one.

USBINTMSK (0003:0B04) Offset : 0x42

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MIDMAIE	MICFIE	MINSPIE	MISOFIE	MISIIE	MISCIE	MIBRIE	MISIEIE	MISUSIE	MIEP4IIE	MIEP3IIE	MIEP2IIE	MIEP1IIE	MIEP1OIE	MIEP0IIE	MIEP0OIE
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	MIDMAIE	DMA End Interrupt Enable.
14	MICFIE	ClearFeature Interrupt Enable.
13	MINSPIE	No SOF Packet Interrupt Enable
12	MISOFIE	SOF Interrupt Enable
11	MISIIE	SetInterface Interrupt Enable
10	MISCIE	SetConfiguration Interrupt Enable
9	MIBRIE	USB Bus Reset Interrupt Enable
8	MISIEIE	Setup Interrupt Enable
7	MISUSIE	Suspend Interrupt Enable
6	MIEP4IIE	EP4 IN Interrupt Enable
5	MIEP3IIE	EP3 IN Interrupt Enable
4	MIEP2IIE	EP2 IN Interrupt Enable
3	MIEP1IIE	EP1 IN Interrupt Enable
2	MIEP1OIE	EP1 OUT Interrupt Enable
1	MIEP0IIE	EP0 IN Interrupt Enable
0	MIEP0OIE	EP0 OUT Interrupt Enable

Interrupt enable register. To enable the interrupt, set each bit to one. To disable the interrupt, clear each bit to zero.

USBCFC (0003:0B06) Offset : 0x43

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MICFC15	MICFC14	MICFC13	MICFC12	MICFC11	MICFC10	MICFC9	MICFC8	MICFC7	MICFC6	MICFC5	MICFC4	MICFC3	MICFC2	MICFC1	MICFC0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	MICFC15:0	Device Initialization Data Write Port. The initialization is required to make the USB device enable USB transfer. Writing this register must be done in prior to core reset (writing USBRST register).

USBST (0003:0B08) Offset : 0x44

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MIRMTF	RSV	MISPD	MISSETUP	MIDID
R/W	-	-	-	-	-	-	-	-	-	-	-	R	-	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	0	0	0

Bits	Register Name	Description
15:5	RSV	Reserved bit.
4	MIRMTF	RemoteWakeupFeature Status. Set when the device receives SetFeature(DEVICE_REMOTE_WAKEUP) request. Clear when the device receives ClearFeature(DEVICE_REMOTE_WAKEUP) request.
3	RSV	Reserved bit.
2	MISPD	Suspend. Suspend status of USB bus. Set when the bus is in the suspend status.
1	MISSETUP	Setup. Setup status. Set when the received SETUP transaction requires decoding operation. When the successive IN or OUT token is received, this bit is automatically cleared. Since the device automatically process the standard requests except of GetDescriptor, SetDescriptor and SynchFrame, this bit is not set for these requests. And the data is not stored in EP0 OUT FIFO when the device receives these requests.
0	MIDID	Device Initialize Done. The flag for device initialization done. Set when 20 bytes of the initialization data are written to USBCFG register. Clear at soft reset by USBRST.

USBINTST (0003:0B0A) Offset : 0x45

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MIDMA	MICFI	MINSPI	MISOFI	MISII	MISCI	MIBRI	MISSETI	MISUSI	MIEP4II	MIEP3II	MIEP2II	MIEP1II	MIEP1OI	MIEP0II	MIEP0OI
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	MIDMA	DMA End Interrupt. Interrupt asserted when DMA transmission is completed.
14	MICFI	ClearFeature Interrupt. Interrupt asserted when ClearFeature(ENDPOINT_HALT) is received.
13	MINSPI	No SOF Packet Interrupt. Interrupt asserted when the device does not receive the next SOF packet within 1ms after a SOF packet is received.
12	MISOFI	SOF Interrupt. Interrupt asserted when a SOF packet is received.
11	MISII	SetInterface Interrupt. Interrupt asserted when SetInterface command is received. AlternateSetting set by HOST can be read from the USBALT register.

Bits	Register Name	Description
10	MISCI	SetConfiguration Interrupt. Interrupt asserted when SetConfiguration command is received. Configuration set by HOST can be read from bit1-0 of USBALT register.
9	MIBRI	USB Bus Reset Interrupt. Interrupt asserted when a USB Bus Reset is received.
8	MISSETI	Setup Interrupt. Interrupt asserted when a SETUP transaction whose request is one of GetDescriptor, SetDescriptor and SynchFrame and the class/vendor requests is received.
7	MISUSI	Suspend Change Interrupt. Interrupt asserted when the USB bus is turned into the suspend state or is resumed (including the device remote wakeup). The actual bus status can be known by reading bit2 of USBST register.
6	MIEP4II	EP4I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP4IIRQSZ at the completion of sending a DATA0 packet.
5	MIEP3II	EP3I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP3IIRQSZ at the completion of sending a DATA0 packet.
4	MIEP2II	EP2I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP2IIRQSZ at the completion of receiving of an ACK handshake.
3	MIEP1II	EP1I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP1IIRQSZ at the completion of receiving of an ACK handshake.
2	MIEP1OI	EP1O Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is greater than or equals to EP1OIRQSZ or the received DATA packet is short or null at the completion of sending of an ACK handshake.
1	MIEP0II	EP0I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP0IIRQSZ at the completion of receiving of an ACK handshake.
0	MIEP0OI	EP0O Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is greater than or equals to EP0OIRQSZ or the received DATA packet is short or null at the completion of sending of an ACK handshake.

Interrupt status registers. The corresponding bit is set when the interrupts occurred. Cleared when this register is read.

USBFRM (0003:0B0C) Offset : 0x46

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	MIFRN10	MIFRN9	MIFRN8	MIFRN7	MIFRN6	MIFRN5	MIFRN4	MIFRN3	MIFRN2	MIFRN1	MIFRN0
R/W	-	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	MIFRN10:0	USB Frame number. The latest frame number received from HOST. Reset at USB Bus Reset, a soft reset by USBRST or when the cable is attached or detached.

USBALT (0003:0B0E) Offset : 0x47

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	MIASI22	MIASI21	MIASI20	RSV	MIASI12	MIASI11	MIASI10	RSV	MIASI02	MIASI01	MIASI00	RSV	RSV	MICONF1	MICONF0
R/W	-	R	R	R	-	R	R	R	-	R	R	R	-	-	R	R
Default	-	0	0	0	-	0	0	0	-	0	0	0	-	-	0	0

Bits	Register Name	Description
15	RSV	Reserved bit

Bits	Register Name	Description
14:12	MIASI22-MIASI20	AlternateSetting of Interface2. AlternateSetting value set to Interface2 by SetInterface command.
11	RSV	Reserved bit
10:8	MIASI12-MIASI10	AlternateSetting of Interface1. AlternateSetting value set to Interface1 by SetInterface command.
7	RSV	Reserved bit
6:4	MIASI02-MIASI00	AlternateSetting of Interface0. AlternateSetting value set to Interface0 by SetInterface command.
3:2	RSV	Reserved bit
1:0	MICONF1- MICONF0	Configuration. Current Configuration number selected by HOST.

This register indicates the current configuration and alternate setting value of each interface assigned by the HOST. Reset at USB Bus Reset, a soft reset by USBRST. And the alternate setting values (bit14-12, bit10-8, bit6-4) are reset at receiving a SetConfiguration request.

USBCTL (0003:0B10) Offset : 0x48

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RFB
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit
0	RFB	Rate Feedback Mode. When set to 1, the data toggle bits of EP2 are changed after each data packet without regard to the presence or the type of handshake packet. When 0, EP2 uses the normal data toggle sequence.

DMAAD0 (0003:0B12) Offset : 0x49

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DAD15	DAD14	DAD13	DAD12	DAD11	DAD10	DAD9	DAD8	DAD7	DAD6	DAD5	DAD4	DAD3	DAD2	DAD1	DAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DAD15-DAD0	DMA start address [15-0]. Specified as offset from start of SDRAM must be 4B aligned when endian conversion is used (ENDI=1). When ENDI=0, the address need not be 4B aligned.

DMAAD1 (0003:0B14) Offset : 0x4A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DAD26	DAD25	DAD24	DAD23	DAD22	DAD21	DAD20	DAD19	DAD18	DAD17	DAD16
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	DAD26-DAD16	DMA start address [26-16]. Specified as offset from start of SDRAM must be 4B aligned when endian conversion is used (ENDI=1). When ENDI=0, the address need not be 4B aligned.

DMASZ (0003:0B16) Offset : 0x4B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DSZ15	DSZ14	DSZ13	DSZ12	DSZ11	DSZ10	DSZ9	DSZ8	DSZ7	DSZ6	DSZ5	DSZ4	DSZ3	DSZ2	DSZ1	DSZ0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DSZ15-DSZ0	DMA transfer packet length. Affected only in DMA multi-packet mode (DMDE=0). When DMDE=1, the transmission length is fixed to MaxPktSize and the value of this register is ignored.

DMACTL (0003:0B18) Offset : 0x4C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	DEP2	DEP1	DEP0	RSV	RSV	DENDI1	DENDI0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DMDE
R/W	-	R/W	R/W	R/W	-	-	R/W	R/W	-	-	-	-	-	-	-	R/W
Default	-	0	0	0	-	-	0	0	-	-	-	-	-	-	-	0

Bits	Register Name	Description																								
15	RSV	Reserved bit																								
14:12	DEP2-DEP0	DMA Endpoint Select. <table><tr><td>DEP[2:0]</td><td>DMA source</td><td>DMA destination</td></tr><tr><td>000</td><td>EP0 OUT</td><td>SDRAM</td></tr><tr><td>001</td><td>SDRAM</td><td>EP0 IN</td></tr><tr><td>010</td><td>EP1 OUT</td><td>SDRAM</td></tr><tr><td>011</td><td>SDRAM</td><td>EP1 IN</td></tr><tr><td>100</td><td>SDRAM</td><td>EP2 IN</td></tr><tr><td>101</td><td>SDRAM</td><td>EP3 IN</td></tr><tr><td>110</td><td>SDRAM</td><td>EP4 IN</td></tr></table>	DEP[2:0]	DMA source	DMA destination	000	EP0 OUT	SDRAM	001	SDRAM	EP0 IN	010	EP1 OUT	SDRAM	011	SDRAM	EP1 IN	100	SDRAM	EP2 IN	101	SDRAM	EP3 IN	110	SDRAM	EP4 IN
DEP[2:0]	DMA source	DMA destination																								
000	EP0 OUT	SDRAM																								
001	SDRAM	EP0 IN																								
010	EP1 OUT	SDRAM																								
011	SDRAM	EP1 IN																								
100	SDRAM	EP2 IN																								
101	SDRAM	EP3 IN																								
110	SDRAM	EP4 IN																								
11:10	RSV	Reserved bit																								
9:8	DENDI1-DENDI0	DMA Endian. <table><tr><td>DENDI[1:0]</td><td></td></tr><tr><td>00</td><td>D3 D2 D1 D0 -> D3 D2 D1 D0</td></tr><tr><td>01</td><td>D3 D2 D1 D0 -> D2 D3 D0 D1</td></tr><tr><td>10</td><td>D3 D2 D1 D0 -> D0 D1 D2 D3</td></tr><tr><td>11</td><td>D3 D2 D1 D0 -> D1 D0 D3 D2</td></tr></table>	DENDI[1:0]		00	D3 D2 D1 D0 -> D3 D2 D1 D0	01	D3 D2 D1 D0 -> D2 D3 D0 D1	10	D3 D2 D1 D0 -> D0 D1 D2 D3	11	D3 D2 D1 D0 -> D1 D0 D3 D2														
DENDI[1:0]																										
00	D3 D2 D1 D0 -> D3 D2 D1 D0																									
01	D3 D2 D1 D0 -> D2 D3 D0 D1																									
10	D3 D2 D1 D0 -> D0 D1 D2 D3																									
11	D3 D2 D1 D0 -> D1 D0 D3 D2																									
7:1	RSV	Reserved bit																								
0	DMDE	DMA transmission mode. <table><tr><td>0</td><td>Automatic multi-packet transmission mode. The transactions with the packet length specified by DSZ register can be achieved automatically. DMA will be terminated by any of the following conditions; - DMAREST becomes 0 - A short packet is received on an OUT endpoint - Clearing DON bit. </td></tr><tr><td>1</td><td>Single packet transmission mode. DMA will be terminated when a DMA transmission with the length of MaxPacketSize is completed regardless of DSZ register value. It is not allowed to read/write data less than MaxPktSize.</td></tr></table>	0	Automatic multi-packet transmission mode. The transactions with the packet length specified by DSZ register can be achieved automatically. DMA will be terminated by any of the following conditions; - DMAREST becomes 0 - A short packet is received on an OUT endpoint - Clearing DON bit.	1	Single packet transmission mode. DMA will be terminated when a DMA transmission with the length of MaxPacketSize is completed regardless of DSZ register value. It is not allowed to read/write data less than MaxPktSize.																				
0	Automatic multi-packet transmission mode. The transactions with the packet length specified by DSZ register can be achieved automatically. DMA will be terminated by any of the following conditions; - DMAREST becomes 0 - A short packet is received on an OUT endpoint - Clearing DON bit.																									
1	Single packet transmission mode. DMA will be terminated when a DMA transmission with the length of MaxPacketSize is completed regardless of DSZ register value. It is not allowed to read/write data less than MaxPktSize.																									

DMATRG (0003:0B1A) Offset : 0x4D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DON
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit

Bits	Register Name	Description
0	DON	DMA start. Writing '1' to this bit starts the DMA transfer. Automatically cleared when DMA completes. Clearing this bit terminates the DMA transfer. At the DMA completion, an interrupt is generated. During a DMA transfer, ARM register access to the same endpoint as the DMA target is not allowed.

DMAREST (0003:0B1C) Offset : 0x4E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DREST15	DREST14	DREST13	DREST12	DREST11	DREST10	DREST9	DREST8	DREST7	DREST6	DREST5	DREST4	DREST3	DREST2	DREST1	DREST0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DREST15-DREST0	DMA transfer packet rest number. Indicates the rest packet number in the current DMA transmission.

DMAST (0003:0B1E) Offset : 0x4F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DSPKT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit
0	DSPKT	Short packet received status. When the DMA is completed with the condition that an OUT endpoint receives a short/null packet, this bit is set to 1. Cleared when the next DMA is invoked.

18.3A Function

Up to 36 windows can be set for AF Engine, and up to 96 windows can be set for AE/AWB Engine. The windows are the same size and organized as an m-by-n matrix. The organization of the matrix, the size and the position of the windows are programmable. The window configuration of the AF Engine is based on a frame formed by the extracted green pixels, while the window configuration of the AE/AWB Engine is based on an original frame. There are some restrictions on the window setting.

AF Engine

- PAXSH must be equal to or greater than (IIRSH + 1).
- Paxels cannot overlap the last pixel in a line. PAXSH, PAXH and PAXHC must be set accordingly.
- Paxels can overlap either the first or the last line in a frame, but not both of them. PAXSV, PAXV and PAXVC must be set accordingly.

AE/AWB Engine

- The number of 2x2 blocks in a Window cannot exceed 256. WINH, WINV, AEWINCH and AEWINCV must be set accordingly.
- The number of sub-sampled pixels in a line cannot exceed 2816. WINH, WINHC and AEWINCH must be set accordingly.
- Windows can overlap either the first or the last line in a frame, but not both of them. WINSV, WINV and WINVC must be set accordingly.

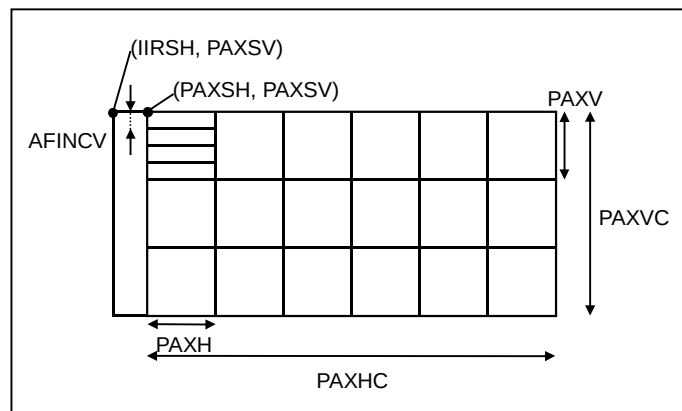


Figure 1 : AF Engine Poxel Configuration (green pixels only)

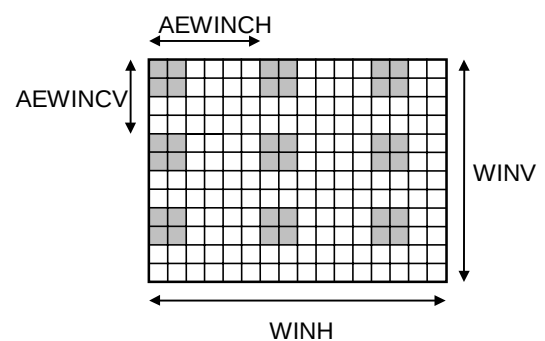
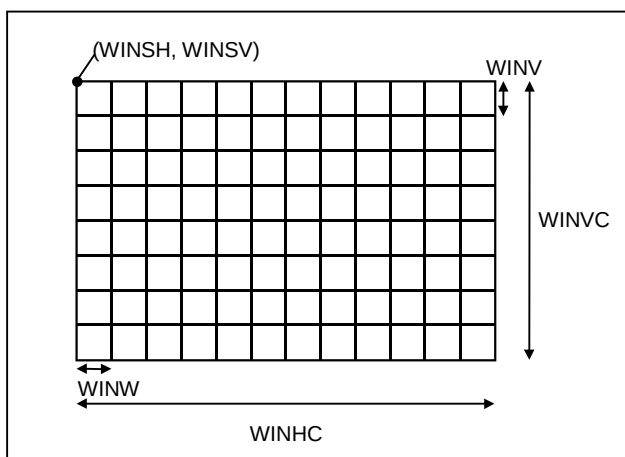


Figure 2 : AE/AWB Engine Window configuration

H3ACTRL (0003:0B80) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AEWEN	RSV	RSV	RSV	AFEN
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:5	RSV	Reserved bit
4	AEWEN	AE/AWB Engine enable. 0: Disable 1: Enable
3:1	RSV	Reserved bit
0	AFEN	AF Engine enable. 0: Disable 1: Enable

AFCTRL (0003:0B82) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	FVMODE	GPOS3	GPOS2	GPOS1	GPOS0
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0

Bits	Register Name	Description
15:5	RSV	Reserved bit
4	FVMODE	[AF Engine] Focus Value(FV) accumulation mode. 0: Sum mode 1: Peak mode
3:0	GPOS(3:0)	[AF Engine] Green pixel extraction. GPOS(0) : an upper left pixel in 2x2 block (even pixel, even line). GPOS(1) : an upper right pixel in 2x2 block (odd pixel, even line). GPOS(2) : a lower left pixel in 2x2 block (even pixel, odd line). GPOS(3) : a lower right pixel in 2x2 block (odd pixel, odd line). 0: other than Green 1: Green

AFPAX1 (0003:0B84) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	PAXV5	PAXV4	PAXV3	PAXV2	PAXV1	PAXV0	PAXH7	PAXH6	PAXH5	PAXH4	PAXH3	PAXH2	PAXH1	PAXH0
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:14	RSV	Reserved bit
13:8	PAXV(5:0)	[AF Engine] Paxel Height (number of green lines). Paxel Height is (PAXV +1) and ranges: 2-64. A value greater than 0 must be set to PAXV.
7:0	PAXH(7:0)	[AF Engine] Paxel Width (number of green pixels). Paxel Width is (PAXH +1) and ranges: 1-256. Setting a small value to PAXH might collapse the data transfer to SDRAM. SDRAM access time can not be longer than the time it takes for the Paxel Width (PAXH+1). SDRAM access time varies depending on CCD and SDRAM clock frequency settings, and also the status of the other peripherals' access to SDRAM.

AFPAX2 (0003:0B86) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	PAXVC1	PAXVC0	RSV	RSV	RSV	RSV	PAXHC3	PAXHC2	PAXHC1	PAXHC0
R/W	-	-	-	-	-	-	R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved bit
9:8	PAXVC(1:0)	[AF Engine] Paxel Count for V direction. Paxel Count is (PAXVC +1) and ranges: 1-4.
7:4	RSV	Reserved bit
3:0	PAXHC(3:0)	[AF Engine] Paxel Count for H direction. Paxel Count is (PAXHC +1) and ranges: 1-9. Setting a value greater than 8 to PAXHC is not allowed.

AFPAX3 (0003:0B88) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	PAXSH11	PAXSH10	PAXSH9	PAXSH8	PAXSH7	PAXSH6	PAXSH5	PAXSH4	PAXSH3	PAXSH2	PAXSH1	PAXSH0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit
11:0	PAXSH(11:0)	[AF Engine] Paxel start position (H direction, green pixels). Range: 1-4,095. PAXSH must be equal to or greater than (IIRSH + 1).

AFPAX4 (0003:0B8A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	PAXSV11	PAXSV10	PAXSV9	PAXSV8	PAXSV7	PAXSV6	PAXSV5	PAXSV4	PAXSV3	PAXSV2	PAXSV1	PAXSV0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:0	PAXSV(11:0)	[AF Engine] Paxel start position (V direction, green lines). Range: 0-4,095.

AFIIRSH(0003:0B8C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	IIRSH11	IIRSH10	IIRSH9	IIRSH8	IIRSH7	IIRSH6	IIRSH5	IIRSH4	IIRSH3	IIRSH2	IIRSH1	IIRSH0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:0	IIRSH(11:0)	[AF Engine] IIR filter start position (H direction, green pixels). Range: 0-4,095. Shift registers of the IIR filter are cleared to zero at the next position of IIRSH.

AFPAX5 (0003:0B8E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	AFINCV3	AFINCV2	AFINCV1	AFINCV0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-	-	-	-	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	-	-	-	-	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:8	AFINCV(3:0)	[AF Engine] Line increments in a Poxel (number of green lines). Increments are (AFINCV +1) and ranges: 1-16.
7:0	RSV	Reserved bit.

AFSDRA1 (0003:0B90) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AFSDRA 21	AFSDRA 20	AFSDRA 19	AFSDRA 18	AFSDRA 17	AFSDRA 16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved bit.
5:0	AFSDRA(21:16)	[AF Engine] Upper 6 bits of SDRAM destination address to store the results of AF engine. There are two continuous buffers, and the results of the two successive frames are stored to the different buffers. Buffer size depends on PAXHC and PAXVC. AFSDRA is the start address of the first buffer, and the address is automatically incremented until data transfer for the two frame ends (address is 32 Byte unit) specified as offset from start of SDRAM.

AFSDRA2 (0003:0B92) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	AFSDRA 15	AFSDRA 14	AFSDRA 13	AFSDRA 12	AFSDRA 11	AFSDRA 10	AFSDRA 9	AFSDRA 8	AFSDRA 7	AFSDRA 6	AFSDRA 5	AFSDRA 4	AFSDRA 3	AFSDRA 2	AFSDRA 1	AFSDRA 0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	AFSDRA(15:0)	[AF Engine] Lower 16 bits of SDRAM destination address to store the results of AF engine. There are two continuous buffers, and the results of the two successive frames are stored to the different buffers. Buffer size depends on PAXHC and PAXVC. AFSDRA is the start address of the first buffer, and the address is automatically incremented until data transfer for the two frame ends (address is 32 Byte unit) specified as offset from start of SDRAM.

AFSDRFLG (0003:0B94) Offset : 0x0A

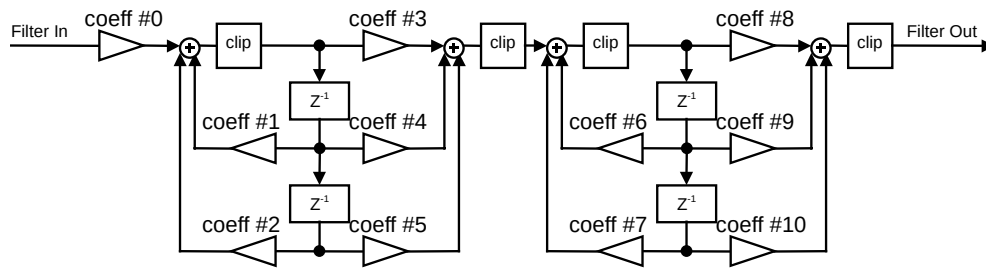
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AFSDR FLG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit.
0	AFSDRFLG	[AF Engine] SDRAM buffer flag to indicate the buffer at which the results of the last frame are stored. 0: Results are stored at the first buffer starting from AFSDRA. 1: Results are stored at the second buffer.

AFCOEFF10 (0003:0B96) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF10 ₁₀	COEFF10 ₉	COEFF10 ₈	COEFF10 ₇	COEFF10 ₆	COEFF10 ₅	COEFF10 ₄	COEFF10 ₃	COEFF10 ₂	COEFF10 ₁	COEFF10 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF10(10:0)	[AF Engine] IIR filter Coefficient #0 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

**Figure 3 : IIR Filter coefficients****AFCOEFF11 (0003:0B98) Offset : 0x0C**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF11 ₁₀	COEFF11 ₉	COEFF11 ₈	COEFF11 ₇	COEFF11 ₆	COEFF11 ₅	COEFF11 ₄	COEFF11 ₃	COEFF11 ₂	COEFF11 ₁	COEFF11 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF11(10:0)	[AF Engine] IIR filter Coefficient #1 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF12 (0003:0B9A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF12 ₁₀	COEFF12 ₉	COEFF12 ₈	COEFF12 ₇	COEFF12 ₆	COEFF12 ₅	COEFF12 ₄	COEFF12 ₃	COEFF12 ₂	COEFF12 ₁	COEFF12 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF12(10:0)	[AF Engine] IIR filter Coefficient #2 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF13 (0003:0B9C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF13 10	COEFF13 9	COEFF13 8	COEFF13 7	COEFF13 6	COEFF13 5	COEFF13 4	COEFF13 3	COEFF13 2	COEFF13 1	COEFF13 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF13(10:0)	[AF Engine] IIR filter Coefficient #3 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF14 (0003:0B9E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF14 10	COEFF14 9	COEFF14 8	COEFF14 7	COEFF14 6	COEFF14 5	COEFF14 4	COEFF14 3	COEFF14 2	COEFF14 1	COEFF14 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF14(10:0)	[AF Engine] IIR filter Coefficient #4 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF15 (0003:0BA0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF15 10	COEFF15 9	COEFF15 8	COEFF15 7	COEFF15 6	COEFF15 5	COEFF15 4	COEFF15 3	COEFF15 2	COEFF15 1	COEFF15 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF15(10:0)	[AF Engine] IIR filter Coefficient #5 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF16 (0003:0BA2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF16 10	COEFF16 9	COEFF16 8	COEFF16 7	COEFF16 6	COEFF16 5	COEFF16 4	COEFF16 3	COEFF16 2	COEFF16 1	COEFF16 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF16(10:0)	[AF Engine] IIR filter Coefficient #6 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF17 (0003:0BA4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF17 10	COEFF17 9	COEFF17 8	COEFF17 7	COEFF17 6	COEFF17 5	COEFF17 4	COEFF17 3	COEFF17 2	COEFF17 1	COEFF17 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.

Bits	Register Name	Description
10:0	COEFF17(10:0)	[AF Engine] IIR filter Coefficient #7 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF18 (0003:0BA6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF18 10	COEFF18 9	COEFF18 8	COEFF18 7	COEFF18 6	COEFF18 5	COEFF18 4	COEFF18 3	COEFF18 2	COEFF18 1	COEFF18 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF18(10:0)	[AF Engine] IIR filter Coefficient #8 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF19 (0003:0BA8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF19 10	COEFF19 9	COEFF19 8	COEFF19 7	COEFF19 6	COEFF19 5	COEFF19 4	COEFF19 3	COEFF19 2	COEFF19 1	COEFF19 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF19(10:0)	[AF Engine] IIR filter Coefficient #9 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF110 (0003:0BAA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF110 10	COEFF110 9	COEFF110 8	COEFF110 7	COEFF110 6	COEFF110 5	COEFF110 4	COEFF110 3	COEFF110 2	COEFF110 1	COEFF110 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF110(10:0)	[AF Engine] IIR filter Coefficient #10 <Table1>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF20 (0003:0BAC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF20 10	COEFF20 9	COEFF20 8	COEFF20 7	COEFF20 6	COEFF20 5	COEFF20 4	COEFF20 3	COEFF20 2	COEFF20 1	COEFF20 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF20(10:0)	[AF Engine] IIR filter Coefficient #0 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF21 (0003:0BAE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF21 ₁₀	COEFF21 ₉	COEFF21 ₈	COEFF21 ₇	COEFF21 ₆	COEFF21 ₅	COEFF21 ₄	COEFF21 ₃	COEFF21 ₂	COEFF21 ₁	COEFF21 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit
10:0	COEFF21(10:0)	[AF Engine] IIR filter Coefficient #1 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF22 (0003:0BB0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF22 ₁₀	COEFF22 ₉	COEFF22 ₈	COEFF22 ₇	COEFF22 ₆	COEFF22 ₅	COEFF22 ₄	COEFF22 ₃	COEFF22 ₂	COEFF22 ₁	COEFF22 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF22(10:0)	[AF Engine] IIR filter Coefficient #2 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF23 (0003:0BB2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF23 ₁₀	COEFF23 ₉	COEFF23 ₈	COEFF23 ₇	COEFF23 ₆	COEFF23 ₅	COEFF23 ₄	COEFF23 ₃	COEFF23 ₂	COEFF23 ₁	COEFF23 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF23(10:0)	[AF Engine] IIR filter Coefficient #3 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF24 (0003:0BB4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF24 ₁₀	COEFF24 ₉	COEFF24 ₈	COEFF24 ₇	COEFF24 ₆	COEFF24 ₅	COEFF24 ₄	COEFF24 ₃	COEFF24 ₂	COEFF24 ₁	COEFF24 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF24(10:0)	[AF Engine] IIR filter Coefficient #4 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF25 (0003:0BB6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF25 ₁₀	COEFF25 ₉	COEFF25 ₈	COEFF25 ₇	COEFF25 ₆	COEFF25 ₅	COEFF25 ₄	COEFF25 ₃	COEFF25 ₂	COEFF25 ₁	COEFF25 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.

Bits	Register Name	Description
10:0	COEFF25(10:0)	[AF Engine] IIR filter Coefficient #5 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF26 (0003:0BB8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF26 10	COEFF26 9	COEFF26 8	COEFF26 7	COEFF26 6	COEFF26 5	COEFF26 4	COEFF26 3	COEFF26 2	COEFF26 1	COEFF26 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF26(10:0)	[AF Engine] IIR filter Coefficient #6 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF27 (0003:0BBA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF27 10	COEFF27 9	COEFF27 8	COEFF27 7	COEFF27 6	COEFF27 5	COEFF27 4	COEFF27 3	COEFF27 2	COEFF27 1	COEFF27 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF27(10:0)	[AF Engine] IIR filter Coefficient #7 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF28 (0003:0BBC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF28 10	COEFF28 9	COEFF28 8	COEFF28 7	COEFF28 6	COEFF28 5	COEFF28 4	COEFF28 3	COEFF28 2	COEFF28 1	COEFF28 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF28(10:0)	[AF Engine] IIR filter Coefficient #8 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF29 (0003:0BBE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF29 10	COEFF29 9	COEFF29 8	COEFF29 7	COEFF29 6	COEFF29 5	COEFF29 4	COEFF29 3	COEFF29 2	COEFF29 1	COEFF29 0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF29(10:0)	[AF Engine] IIR filter Coefficient #9 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AFCOEFF210 (0003:0BC0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	COEFF210 ₁₀	COEFF210 ₉	COEFF210 ₈	COEFF210 ₇	COEFF210 ₆	COEFF210 ₅	COEFF210 ₄	COEFF210 ₃	COEFF210 ₂	COEFF210 ₁	COEFF210 ₀
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	COEFF210(10:0)	[AF Engine] IIR filter Coefficient #10 <Table2>. Decimal 6 bit: $-16 \leq \text{coeff} \leq 15 + 63/64$

AEWCTRL (0003:0BC2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AVE2LMT ₇	AVE2LMT ₆	AVE2LMT ₅	AVE2LMT ₄	AVE2LMT ₃	AVE2LMT ₂	AVE2LMT ₁	AVE2LMT ₀
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	AVE2LMT(7:0)	[AE/AWB Engine] Saturation limit. If any of the pixels in a 2x2 block are equal to or greater than AVE2LMT, the block is treated as a saturated data, and not counted into CNT2. If a sub-sampled 2x2 block is saturating, the pixels equal to or greater than AVE2LMT are clipped to AVE2LMT, and accumulated to AVE2.

AEWWIN1 (0003:0BC4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	WINV4	WINV3	WINV2	WINV1	WINV0	RSV	WINH6	WINH5	WINH4	WINH3	WINH2	WINH1	WINH0
R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	0	0	0	0	-	0	0	0	0	0	0	0

Bits	Register Name	Description
15:13	RSV	Reserved bit.
12:8	WINV(4:0)	[AE/AWB Engine] Window Height (even number). Window Height is ((WINV + 1) x 2) and ranges: 4-64. A value greater than 0 must be set to WINV.
7	RSV	Reserved bit.
6:0	WINH(6:0)	[AE/AWB Engine] Window Width (even number). Window Width is ((WINH + 1) x 2) and ranges: 2-256. Setting a small value to WINH might collapse the data transfer to SDRAM. SDRAM access time can not be longer than the time it takes for the Window Width ((WINH+1)x2). SDRAM access time varies depending on CCD and SDRAM clock frequency settings, and also the status of the other peripherals' access to SDRAM.

AEWWIN2 (0003:0BC6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	WINVC2	WINVC1	WINVC0	RSV	RSV	RSV	RSV	WINHC3	WINHC2	WINHC1	WINHC0
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:8	WINVC(2:0)	[AE/AWB Engine] Window Count for V direction. Window Count is (WINVC + 1) and ranges: 1-8.

Bits	Register Name	Description
7:4	RSV	Reserved bit
3:0	WINHC(3:0)	[AE/AWB Engine] Window Count for H direction. Window Count is (WINHC +1) and ranges: 1-12. Setting a value greater than 11 to WINHC is not allowed.

AEWWIN3 (0003:0BC8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	WINSH10	WINSH9	WINSH8	WINSH7	WINSH6	WINSH5	WINSH4	WINSH3	WINSH2	WINSH1	WINSH0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	WINSH(10:0)	[AE/AWB Engine] Window start position (H direction, even number). Window start position is (WINSH x2) and ranges: 0-4,094.

AEWWIN4 (0003:0BCA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	WINSV10	WINSV9	WINSV8	WINSV7	WINSV6	WINSV5	WINSV4	WINSV3	WINSV2	WINSV1	WINSV0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved bit.
10:0	WINSV(10:0)	[AE/AWB Engine] Window start position (V direction, even number). Window start position is (WINSV x2) and ranges: 0-4,094.

AEWWIN5 (0003:0BCC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	AEWINCV3	AEWINCV2	AEWINCV1	AEWINCV0	RSV	RSV	RSV	RSV	AEWINCH3	AEWINCH2	AEWINCH1	AEWINCH0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:8	AEWINCV(3:0)	[AE/AWB Engine] Sampling point increments in a Window (V direction, even number). Increments are ((AEWINCV +1) x2) and ranges: 2-32.
7:4	RSV	Reserved bit.
3:0	AEWINCH(3:0)	[AE/AWB Engine] Sampling point increments in a Window (H direction, even number). Increments are ((AEWINCH +1) x2) and ranges: 2-32.

AEWSDRA1 (0003:0BCE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AEWSDRA21	AEWSDRA20	AEWSDRA19	AEWSDRA18	AEWSDRA17	AEWSDRA16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved bit

Bits	Register Name	Description
5:0	AEWSDRA(21:16)	[AE/AWB Engine] Upper 6 bits of SDRAM destination address to store the results of AE/AWB Engine. There are two continuous buffers, and the results of the two successive frames are stored to the different buffers. Buffer size depends on WINHC and WINVC. AEWSDRA is the start address of the first buffer, and the address is automatically incremented until data transfer for the two frame ends (address is 32 Byte unit) specified as offset from start of SDRAM.

AEWSDRA2 (0003:0BD0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	AEWSDRA15	AEWSDRA14	AEWSDRA13	AEWSDRA12	AEWSDRA11	AEWSDRA10	AEWSDRA9	AEWSDRA8	AEWSDRA7	AEWSDRA6	AEWSDRA5	AEWSDRA4	AEWSDRA3	AEWSDRA2	AEWSDRA1	AEWSDRA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	AEWSDRA(15:0)	[AE/AWB Engine] Lower 16 bits of SDRAM destination address to store the results of AE/AWB Engine. There are two continuous buffers, and the results of the two successive frames are stored to the different buffers. Buffer size depends on WINHC and WINVC. AEWSDRA is the start address of the first buffer, and the address is automatically incremented until data transfer for the two frame ends (address is 32 Byte unit) specified as offset from start of SDRAM.

AEWSDRFLG (0003:0BD2) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AEWSDRFLG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit.
0	AEWSDRFLG	[AE/AWB Engine] SDRAM buffer flag to indicate the buffer at which the results of the last frame are stored. 0: Results are stored at the first buffer starting from AEWSDRA. 1: Results are stored at the second buffer.

3A_TEST (0003:0BFE) Offset : 0x3f

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	3ATST1	RSV	RSV	RSV	3ATST0
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:5	RSV	Reserved bit.
4	3ATST1	Test bit. Fixed "0". Do not use.
3:1	RSV	Reserved bit
0	3ATST0	Test bit. Fixed "0". Do not use.

19.Address Trace Module

ATMCTRL (0003:0D00) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	Low16	RSV	RSV	RSV	RSV	RSV	RSV	RSV	a_lock
R/W	-	-	-	-	-	-	-	R/W	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	0

Bits	Register Name	Description
8	low16	0 : default / 32bit ARM address 1 : only lower 16bit of the stored ARM address is displayed = example = The last ARM address before setting a_lock : 0x100040 The 2 nd last ARM address before setting a_lock : 0x10003C [low16=0] ... a pair of ATMxx registers shows 32bit width ARM address. ATM00 : 0x0010 ATM01 : 0x0040 [low16=1] ... ATMxx registers shows lower 16bit of ARM address. ATM00 : 0x0040 (upper 16bit is omitted.) ATM01 : 0x003C (upper 16bit is omitted.)
0	a_lock	0 : stores current ARM address into register 1 : stop storing current ARM address = remark = Debug program must set a_lock bit at the check point. When using debugger tool, a_lock must set without opening any memory window. Because memory window overwrites the data in ATM00-61 with it's unique command code.

ATM00-ATM61 (0003:0D04-0003:0D7E) Offset : 0x02-0x3F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ATMn[15:0] (n=00,01,...61)															
R/W	R															
Default	0x0000															

Bits	Register Name	Description
15:0	ATMn (n=00,01,..61)	ARM address history (1) Order is below [low16=0] - The last ARM address before setting a_lock is stored in ATM00 (upper 16bit) and ATM01(lower 16bit). - The 2nd last ARM address before setting a_lock is stored in ATM02 (upper 16bit) and ATM03(lower 16bit). - The 31st last ARM address before setting a_lock is stored in ATM60 (upper 16bit) and ATM61(lower 16bit). [low16=1] - The last ARM address before setting a_lock is stored in ATM00 (lower 16bit). - The 2nd last ARM address before setting a_lock is stored in ATM01 (lower 16bit). - The 62nd last ARM address before setting a_lock is stored in ATM61 (lower 16bit). (2) Only Instruction address is stored in this register. No data or code is stored. (3) On/Off of I-Cache is NOT affect the stored ARM address.

20. I²C Bus Interface

TXDATA (0003:0D80) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ACK	TXD7	TXD6	TXD5	TXD4	TXD3	TXD2	TXD1	TXD0
R/W	-	-	-	-	-	-	-	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit. Not in use.
8	ACK	Acknowledge control (Setting of acknowledge level) "0" is written at the time of outputting Acknowledge. "1" is written at the time of sending data from DM270 when Not Acknowledge is output.
7:0	TXD7-TXD0	Transmission address/data. "0xFF" is written at the time of receiving from slave.

RXDATA (0003:0D82) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ACKS	RXD7	RXD6	RXD5	RXD4	RXD3	RXD2	RXD1	RXD0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit. Not in use.
8	ACKS	Acknowledge status register After transfer completed (occurrence of interrupt), acknowledge value can be read.
7:0	RXD7-RXD0	Receiving data register After transfer completed (occurrence of interrupt), received data can be read. At the time of transmitting, after transfer completed, values written in TXDATA (0003:0D80) can be read.

SCS (0003:0D84) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CLKSEL	XMIT	COND1	COND0
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:4	RSV	Reserved bit. Not in use.
3	CLKSEL	Selection of transfer clock 0 : 100KHz 1 : 400KHz
2	XMIT	On reading. Status on transfer can be read. 0 : Transfer on standby 1 : On transmitting On writing. Trigger for transfer 0 : No transfer 1 : Transfer starts.
1:0	COND1-COND0	Selection of transfer condition 00 : Normal transfer 01 : Transfer with start condition 10 : Transfer with stop condition 11 : Reserved

21. Instruction Cache

ICAMODE (0003:0E00) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	REQ4W	RSV	RSV	RSV	RSV	RSV	RSV	ICACLR	ICAENB
R/W	-	-	-	-	-	-	-	R/W	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	0	-	-	-	-	-	-	0	0

Bits	Register Name	Description
8	REQ4W	Enable 4 burst Instruction read from SDRAM. 1: 4 word burst access request 0: Single word access request
1	ICACLR	Cache Clear bit. When write '1' to this bit, ICAENB MUST be '0'. 1: ICACHE Clear start 0: ICACHE Clear done
0	ICAENB	Cache enable bit. 1: ICACHE is enable 0: ICACHE is disable (bypass)

When CACLR bit is set to 1, clearing of cache starts. When this bit is read during clearing operation, 1 is returned, and back to 0 after clearing completed. ICAENB must be 0 when writing 0 to ICACLR.

* After reset, ensure that cache-clearing operation is conducted by writing 1 to ICACLR for initialization of TAG RAM.

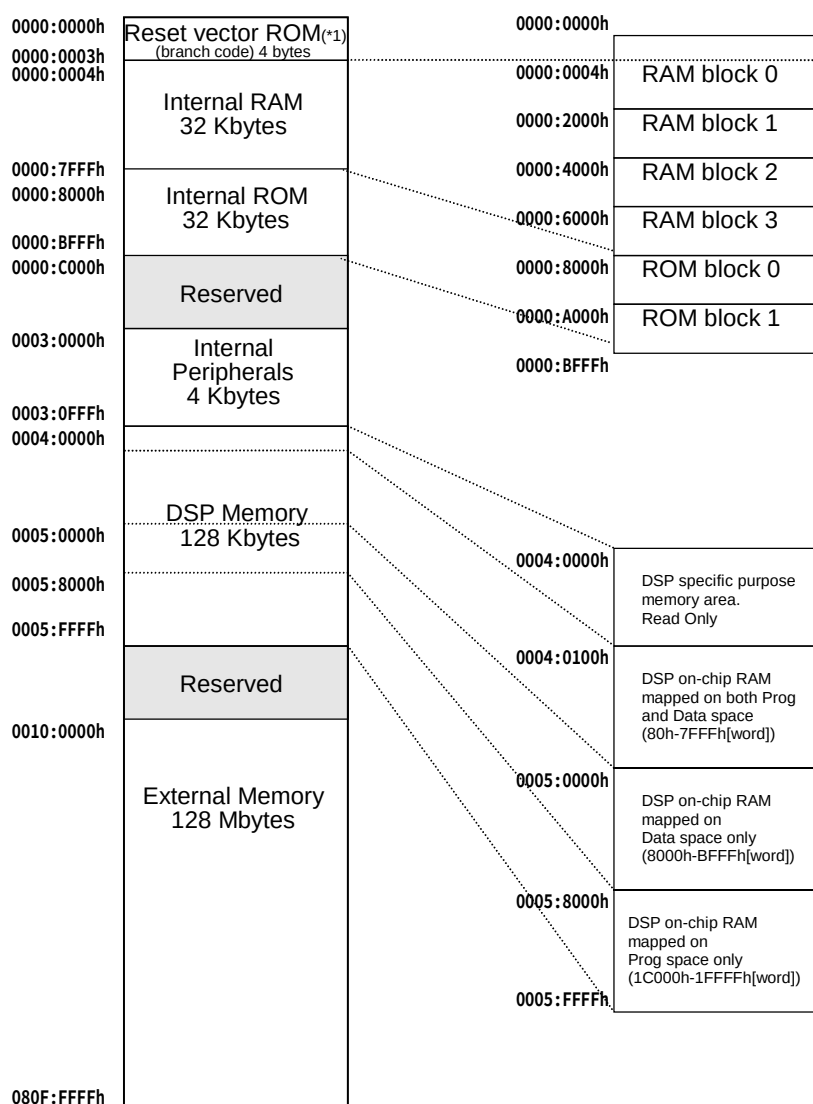
When the program located at external memory is updated, cache should be cleared by writing 1 to ICACLR.

REQ4W enable burst transfer (4 burst) from SDRAM to ICACHE, when the program is located in SDRAM. (When 32-bit SDRAM is used) REQ4W can be set, only when ICAENB is 1. (REQ4W and ICAENB can be set to 1 at same time.)

22. Appendix

Table 2 : ARM Memory Space

Area Name	Start address	End address	Size in bytes
Reset vector ROM ^(*1)	0x0000:0000	0x0000:0003	4
ARM Internal RAM	0x0000:0004	0x0000:7FFF	32 K
ARM Internal ROM	0x0000:8000	0x0000:BFFF	16 K
Internal Peripheral	0x0003:0000	0x0003:0FFF	4 K
DSP Memory	0x0004:0000	0x0005:FFFF	128 K
External Memory	0x0010:0000	0x080F:FFFF	128 M



(*1): Reset vector ROM is used when BTSEL pin is set '0'. In this case, the first 4 byte of RAM block 0 (0000:0000h-0000:0004h) is not used. On the other hand, When BTSEL = '1', the area of Reset vector ROM is replaced by the first 4 byte of the RAM.

Figure 4 : ARM Memory Space Image

INDEX

1. TIMER	6
TMMD0 (0003:0000) OFFSET : 0x0	6
TMMD1 (0003:0080) OFFSET : 0x0	6
TMMD2 (0003:0100) OFFSET : 0x0	6
TMMD3 (0003:0180) OFFSET : 0x0	6
TMPS0 (0003:0004) OFFSET : 0x2	6
TMPS1 (0003:0084) OFFSET : 0x2	6
TMPS2 (0003:0104) OFFSET : 0x2	6
TMPS3 (0003:0184) OFFSET : 0x2	6
TMVAL0 (0003:0006) OFFSET : 0x3	7
TMVAL1 (0003:0086) OFFSET : 0x3	7
TMVAL2 (0003:0106) OFFSET : 0x3	7
TMVAL3 (0003:0186) OFFSET : 0x3	7
TMTRG0 (0003:0008) OFFSET : 0x4	7
TMTRG1 (0003:0088) OFFSET : 0x4	7
TMTRG2 (0003:0108) OFFSET : 0x4	7
TMTRG3 (0003:0188) OFFSET : 0x4	7
TMCNT0 (0003:000A) OFFSET : 0x5	7
TMCNT1 (0003:008A) OFFSET : 0x5	7
TMCNT2 (0003:010A) OFFSET : 0x5	7
TMCNT3 (0003:018A) OFFSET : 0x5	7
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TXDATA0 (0003:0200) OFFSET : 0x0	8
TXDATA1 (0003:0280) OFFSET : 0x0	8
RXDATA0 (0003:0202) OFFSET : 0x1	8
RXDATA1 (0003:0282) OFFSET : 0x1	8
SIOEN0 (0003:0204) OFFSET : 0x2	8
SIOEN1 (0003:0284) OFFSET : 0x2	8
SIOMODE0 (0003:0206) OFFSET : 0x3	9
SIOMODE1 (0003:0286) OFFSET : 0x3	9
DMATRG0 (0003:0208) OFFSET : 0x4	9
DMAMODE0 (0003:020A) OFFSET : 0x5	9
DMASTAD0 (0003:020C) OFFSET : 0x6	10
DMASTAD1 (0003:020E) OFFSET : 0x7	10
DMASTAT0 (0003:0210) OFFSET : 0x8	10
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DTRR0 (0003:0300) OFFSET : 0x0	11
DTRR1 (0003:0380) OFFSET : 0x0	11
BRSR0 (0003:0302) OFFSET : 0x1	11
BRSR1 (0003:0382) OFFSET : 0x1	11
MSR0 (0003:0304) OFFSET : 0x2	12
MSR1 (0003:0384) OFFSET : 0x2	12
RFCR0 (0003:0306) OFFSET : 0x3	12
RFCR1 (0003:0386) OFFSET : 0x3	12
TFCR0 (0003:0308) OFFSET : 0x4	13
TFCR1 (0003:0388) OFFSET : 0x4	13
LCR0 (0003:030A) OFFSET : 0x5	13
LCR1 (0003:038A) OFFSET : 0x5	13
SR0 (0003:030C) OFFSET : 0x6	14
SR1 (0003:038C) OFFSET : 0x6	14
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WDTRST (0003:0402) OFFSET : 0x1	15
WDTPRSL (0003:0404) OFFSET : 0x2	15

WDTVAL (0003:0406) OFFSET : 0x3	15
WDTEXRST (0003:0408) OFFSET : 0x4	16
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MMCCLK (0003:0482) OFFSET : 0x1	17
MMCST0 (0003:0484) OFFSET : 0x2	18
MMCST1 (0003:0486) OFFSET : 0x3	19
MMCIM (0003:0488) OFFSET : 0x4	19
MMCTOR (0003:048A) OFFSET : 0x5	20
MMCTOD (0003:048C) OFFSET : 0x6	20
MMCBLEN (0003:048E) OFFSET : 0x7	21
MMCNBLK (0003:0490) OFFSET : 0x8	21
MMCNBLC (0003:0492) OFFSET : 0x9	21
MMCDRR (0003:0494) OFFSET : 0xA	21
MMCDXR (0003:0496) OFFSET : 0xB	22
MMCCMD (0003:0498) OFFSET : 0xC	22
MMCARGL (0003:049A) OFFSET : 0xD	23
MMCARGH (0003:049C) OFFSET : 0xE	23
MMCRSP0 (0003:049E) OFFSET : 0xF	23
MMCRSP1 (0003:04A0) OFFSET : 0x10	23
MMCRSP2 (0003:04A2) OFFSET : 0x11	24
MMCRSP3 (0003:04A4) OFFSET : 0x12	24
MMCRSP4 (0003:04A6) OFFSET : 0x13	24
MMCRSP5 (0003:04A8) OFFSET : 0x14	24
MMCRSP6 (0003:04AA) OFFSET : 0x15	24
MMCRSP7 (0003:04AC) OFFSET : 0x16	25
MMCDRSP (0003:04AE) OFFSET : 0x17	25
MMCETOK (0003:04B0) OFFSET : 0x18	26
MMCCIDX (0003:04B2) OFFSET : 0x19	26
MMCKC (0003:04B4) OFFSET : 0x1A	26
MMCTORC (0003:04B6) OFFSET : 0x1B	26
MMCTODC (0003:04B8) OFFSET : 0x1C	27
MMCBLNC (0003:04BA) OFFSET : 0x1D	27
DMATRG (0003:04BC) OFFSET : 0x1E	27
DMAMODE (0003:04BE) OFFSET : 0x1F	27
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DMASTAT0 (0003:04C4) OFFSET : 0x22	28
DMASTAT1 (0003:04C6) OFFSET : 0x23	28
DMATOR (0003:04C8) OFFSET : 0x24	29
SDIOCTL (0003:04CA) OFFSET : 0x25	29
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FIQENTRY1 (0003:0512) OFFSET : 0x09	32
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IRQENTRY1 (0003:051A) OFFSET : 0x0D	33
FISEL0 (0003:0520) OFFSET : 0x10	33
FISEL1 (0003:0522) OFFSET : 0x11	33
FISEL2 (0003:0524) OFFSET : 0x12	34

EINT0 (0003:0528) OFFSET : 0x014	34
EINT1 (0003:052A) OFFSET : 0x015	34
EINT2 (0003:052C) OFFSET : 0x016	35
INTRAW (0003:0530) OFFSET : 0x018	35
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EABASE1 (0003:053A) OFFSET : 0x01D	36
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INTPRI3 (0003:0546) OFFSET : 0x023	37
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INTPRI12 (0003:0558) OFFSET : 0x02C	39
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INTPRI15 (0003:055E) OFFSET : 0x02F	39
INTPRI16 (0003:0560) OFFSET : 0x030	39
INTPRI17 (0003:0562) OFFSET : 0x031	40
INTPRI18 (0003:0564) OFFSET : 0x032	40
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INV1 (0003:0588) OFFSET : 0x4	42
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BITSET0 (0003:058C) OFFSET : 0x6	42
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BITSET2 (0003:0590) OFFSET : 0x8	43
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BITCLR1 (0003:0594) OFFSET : 0xA	43
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VIDEOWIN0_OFFSET (0003:068C) OFFSET : 0x06	54

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VIDEOWIN1_AD_LOWER (0003:0698) OFFSET : 0x0C	56
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VIDEOWIN0_XL (0003:06A8) OFFSET : 0x14	58
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VIDEOWIN1_YP (0003:06AE) OFFSET : 0x17	59
VIDEOWIN1_XL (0003:06B0) OFFSET : 0x18	59
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OSDWIN0_XP (0003:06B4) OFFSET : 0x1A	60
OSDWIN0_YP (0003:06B6) OFFSET : 0x1B	60
OSDWIN0_XL (0003:06B8) OFFSET : 0x1C	60
OSDWIN0_YL (0003:06BA) OFFSET : 0x1D	60
OSDWIN1_XP (0003:06BC) OFFSET : 0x1E	61
OSDWIN1_YP (0003:06BE) OFFSET : 0x1F	61
OSDWIN1_XL (0003:06C0) OFFSET : 0x20	61
OSDWIN1_YL (0003:06C2) OFFSET : 0x21	61
CUR_XP (0003:06C4) OFFSET : 0x22	62
CUR_YP (0003:06C6) OFFSET : 0x23	62
CUR_XL (0003:06C8) OFFSET : 0x24	62
CUR_YL (0003:06CA) OFFSET : 0x25	62
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W0BMP23 (0003:06D2) OFFSET : 0x29	63
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W0BMP89 (0003:06D8) OFFSET : 0x2C	64
W0BMPAB (0003:06DA) OFFSET : 0x2D	64
W0BMPCD (0003:06DC) OFFSET : 0x2E	64
W0BMPEF (0003:06DE) OFFSET : 0x2F	65
W1BMP01 (0003:06E0) OFFSET : 0x30	65
W1BMP23 (0003:06E2) OFFSET : 0x31	65
W1BMP45 (0003:06E4) OFFSET : 0x32	65
W1BMP67 (0003:06E6) OFFSET : 0x33	66
W1BMP89 (0003:06E8) OFFSET : 0x34	66
W1BMPAB (0003:06EA) OFFSET : 0x35	66
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CONTROL_MISC (0003:06F4) OFFSET : 0x3A	67
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HDW (0003 :0704) OFFSET : 0x2	70
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PPLN (0003 :0708) OFFSET : 0x4	70
LPFR (0003 :070A) OFFSET : 0x5	71

SPH (0003 :070C) OFFSET : 0x6	71
LPH (0003 :070E) OFFSET : 0x7	71
SLV0 (0003 :0710) OFFSET : 0x8	71
SLV1 (0003 :0712) OFFSET : 0x9	71
LNv (0003 :0714) OFFSET : 0xA	72
CULH (0003 :0716) OFFSET : 0xB	72
CULV (0003 :0718) OFFSET : 0xC	72
HSIZE (0003 :071A) OFFSET : 0xD	72
SDOFST(0003 :071C) OFFSET : 0xE	73
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CADL(0003 :0720) OFFSET : 0x10	74
CCLAMP (0003 :0722) OFFSET : 0x11	74
CDCSUB (0003 :0724) OFFSET : 0x12	74
CCOLP (0003 :0726) OFFSET : 0x13	74
COFST0 (0003 :0728) OFFSET : 0x14	75
COFST1 (0003 :072A) OFFSET : 0x15	75
CMFIL (0003 :072C) OFFSET : 0x16	75
CGAIN (0003 :072E) OFFSET : 0x17	75
COFSTA (0003 :0730) OFFSET : 0x18	76
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OUTCLIP (0003 :0734) OFFSET : 0x1A	76
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VSRT (0003:0790) OFFSET : 0x08	81
VSIZ (0003:0792) OFFSET : 0x09	81
PVSET2 (0003:0794) OFFSET : 0x0A	81
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WB_GAIN1 (0003:079C) OFFSET : 0x0E	83
SMTH (0003:079E) OFFSET : 0x0F	83
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MTX_GAIN2 (0003:07AC) OFFSET : 0x16	85
MTX_GAIN3 (0003:07AE) OFFSET : 0x17	86
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